Pricing Interest Rate Derivatives Using The Forward Market Model

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Abstract

In this dissertation, we look at the LIBOR Market Model (LMM) and the generalized Forward Market Model (FMM). We first build up from classical short rate models and then proceed to look at the construction of the LMM along with its shortcomings. We then define and model forward risk-free rates that will be replacing IBORs and show that the LMM can be extended to model both forward-looking and backward-looking rates with the help of extended zero-coupon bond. This extension of the LMM is called the generalized Forward Market Model.

Declaration

Signature

Date

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List of abbreviations

ARRC Alternative reference rates commitee

BGM Brace Gatarek and Musiela

CBOE Chicago bond options exchange

CIR Cox-Ingersoll-Ross

CCP Central counterparty

CME Chicago Mercantile Exchange

Flr Black floor payoff

FMM Forward market model

FRA Forward rate agreement

FX Foreign exchange.

GBM Geometric Brownian motion

HJM Heath-Jarrow-Morton

IBOR Interbank offered rates

IRS Interest rate swap

ATM At-the-money

ITM In-the-money

OTM Out-the-money

ISDA International swaps and derivatives association

LMM LIBOR market model

OTC Over-the-counter

OIS Overnight index swap

PAI Price alignment interest

PDE Partial differential equation

PFS European payer swaption payoff

RFR Risk free rates

SARON Swiss average overnight rate

SDE Stochastic differential equation

SOFR Secured overnight funding rate

SONIA Sterling overnight index average

ZCB Zero-coupon bond

List of symbols

$\mathbb{Q}$ - Risk neutral measure

$\mathbb{Q}_0$ - Real world measure

$\sim$ - Equivalence

r_t - Short rate

C_t - Cash account

$S(t)$ - Underlying asset

$\pi(t)$ - Portfolio value

$\mathbb{E}^N$ - Expectation under the measure N

$D_{t,T}$ - The discount factor between t and T.

$\mathcal{N}$ - The notional amount

κ - The strike price

τ - Compounding period

f_t - The instantaneous forward rate

H_T - Time T payoff

$\mu(t, r_t)$ - Drift of the short rate

$\sigma(t, r_t)$ -Short rate volatility

$f_{t,T}$ - Forward rate

$P_{t,T}$ - Zero-coupon bond

Φ - The standard normal distribution function

ρ - Correlation coefficient

$\mathcal{F}_t$ - The filtration at time t

$W(t)$ - The Wiener process

CHAPTER 1

Introduction

1.1 Theory of derivatives

We start off by asking the question, what is a financial derivative? We can describe a *derivative* as a financial contract between two parties that derives its value from an underlying asset. This underlying asset can be any random variable like securities, currency, interest rates or even the amount of snow in a particular area. Derivatives are traded either on exchange markets or in the over-the-counter (OTC) markets. Derivatives were initially invented as a risk management tool in the commodities markets. So commodity derivatives were the first types of derivatives to be introduced. By commodity, we are referring to things like wheat, gold or silver etc. Financial derivatives followed. The term financial derivatives refers to a range of financial instruments including stocks, bonds, treasury bills, interest rates, etc. Examples of financial derivatives are futures, forwards, options which will be formally defined in the coming chapters. Theoretical work in the area of interest rate derivatives has produced a variety of models and techniques to value these options, some of which are widely used by practitioners.

A derivative exchange defines and standardizes contracts that can then be traded by the market participants. Derivative exchanges have been in existence for a long time. The largest exchange group is known as the Chicago Mercantile Exchange group which is formed by the Chicago Board of Trade (est 1848). In comparison to custom-made contracts, exchange-traded derivatives are more liquid and have low transaction costs. Some examples of exchange traded derivatives are Dow Jons, S&P 500, Nikki 225, NIFTY option which are traded on New York Stock Exchange, Tokyo Stock Exchange, National Stock Exchange, Bombay Stock Exchange and so on. As previously mentioned, some derivatives are traded in the OTC markets. These derivatives, however, expose the parties to risk such as counter-party risk, legal risk and operational risk. The OTC market for interest rate derivatives, however, has been growing rapidly in recent years and it has become one of the biggest and liquid option markets.

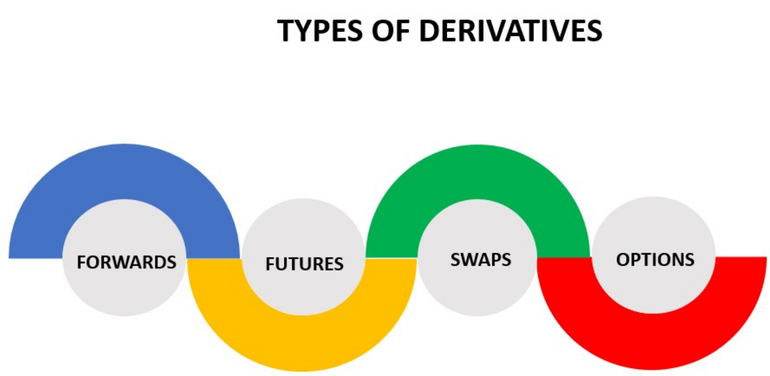


Figure 1.1: Derivatives classification.

Figure 1.1 shows how derivatives evolved through time. Forwards were the first to be introduced in the market. They were then followed by the emergence of futures. Swaps were then introduced in the market and they gained popularity quickly. The first OTC swap is thought to have been between IBM and the World Bank. Options were the last to be introduced in the market.

Since the integration of capital markets globally, there has been a rise in financial risk due to the frequent changes in the interest rates, currency exchange rate and stock prices. Financial derivatives were invented to manage and overcome the risks that are a result of the fluctuations of the previously mentioned variables. Derivatives can also be used to serve the following purposes:

(i) **Hedging or to reduce the risk one is exposed to**. Since the derivative price and the value of the underlying asset are linked, the contracts are primarily used for hedging risks. For example, an investor may purchase a derivative contract whose value moves in the opposite direction to the value of an asset the investor owns. In this way, profits in the derivative contract may offset losses in the underlying asset.

(ii) **Speculation**. Speculators trade based on their educated guesses on where they believe the market is headed. For example, if a speculator thinks that a stock is overpriced, they may sell short the stock and wait for the price to decline, at which point it can be bought back for a profit.

(iii) **Arbitrage**. This is the condition under which two equivalent assets or derivatives or combination of assets and derivatives or combination of assets and derivatives sell for different prices. This grants an arbitrageur to buy at a low price, and sell at a high price, and earn a risk-free profit from this transaction without investing any money.

1.2 Theory of short rates

The modelling of the term structure of interest rates has resulted in a lot approaches in the literature and new approaches continue to be explored and proposed by both academics and practitioners.

The first generation of models developed were based on the spot rate. This choice was due to a combination of mathematical convenience and tractability, or numerical ease of implementation. Furthermore, the most widely used of these models are one-factor models, in which the entire yield curve is specified by a single stochastic state variable, in this case the spot or short-term rate.

Despite allowing negative interest rates with positive probability, affine (Gaussian) models remain popular amongst market participants because they are easily implemented and they can also provide explicit numerical solutions when used to price derivatives. These models, however, are one factor models and the user of the models cannot freely choose the volatility structure, which is a major drawback. Multi-factor short rate models have properties that make them more desirable over single-factor models with the one stand out property being able to describe the intrinsic volatility structure and the Heath-Jarrow-Morton framework being the best multi-factor model on the spectrum.

Market models such as the HJM and LMM were proposed to address the issues posed by affine term structure models.

Brace, Gatarek, and Musiela (1997) proposed an alternative model to deal with the drawbacks of the HJM. We call this model the LIBOR Market Model (LMM) and the driving variable of the model is the forward rates which market participants use along with LIBOR discounting. Calibrating the model to the value of instruments such as interest rate caps is also easy.

1.3 Delimitation

There are many interest rate models in the literature representing different characteristics and assumptions. We discuss the different classes of interest rate models such as classical short rates and market models. Interest rate models that have distributions that are normal and lognormal account for several models. Hence, when pricing interest rate derivatives in the empirical study later on, we restrict ourselves to the selection of the LMM and the generalized forward market models. When pricing the simple exotic options, we will not be addressing the analytical pricing formulas.

Moreover, this dissertation concentrates only on the comparative study of the valuation of the interest models. Therefore, we will only be focusing on implementing the models and calibration theory of the aforementioned models is not considered. We also assume a 'smile-less' world. Calibration theory for market models are available in a number of various textbooks that mainly deal with interest rate derivatives valuation. For pricing purposes, the year fractions and volatilities are assumed to be constant. The interest rate products that are being priced here follow a European style. American style products are not discussed in the dissertation.

In the dissertation we first give an introduction to the theory of interest modelling. We then introduce classical short rate models and then proceed to market model and close off by giving a comparative study of the results. Below is a more detailed breakdown of what the reader can expect from the chapters in the dissertation.

- Chapter 2 *Principles of interest rates.* Introduces the reader to the basic theory of interest rates and forward rates relevant which will be important in the chapters that are to follow.

- Chapter 3 *Traditional derivatives pricing* introduces the theory of term structure modelling and the theory of pricing interest rate caps/floors and swaptions.

- Chapter 4 *Classical short rate models* introduces short rate models. We look at the equilibrium models, arbitrage-free models and conclude by discussing the HJM framework.

- Chapter 5 *LIBOR Market Model.* We discuss the theory of the LMM and how it is used to price interest rate derivatives such as caps and swaptions.

- Chapter 6 *The generalized Forward market model.* We discuss the theory of the generalized forward market model and its characteristics and how it is used to value interest rate derivatives such as caps and swaptions.

- Chapter 7 *Results.* Discuss the results obtained from the study.

- Chapter 8 *Conclusion.* Concludes the dissertation and also suggest alternative research areas.

CHAPTER 2

Principles of interest rates

This chapter introduces the theory of short rates, key definitions, forward rates and forward rate agreements.

2.1 Fundamental concepts

When a deposit is made into a bank account, the expectation is that the money should grow over time. It is common knowledge and wisdom that lending money requires some sort of compensation, making obtaining a particular amount tomorrow not equal to receiving the exact same amount today. In order for us to express this mathematically, we have to introduce some definitions and concepts. The first definition we look at is that of a cash account.

Definition 2.1. *The cash account is the value of a unitary investment at $t \geq 0$ made* *at $t = 0$. We can denote it by C_t and has dynamics given by*

$$dC_t = r_t C_t dt, \quad C(0) = 1$$

and has a solution given by

$$C_t = \exp\left(\int_0^t r_u du\right).$$

It is worth noting that the cash account C_t has the property of not being affected by a change of measure to a risk-adjusted probability measure and it is the only asset in finance with this property.

Definition 2.2. *A zero-coupon bond is a type of fixed income instrument that makes only one payment at maturity, which includes both interest accrued and the repayment of principal. The contract value at time $t < T$ is denoted by $P_{t,T}$. The time-t price $P_{t,T}$ of a zero-coupon bond is defined as the expectation*

$$P_{t,T} = \mathbb{E}^{\mathbb{Q}}\{e^{-\int_t^T r(u)du}\}$$

under the risk-neutral measure $\mathbb{E}^{\mathbb{Q}}$.

Zero-coupon bonds play a key role in interest rate modelling. These zero-coupon can be used to express all interest rates. On the subject of interest rates, they can be categorised into government rates, which are related to the government bonds and interbank rates, which are related to the rates that banks charge each other for swap transactions between them. Although we will be mainly interested the interbank sector, it is worth noting that the mathematical modelling can still be applied to government rates. Another important thing to note is that interest rates cannot be directly traded and that they are just quoted in the markets and zero-coupon bonds are just theoretical instruments and they cannot be observed directly.

Definition 2.3. *A stochastic process $D_{t,T}$ for $t \in [0, T]$ is the amount equivalent to one* *unit of currency that will be paid at time T defined by*

$$D_{t,T} = \frac{C_t}{C_t} = e^{-\int_t^T r_u du}$$

and it is called the (stochastic) discount factor.

Definition 2.4. *The Continuously-compounded spot interest rate for $t \in [0, T]$ is denoted by $R(t, T)$ and is defined as*

$$R(t, T) := -\frac{\ln P_{t,T}}{\tau(t, T)}$$

Definition 2.5. *The Simply-compounded spot interest rate for $t \in [0, T]$ is denoted by $L_{t,T}$ and is defined as*

$$L_{t,T} := \frac{1 - P_{t,T}}{\tau(t, T) P_{t,T}}.$$

The LIBOR rate is the most important simply-compounded spot rate as it is the rate at which banks lend each other and hence we denote the sinply-compounded spot interest rate by $L_{t,T}$.

The LIBOR bond prices is given as follows

$$P_{t,T} = \frac{1}{1 + L_{t,T}\tau(t, T)}.$$

We also consider another compounding method known as annual compounding.

Definition 2.6. *The Annually-compounded spot interest rate for $t \in [0, T]$ is denoted by $Y(t, T)$ and is given by*

$$Y(t, T) := \frac{1}{(P_{t,T})^{\frac{1}{\tau(t,T)}}} - 1.$$

The bond prices in terms of these rates is:

$$P_{t,T} = \frac{1}{(1 + Y(t, T))^{\tau}(t, T)}.$$

Definition 2.7. *The n-times-per-year compounded spot interest denoted by $Y^n(t,T)$ and is defined by*

$$Y^n(t,T) := \frac{n}{[P_{t,T}^{\frac{1}{n\tau(t,T)}}]} - n.$$

All the aforementioned spot interest rates are equivalent in infinitesimal time intervals. We can therefore define the short rate as follows:

Definition 2.8. *The instantaneous forward rate at time t with maturity T is given by:*

$$f_{t,T} = -\frac{\partial \ln P_{t,T}}{\partial T}.$$

Definition 2.9. *The instantaneous short interest rate r_t is the limit of each of the different spot rates for $t \in [0,T]$ with $T \to t^+$. Mathematically, it can be defined by:*

This definition can help us define the cash account mathematically so, whose value grows according to the instantaneous rate.

2.2 Forward rates

In this section we are introducing forward rates that are fundamental in interest rate modelling, more especially for market models as we shall see in the coming chapters. We begin with the following definition.

Definition 2.10. *Let $t \leq T \leq U$. The continuously-compounded forward interest rate is denoted by $R_{t,T,U}$ and is given by*

$$R_{t,T,U} := -\frac{\ln P_{t,U} - \ln P_{t,T}}{\tau(T,U)}.$$

With some algebra we get

$$e^{R\tau(T,U)} = \frac{P_{t,T}}{P_{t,U}}.$$

In the market, the preferred notation is the simple rate one, whereas the continuous one is used in theoretical contexts and does not regard the model we are exposing.

Definition 2.11. *A forward rate agreement (FRA) is an over-the-counter contract between two parties that determines the rate of interest to be paid between the expiry T and maturity U. One party (The holder) receives a fixed rate κ and pays a floating rate $L(T,U)$. It has a U-payoff given by*

$$\mathcal{N}\tau(T,U)(\kappa - L(T,U)),$$

where $\mathcal{N}$ is the contract nominal value. Furthermore, the LIBOR rate can substituted with its expression so that the definition becomes

$$\mathcal{N}\left(\tau(T,U)\kappa - \frac{1}{P_{T,U}} + 1\right).$$

It can be proved through the use of arbitrage-free arguments to give a time t valuation of the above FRA is given by

$$FRA(t,T,U,\tau_{T,U},\mathcal{N},\kappa) = \mathcal{N}[P_{t,U}\tau_{T,U}\kappa - P_{t,T} + P_{t,U}]. \tag{2.1}$$

We now introduce another interest rate known as the simply-compounded forward interest rate. This is the unique value that makes the FRA to be fair at time t. We get this value by setting the FRA value to zero. The resulting rate defines the simply-compounded forward rate.

Definition 2.12. *Let $t \leq T \leq U$. The simply-compounded forward interest rate at time t is denoted by $F_{t,T,U}$ and is defined by*

$$F_{t,T,U} := \frac{1}{\tau_{T,U}}\left(\frac{P_{t,T}}{P_{t,U}} - 1\right). \tag{2.2}$$

With some algebra we get the following equation

$$1 + \tau(T,U)F = \frac{P_{t,T}}{P_{t,U}}.$$

The value of FRA in terms of the forward rate is given as

$$\begin{aligned} FRA(t,T,U,\tau_{T,U},\mathcal{N},\kappa) &= \mathcal{N}\left[P_{t,U}\tau_{T,U}\kappa + P_{t,U}\left(1 - \frac{P_{t,T}}{P_{t,U}}\right)\right] \\ &= \mathcal{N}\left[P_{t,U}\tau_{T,U}\kappa + P_{t,U}\left(-\tau_{T,U}F_{t,T,U}\right)\right] \\ &= \mathcal{N}[P_{t,U}\tau_{T,U}(\kappa - F_{t,T,U}]. \end{aligned} \tag{2.3}$$

When we compare the payoff and the value of the above FRA, the forward rate $F_{t,T,U}$ can be thought of as a forecast of what the spot rate $L(T,U)$ will be at a future time T. It is worth noting that $F_{t,T,U}$ is stochastic at time t.

We can also mention another type of interest rate, which is analogous to the instantaneous short interest rate in the future.

Definition 2.13. *The instantaneous forward interest rate for $t \in [0,T]$ is denoted by f_t and is defined by*

$$\begin{aligned} f_t = & \lim_{U \to T^+} F_{t,T,U} \\ = & \lim_{U \to T^+} R_{t,T,U} \\ = & -\frac{\partial \ln P_{t,T}}{\partial T}. \end{aligned}$$

In this chapter, we introduced fundamental concepts pertaining to interest rate modelling such as the cash account, instantaneous short rates, and forward rates, which will be important in the chapters to follow. A thorough understanding of these concepts will help the reader understand the chapters to follow much better. In the next chapter, we look at how the short rates can be used to price interest rate derivatives.

CHAPTER 3

Traditional derivatives pricing

Many financial instruments cannot be priced by simply discounting the future cash flows since the instruments have uncertain future payoffs. Instead, they can be priced using dynamic interest rate models which generate and assign probabilities to events for future term structures and by so doing, enabling us to price instruments with non-deterministic cash flows. In their paper, Black and Scholes (1973) made the fundamental economic assumption of the absence of arbitrage in the financial market that is being considered. Loosely speaking, arbitrage-free implies that it is impossible to invest $\mathcal{N} = 0$ at $t = 0$ and get a payoff $H_T > 0$ at time $T \geq t$ with positive probability. The key idea behind the model is to hedge the option by buying and selling the underlying asset in just the right way and as a consequence, to eliminate risk. By using this trick, Black and Scholes could then conclude that the portfolio instantaneous return was equal to the instantaneous risk-free rate, which resulted to their much celebrated differential equation and their option pricing formula.

3.1 Traditional pricing theory

We say that a variable follows a *stochastic process* if over time, its dynamics involve some randomness or uncertainty about it. When expressed in terms of time, a stochastic process is said to be in discrete-time if its index set contains a finite or countable number of elements, such as a finite set of numbers, the set of integers, or the natural numbers. Time is said to be continuous if the index set is some interval of the real line. Discrete-time stochastic processes and continuous-time stochastic processes are the two types of stochastic processes. We introduce two major continuous stochastic processes: Wiener process and geometric Brownian motion.

Definition 3.1. *A continuous stochastic process W is called a Wiener process if the following are satisfied:*

(i) $W(0) = 0$.

(ii) W *has stationary independent increments.*

(iii) For each time $t > 0$*, the process* $W(t)$ *has a normal distribution.*

(iv) The function $t \to W(t)$ *is continuous in* t*, with probability 1.*

Definition 3.2. *A geometric Brownian motion* $S(t)$ *is the solution of an SDE with linear drift and diffusion coefficients*

$$dS(t) = \mu S(t)dt + \sigma S(t)dW(t) \tag{3.1}$$

where μ *is the drift and* σ *is the volatility.*

The SDE has the following analytic solution

$$S(t) = S(0) \exp\left(\left(\mu - \frac{\sigma^2}{2}\right) t + \sigma W(t),\right) \tag{3.2}$$

which is referred to as the predictor-correct method. Alternatively we can find rewrite the discrete version of (3.1) as follows

$$S(t_{i+1}) = S(t_i) + \mu S_{t_i}\delta t_i + \sigma S(t_i)\delta W(t_i). \tag{3.3}$$

Definition 3.3. *An Ito process or stochastic integral $X(t)$ is a stochastic process that has the following form*

$$dX(t) = \mu(t)X(t)dt + \sigma(t)X(t)dW(t). \tag{3.4}$$

Theorem 3.1. *Let $X(t)$ be an Ito process satisfying Equation (3.1) and let $f(x,t)$ be a function that is twice differentiable; then $f(X(t),t)$ is an Ito process, and*

$$d(f(X(t),t)) = \frac{\partial f}{\partial t}(X(t),t)dt + \frac{\partial f}{\partial X(t)}dX(t) + \frac{1}{2}\frac{\partial^2 f}{\partial^2 X(t)}dX^2(t), \tag{3.5}$$

where $dX(t)^2$ is defined by

$$dt^2 = 0 \tag{3.6}$$

$$dtdW(t) = 0 \tag{3.7}$$

$$dW(t)^2 = dt \tag{3.8}$$

$$\tag{3.9}$$

Definition 3.4. *Let $M(t)$ be a stochastic process. $M(t)$ is a martingale with respect to a probability measure $\mathbb{Q}_0$ iff*

(i) $\mathbb{E}^{\mathbb{Q}_0}\{|M(t)|\}$ *is bounded*

(ii) $\mathbb{E}^{\mathbb{Q}_0}\{M(t)|\mathcal{F}_u\} = M(u), \ \forall \ u \leq t.$

We now introduce some important definitions in Stochastic calculus including the

Radon-Nikodym derivative which describes how one can move from one measure to another.

Definition 3.5. *A contingent claim is a random variable X that represents the time T payoff from a seller to a buyer. A contigent claim is said to be marketable or attainable if there exists a self-financing trading strategy ω such that $\pi_T(\omega) = X$.*

Consider a financial market where the price fluctuation of stocks is given by a stochastic process $X = X(t),\ 0 \leq t \leq T$ on some probability space $(\Omega, \mathcal{F}, \mathbb{Q}_0)$. If this market is complete then any contingent claim H, viewed as a random variable on $(\Omega, \mathcal{F}, \mathbb{Q}_0)$, can be generated by a dynamic portfolio strategy based on the underlying stock process X. We say that this contingent claim is *attainable.*

Definition 3.6. *An arbitrage opportunity is the possibility to make a profit in a financial market without risk and without net investment of capital.*

Proposition 3.1. *Let $\mathbb{Q}$ be the risk-neutral measure and assume that H is an attainable contingent claim. Then, for each time $t,\ 0 \leq t\ \leq T$, there exists a unique π_t associated with H, i.e.,*

$$\pi_t = \mathbb{E}^{\mathbb{Q}}\{D_{t,T}H|\mathcal{F}_t\}. \tag{3.10}$$

Definition 3.7. *A financial market is Complete if and only if every contingent claim is attainable.*

Definition 3.8. *Suppose we have two equivalent measures, $\mathbb{Q}_0$ and $\mathbb{Q}$. It is possible to express $\mathbb{Q}_0$ through $\mathbb{Q}$ with the random variable $d\mathbb{Q}_0/d\mathbb{Q}$, known as the Radon-Nikodym derivative. The expectation of a variable S with respect to $\mathbb{Q}_0$ is given by*

$$\mathbb{E}^{\mathbb{Q}_0}\{S\} = \mathbb{E}^{\mathbb{Q}}\left\{\frac{d\mathbb{Q}_0}{d\mathbb{Q}}S\right\}.$$

We can also define a process on the filtration $\mathcal{F}_T$ through the Radon-Nikodym derivative as follows

$$\gamma(t) = \mathbb{E}^{\mathbb{Q}}\left\{\frac{d\mathbb{Q}_0}{d\mathbb{Q}}|\mathcal{F}_t\right\}.$$

This is another version of the Radon-Nikodym derivative because it also transfers the Radon-Nikodym derivative for random variables to Ito processes. It also satisfies the following property:

$$\mathbb{E}^{\mathbb{Q}}\left\{S\middle|\mathcal{F}_u\right\} = \gamma(t)^{-1}\mathbb{E}^{\mathbb{Q}_0}\left\{\gamma(t)S\middle|\mathcal{F}_u\right\}, \quad u \leq t \leq T.$$

When using the Radon-Nikodym derivative to move from one measure to another, only the drift parameter is affected, the volatility remains the same. Girsanov's theorem formalises the mapping of a SDE from one measure to another. We now introduce the *Girsanov's theorem.*

Theorem 3.2. *The Cameron-Martin-Girsanov Theorem, Girsanov Theorem states that if $\zeta = \{\zeta(t) : t \in [0,T]\}$ is a $\mathcal{F}_t$ predictable process such that $\mathbb{E}^{\mathbb{Q}_0}\left\{e^{\frac{1}{2}\int_0^T \zeta(t)^2 dt}\right\}$ is bounded. Then there exists a measure $\mathbb{Q}$ such that*

(i) $\mathbb{Q}$ *is equivalent to* $\mathbb{Q}_0$

(ii) $d\mathbb{Q}_0/d\mathbb{Q} = e^{-\int_0^T \zeta(t)dW(t) - \frac{1}{2}\int_0^T \zeta(t)^2 dt}$

(iii) $\tilde{W}(t) = W(t) = \int_0^T \zeta(u)du$ *is a* $\mathbb{Q}$*-Brownian motion.*

Theorem 3.3. *Let V be the value of a derivative, S be an underlying variable at t and $dC = rCdt$ be the evolution of the underlying cash account. If S follows the geometric Brownian motion with the constant drift r then the evolution of V is*

$$\frac{\partial V}{\partial dt} + rS\frac{\partial V}{\partial dS} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial^2 S(t)} = rV. \tag{3.11}$$

Proof. By equation (3.5),

$$\begin{aligned} dV &= \frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial S}dS + \frac{1}{2}\frac{\partial^2 V}{\partial^2 S}dS^2 \\ &= \left(\frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial S}dS + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial^2 S}dS^2\right) dt + \sigma S \frac{\partial V}{\partial S}dW \end{aligned}$$

since

$$dS = \mu S dt + \sigma S dW(t) \tag{3.12}$$

and

$$\begin{aligned} dS^2 &= \mu^2 S^2 dt^2 + \mu\sigma S^2 dt dW + \sigma^2 S^2 dW^2 \\ &= \sigma^2 S^2 dt \end{aligned}$$

by Equations (3.6), (3.7), and (3.8). Now consider a portfolio consisting of the derivative V, say a call option and β stocks. Then the portfolio has a cost given by $V + \beta S$. By the same argument as above, we see that

$$d(V+\beta S) = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S}dS + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial^2 S}dS^2 \beta\mu S\right) dt + \sigma S \frac{\partial V}{\partial S}dW + \sigma S \left(\frac{\partial V}{\partial S} + \beta\right) dW(t). \tag{3.13}$$

Now we let $\beta = -\frac{\partial V}{\partial S}$ to hedge away all risk in portfolio.

$$d(V + \beta S) = \left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial^2 S}dS^2 \beta\mu S\right) dt. \tag{3.14}$$

It is clear to see that the portfolio is free of risk or randomness. This result proves to be critical because since it is risk free, it must grow at a risk-free rate r over time. So,

$\frac{d}{dt}(V + \beta S) = r(V + \beta S) = r(V - S\frac{\partial V}{\partial S})$ and

$$r\left(V - S\frac{\partial V}{\partial S}\right) = \left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial^2 S}\right).$$

Moving the variables around results in the Black-Scholes equation:

$$\frac{\partial V}{\partial dt} + rS\frac{\partial V}{\partial dS} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial^2 S(t)} = rV$$

□

Definition 3.9. *A numeraire is any asset that is strictly positive for all $t \in [0, T]$. A numeraire can be used to denominate all prices in an economy.*

In the risk-neutral approach to valuation of derivatives, the value $\pi(t) = \pi(t, T, S(t))$ of an asset $S(t)$ at time-t is given by choosing a numeraire $N(t)$ and taking an expectation with respect to an equivalent martingale measure $\mathbb{Q}^N$ under which the discounted value of the derivative is a martingale. Hence $\pi(t)$ is defined from the expression

$$\frac{\pi(t)}{N(t)} = \mathbb{E}^{\mathbb{Q}^N}\left\{\frac{\pi(T)}{N(T)}\middle|\mathcal{F}_t\right\}. \tag{3.15}$$

If we choose the cash account C_t as the numeraire, then (3.15) can be written as

$$\begin{aligned}\pi(t) &= \mathbb{E}^{\mathbb{Q}^C}\left\{\frac{C_t}{N(T)}\pi(T)\middle|\mathcal{F}_t\right\} \\ &= \mathbb{E}^{\mathbb{Q}^C}\left\{e^{-\int_t^T r_u du}\pi(T)\middle|\mathcal{F}_t\right\}\end{aligned} \tag{3.16}$$

where $\mathbb{Q}^C$ is the equivalent martingale measure that makes $\frac{\pi(t)}{N(t)}$ a martingale. The exponent can be pulled out of the expectation if the interest rate is deterministic so that Equation (3.16) becomes

$$\pi(t) = e^{-\int_t^T r_u du} \mathbb{E}^{\mathbb{Q}^C}\left\{ \pi(T) \middle| \mathcal{F}_t \right\}. \tag{3.17}$$

Lastly, when interest rates are constant, i.e. $r(u) = r$ just like in the Black and Scholes model, Equation (3.17) further simplifies to

$$\pi(t) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}^C}\left\{ \pi(T) \middle| \mathcal{F}_t \right\}. \tag{3.18}$$

If we set the asset $S(t)$ to be the interest rate r_t, we can write the value of the derivative as $\pi(t) = \pi(t, T, r_t)$. Furthermore, for term structure modelling, interest rates cannot be assumed to be deterministic as this is too restrictive. Hence the value of a derivative with an interest rate as the underlying asset can only be given by Equation (3.15)

$$\pi(t) = \mathbb{E}^{\mathbb{Q}^C}\left\{ e^{-\int_t^T r_u du} \pi(T, T, r_t) \middle| \mathcal{F}_t \right\} \tag{3.19}$$

when the cash account C_t is taken as the numeraire. Unfortunately, Equation (3.19) cannot be simplified to Equation (3.17) or (3.18). It is hard to evaluate the expectation in Equation (3.19) due to the fact that it involves two terms that each depend on the underlying variable.

The realisation that the cash account, C_t is not the only viable choice as the numeraire or the risk-neutral measure for pricing was an essential step forward in term-structure modelling.

The zero-coupon bond $P_{t,T}$ can also be thought of as a numeraire. In fact, by taking $P_{t,T}$ as the numeraire, Equation (3.19) can now be evaluated. The equivalent martingale measure associated with using $P_{t,T}$ as the numeraire is the *T-forward measure*. We use the fact that $P_{t,T} = 1$. In Equation (3.15), let $N(t) = P_{t,T}$. Then

$$\frac{\pi(t)}{P_{t,T}} = \mathbb{E}^{\mathbb{Q}^T}\left\{ \frac{\pi(T)}{P_{t,T}} \middle| \mathcal{F}_t \right\} \tag{3.20}$$

then

$$\pi(t) = P_{t,T}\mathbb{E}^{\mathbb{Q}^T}\left\{\pi(T)\big|\mathcal{F}_t\right\} \tag{3.21}$$
$$= \mathbb{E}^{\mathbb{Q}^C}\left\{e^{-\int_t^T r_u du}\big|\mathcal{F}_t\right\}\mathbb{E}^{\mathbb{Q}^T}\left\{\pi(T)\big|\mathcal{F}_t\right\}.$$

Equation (3.21) is the key result. We see that by using $P_{t,T}$, we can convert the expectation under $\mathbb{Q}^C$ of a product of two terms in Equation (3.19) into the product of two expectations in Equation (3.21), one under $\mathbb{Q}^C$ and the other under $\mathbb{Q}^T$, which can be easily evaluated. The key take away from this is that $\pi(t)$ can be written as

$$\pi(t) = \mathbb{E}^{\mathbb{Q}^C}\left\{e^{-\int_t^T r_u du}\pi(T)\big|\mathcal{F}_t\right\} \tag{3.22}$$
$$= P_{t,T}\mathbb{E}^{\mathbb{Q}^T}\left\{\pi(T)\big|\mathcal{F}_t\right\}. \tag{3.23}$$
$$\tag{3.24}$$

3.2 Pricing interest rate derivatives

Plain Vanilla Interest Rate Swap is an agreement between two parties (known as counterparties) where one stream of future interest payments is exchanged for another based on a specified principal amount. Interest rate swaps often exchange a fixed payment for a floating payment that is linked to an interest rate (most often the 3M LIBOR). The value date of interest rate swaps can be effective on spot or any specified date in the future.

Suppose that we have that we have n floating leg payoffs for the tenor structure $T = T_0, \dots, T_n$ with the corresponding set of year fractions $\tau = \tau_0, \dots, \tau_n$. Let $L(T_{i-1}, T_i)$ be a floating forward rate and n be a number of fixing times of the floating leg. Let κ be the fixed rate and n be a number of fixing times of the fixed rate. Then for every time

$T_i \in T_{\alpha+1}, \ldots, T_\eta$ one party gives the floating leg payments defined by

$$N\tau_i L(T_{i-1}, T_i)$$

and receives the fixed leg payments given by

$$N\tau_i\kappa,$$

where N is the notional value. We have two types of IRS: a *payer IRS*, and a *receiver IRS*. The discounted payoff at a time $t \leq T_\alpha$ of a IRS is given by

$$\sum_{i=\alpha+1}^{\eta} D_f(t, T_i) N\tau_i v \left(L(T_{i-1}, T_i) - \kappa\right),$$

where $v = -1$ is a receiver IRS and $v = 1$ is a payer IRS.

Figure 3.1: A graph showing how swaps work.

In a swap contract, the payer recieves the floating interest rate, usually the LIBOR rate, and pays a fixed interest rate. We now denote *payer IRS* and *receiver IRS* as PFS and RFS respectively. The arbitrage-free value at $t < T_\alpha$ of a PFS with a unit notional amount is:

$$PFS(t,T,\kappa) = \mathbb{E}^{\mathbb{Q}}\left\{\sum_{i=\alpha+1}^{\beta} D_f(t,T_i)N\tau_i(F_t(T_{i-1},T_i)-\kappa)\middle|\mathcal{F}_t\right\} \tag{3.25}$$

$$= \sum_{i=\alpha+1}^{\beta} P_t(T_i)\tau_i E^{Q_i}\left\{L_t(T_{i-1},T_i)-\kappa\middle|\mathcal{F}_t\right\} \tag{3.26}$$

$$= P_t(T_\alpha) - P_t(T_\beta) - \kappa\sum_{i=\alpha+1}^{\beta} P_t(T_i)\tau_i \tag{3.27}$$

$$= \sum_{i=\alpha+1}^{\beta} P_t(T_i)\tau_i F_t(T_{i-1},T_i) - \kappa. \tag{3.28}$$

Proof. The formula (3.27) can be obtained analogously to the price of a FRA: the floating payment determined at T_{i-1} and payed at T_i can be rewritten as

$$\frac{1}{P_{T_{i-1}}(T_i)} - 1.$$

That is the equivalent of

$$P_t(T_{i-1}) - P_t(T_i) \tag{3.29}$$

at time $t < T_\alpha$, so that the arbitrage-free value at t of the whole floating side is given by

$$\sum_{i=\alpha+1}^{\beta}(P_t(T_{i-1}) - P_t(T_i)) = P_t(T_\alpha) - P(t,T_\beta);$$

on the other hand the amount $\tau_i\kappa$ at time T_i equals to have $P_t(T_i)\tau_i\kappa$ units of currency at time t, so that the arbitrage-free value at t of the whole fixed side is

$$\sum_{i=\alpha+1}^{\beta} P_t(T_i)\tau_i\kappa = \kappa\sum_{i=\alpha+1}^{\beta} P_t(T_i)\tau_i.$$

Equivalently we could have obtained (3.29) by the risk neutral pricing formula 3.10

$$
\begin{aligned}
&\mathbb{E}^{\mathbb{Q}}\left\{D_{t,T}\left(\frac{1}{P_{T_{i-1}}(T_i)}-1\right)\middle|\mathcal{F}_t^W\right\} = \\
&= \mathbb{E}^{\mathbb{Q}}\left\{e^{-\int_t{}_i^T r_s ds}\left(\frac{1}{P_{T_{i-1}}(T_i)}-1\right)\middle|\mathcal{F}_t^W\right\} - \mathbb{E}^{\mathbb{Q}}\left\{D_{t,T}\middle|\mathcal{F}_t^W\right\} \\
&= \mathbb{E}^{\mathbb{Q}}\left\{e^{-\int_t{}^{T_i} r_s ds}\mathbb{E}^{\mathbb{Q}}\left\{e^{\int_{T_{i-1}}^{i} r_s ds}\middle|\mathcal{F}_t^W\right\} - P_t(T_i)\right\}, \\
&= \mathbb{E}^{\mathbb{Q}}\left\{e^{-\int_t^{T_{i-1}} r_s ds}\middle|\mathcal{F}_t^W\right\} - P_t(T_i) = P_t(T_{i-1}) - P_t(T_i).
\end{aligned}
$$

Then the formula (3.28) can be obtained by substituting the expression for forward rates (2.2), as we made in (2.3). □

Definition 3.10. *The forward swap rate $S_{\alpha,\beta}(t)$ at time t is the fixed rate that the swap receiver demands in exchange for the uncertainty of having to pay the floating rate over time. Mathematically this is given by*

$$
S_{\alpha,\beta}(t) = \frac{P_t(T_\alpha) - P_t(T_\beta)}{\sum_{i=\alpha+1}^{\beta} \tau_i P_t(T_i)}.
$$

Definition 3.11. *An interest rate cap is a financial contract that can be viewed as a payer interest rate swap where each exchange payment is executed if and only if it has a positive value. The discounted payoff of a cap at time t is given by*

$$
\sum_{i=\alpha+1}^{\beta} D_f(t, T_i) N \tau_i \left(L(T_{i-1}, T_i) - \kappa\right)_+ \tag{3.30}
$$

and it is associated with the tenor structure $T = T_\alpha, \ldots, T_\beta$ which has a corresponding set of year fractions $\tau = \tau_\alpha, \ldots, \tau_\beta$, and a notional amount N.

We can define a floor in an analogous way as follows.

Definition 3.12. *An interest rate floor is a financial contract that can be viewed as a receiver interest rate swap where the holder pays the floating leg and receives the constant* *rate. The discounted payoff is given by*

$$\sum_{i=\alpha+1}^{\beta} D_f(t,T_i) N \tau_i \left(\kappa - L(T_{i-1},T_i)\right)_+ .$$

Each such term defines a contract that is termed *caplet*. The *floorlet* contracts are defined in an analogous way.

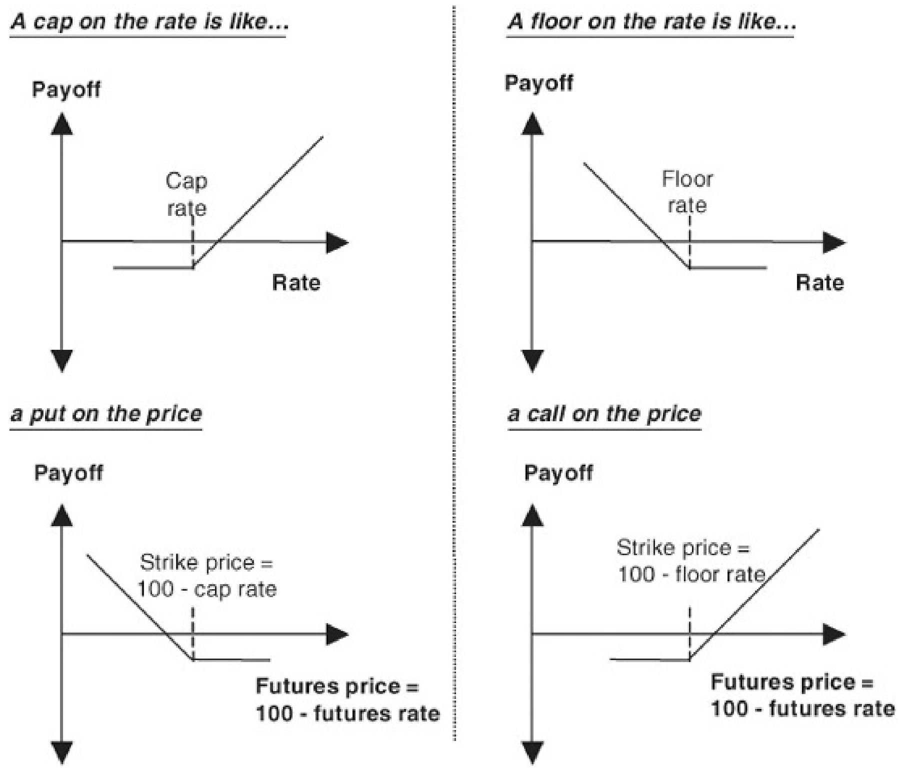


Figure 3.2: Floors. Source:https://www.oreilly.com.

In Figure 3.2, we see a graph of a cap and a floor payoff with a strike price $\kappa = 100$. In the case of a cap, the payoff is negative if the rate is below the strike price and it becomes positive if the rate is above the strike price. The floor case can be explained analogously so.

Market participants preferably use the Black's formula to price caps/floors. This formula was originally developed in 1976 to price commodity options and it is often referred to as the Black76 model. The price at time t of the cap with an associated tenor structure T and a Notional N is given by

$$\mathbf{Cap}^{Black}(t, T, \tau, N, \kappa, v) = N \sum_{i=\alpha+1}^{\beta} P_t(T_i)\tau_i \mathrm{Black}\left(\kappa, F_t(T_{i-1}, T_i), v_i\right), \tag{3.31}$$

where, denoting by Φ the standard CDF,

$$\begin{aligned}
\mathrm{Black}(\kappa, F(t;T_{i-1},T_i), v_i) &= F(t;T_{i-1},T_i)\Phi(d_1(\kappa, F(t;T_{i-1},T_i), v)) - \kappa\Phi(d_1(\kappa, F(t;T_{i-1},T_i), v)),\\
d_1(\kappa, F, v) &= \frac{\ln F/\kappa + v^2/2}{v},\\
d_2(\kappa, F, v) &= \frac{\ln F/\kappa - v^2/2}{v},\\
v_i &= v\sqrt{T_{i-1}},
\end{aligned}$$

with the common volatility parameter σ that is retrieved from market quotes. Analogously, the Black's formula for floorlets is

$$\mathbf{Flr}^{Black}(t, T, \tau, N, \kappa, v_i) = N \sum_{i=\alpha+1}^{\beta} P_0(T_i)\tau_i \mathrm{Black}(\kappa, F(0, T_{i-1}, T_i) v_i),$$

There are several caps (floors) whose implied volatilities are quoted by the market. For example, we have quotes with $\alpha = 0$, T_0 equal to three months, and all other T_i's equally three-month spaced.

We also have nonstandard caps/floors that are available in the market such as Ratchet Caps and Flexi caps. These are often referred to as simple exotic options because their payoff is European style. These caps differ from the standard cap by incorporating conditions to determine how the cap rate κ is set for each caplet. Consider a ratchet cap. The cap rate in this type of cap is given by the LIBOR rate at the previous reset date plus a spread. In a sticky cap it is given by the previous capped rate plus a spread.

Definition 3.13. *An interest rate ratchet cap is a financial contract consisting of a series of caplets paying at $\{T_{\alpha+1}, \ldots, T_\beta\}$ where the strike is equal to the sum of the LIBOR rate at the previous reset date a specified spread $\epsilon \in \mathbb{R}$. The discounted payoff is given by*

$$\sum_{i=\alpha+1}^{\beta} D_f(t, T_i) N \tau_i (L(T_{i-1}, T_i) - (L(T_{i-2}, T_{i-1}) + \epsilon))_+.$$

Definition 3.14. *An interest rate sticky cap is a financial contract consisting of a series of caplets paying at $\{T_{\alpha+1}, \ldots, T_\beta\}$ where the strike is equal to the sum of the cap rate at the previous reset date a specified spread $\epsilon \in \mathbb{R}$. The discounted payoff is given by*

$$\sum_{i=\alpha+1}^{\beta} D_f(t, T_i) N \tau_i (L(T_{i-1}, T_i) - (\min(L(T_{i-2}, T_{i-1}), \kappa) + \epsilon))_+.$$

We cannot use the Black76 formula to price this cap. One has to resort to approximations for the analytical formula for pricing such caps. In this thesis, we will not be addressing the analytical pricing of the ratchet and sticky caps.

We also have another popular interest rate derivative called swaption. This is an option where the underlying asset is the interest rate swap. We have a payer swaption and a receiver swaption.

Definition 3.15. *A European payer swaption with strike κ is a contract that grants the holder. The holder is the payer of the fixed interest rate and the receiver of the floating interest rate. The holder implements the contract if the fixed rate is more than the strike rate at the swap maturity T_α. We also call it a Put swaption.*

Proposition 3.2. *The payer swaption payoff at its first reset date T_α, which is also the* *swaption maturity, is*

$$(PFS(T_\alpha, T, \kappa))_+ = N \sum_{i=\alpha+1}^{\beta} P_{T_\alpha}(T_i)\tau_i(F_{T_\alpha}(T_{i-1}, T_i) - \kappa)_+ \tag{3.32}$$

$$= (S_{\alpha,\beta}(T_\alpha) - \kappa)_+ \sum_{i=\alpha+1}^{\beta} P_{T_\alpha}(T_i)\tau_i \tag{3.33}$$

Proof. We can obtain the formula (3.32) by taking the positive part of the T_α-price of a PFS in the form (3.28), whilst the formula (3.33) is obtained from the other version 3.27 of the same price as follows:

$$(PFS(T_\alpha, T, \kappa))_+ = (P_{T_\alpha}(T_\alpha) - P_{T_\alpha}(T_\beta) - \kappa \sum_{i=\alpha+1}^{\beta} P_{T_\alpha}(T_i)\tau_i)_+$$

$$= (S_{\alpha,\beta}(T_\alpha) - \kappa)_+ \sum_{i=\alpha+1}^{\beta} P_{T_\alpha}(T_i)\tau_i$$

□

Market participants use the Black's formula for swaptions when pricing swaptions. The price at time t of the payer swaption is given by:

$$PS^{Black}(t, T_\alpha, T, \kappa, v_{\alpha,\beta}) = \sum_{i=\alpha+1}^{\beta} P_{T_\alpha}(T_i)\tau_i \text{Black}(\kappa, S_{\alpha,\beta}(t), v_{\alpha,\beta}\sqrt{T_\alpha - t}). \tag{3.34}$$

We can define a *Receiver Swaption* in an analogous way.

In this chapter, we introduced the interest rate derivatives that we will be looking at pricing in this dissertation such as interest rate swaps, caps/floors, and swaptions. We also discussed the arbitrage-free theory of those pricing interest rate derivatives. As the name suggests, the underlying product in these interest rate derivatives is the interest short rate introduced is Chapter 2, so the modelling of these short rate is imperative in finance. In the next part of the dissertation, we look at some of the affine term structure models that have been introduced in the literature and conclude with the HJM framework.

CHAPTER 4

Classical short rate models

In finance, fund managers and other market participants are heavily reliant on the movements of interest rates over some period to make a decision. A short rate model is a mathematical model used illustrate the evolution of interest rates over time by determining the evolution of the short rate r_t over time. One factor models for the short rate r were the first to be introduced in the literature. Since the bonds and rates are readily defined, it makes modelling the dynamics of r convenient. In general, short rate models assume that the dynamics of the short rate can be given by a stochastic process:

$$dr_t = \mu(t, r_t)dt + \sigma(t, r_t)dW(t) \tag{4.1}$$

where μ and σ are the drift and volatility functions of the short rate and W is a one dimensional Wiener process under the risk neutral measure $\mathbb{Q}$.

Short rate models can be classified as affine term structure models such as equilibrium models and arbitrage-free models, yield curve models such as the Heath-Jarrow-Morton model (HJM), and market models. In this chapter, we discuss some of the famous equilibrium and arbitrage-free models and close off with the HJM model. Market models will be discussed in the coming chapters.

4.1 Equilibrium models

Equilibrium models typically begin with an assumption for short-term interest rates, which are usually derived from more general assumptions about the state variables that describe the overall economy. Using the assumed process for short-term rates, one can determine the yield on longer-term bonds by looking at the expected path of interest rates until the bond's maturity. One of the primary advantages of equilibrium models is that the prices of many popular securities have closed-form analytic solutions. The first model we look at is the Vasicek model.

The Vasicek model

Vasicek (1977) proposed the first equilibrium model. In his derivation he assumed that the short rate r_t follows an Ornstein process. He proceeded to show that the dynamics are given by

$$dr_t = k[\theta - r_t]dt + \sigma dW(t), \qquad r(0) = r_0$$

where r_0, k, θ, and σ are positive constants. A special feature of the Vasicek's model is that the stochastic differential Equation (4.1) has a closed form solution that can be found utilizing the method of variations of constants.

The homogeneous equation

$$dr_t = -\lambda r_t dt$$

has the solution given by

$$r_t = Ke^{-\lambda t} \tag{4.2}$$

where K is an arbitrary constant. We require a particular solution to the inhomogenous equation in the form (4.2) with the constant K replaced by an unknown function $\Phi(t)$,

$$r_1(t) = \Phi(t)e^{-\lambda t}$$

$\Phi(t)$ has to satisfy the ordinary differential equation:

$$d\Phi(t) = \lambda \mu e^{\lambda t} dt + \sigma e^{\lambda t} dW(t).$$

Consequently

$$\Phi(t) = \mu e^{\lambda t} + \sigma \int_0^t e^{\lambda v} dW(v).$$

The solution to our problem is the sum of the solution (4.2) with $K = r_0 - \mu$ and the particular solution $r_1(t)$:

$$r_t = r(v)e^{-\lambda(t-v)} + \mu(1 - e^{-\lambda(t)(t-v)}) + \sigma \int_0^t e^{-\lambda(t)} dW(v). \tag{4.3}$$

The Vasicek model has a mean reversion but it does, however, allow negative interest rates which is not really expected in the market. It is worth noting that this negative interest rates can only be exhibited under extreme conditions.

The Dothan model

$\mathbb{Q}_0$ be given by

$$dr_t = ar_t dt + \sigma r_t dW(t) \tag{4.4}$$

where $r_0,\ a$ and σ are non-negative constants.

$$dr_t = \bar{a}r_t dt + \sigma r_t dW(t). \tag{4.5}$$

The dynamics (4.5) are integrated as follows

$$r_t = r(v)e^{(a-\frac{1}{2}\sigma^2)dt+\sigma dW(t)}$$

for $v \leq t$ so that

$$\mathbb{E}^{\mathbb{Q}}\{r_t|\mathcal{F}_v\} = r^2(v)e^{2adt} \tag{4.6}$$

$$\mathrm{Var}\{r_t|\mathcal{F}_v\} = r^2(v)e^{2adt}(e^{\sigma^2 dt} - 1). \tag{4.7}$$

Although this is positive, we observe that the model has the mean-reversion property if and only if $a < 0$ and the mean reversion level is necessarily zero. The exponential Vasicek model was introduced in an attempt to address this mean-reversion restriction. The Dothan model is the only lognormal short rate model that allows for an analytical formula for the zero coupon bond price and it is commonly referenced in the bond pricing literature.

The main drawback of this model is that just like all lognormal models for the interest rate short rate, $\mathbb{E}^{\mathbb{Q}}\{C_t\} = \infty$, in other words the cash account explodes. This drawback can, however, be overcome by resorting to approximating trees since one deals with a finite number of states, and hence with finite expectations.

The Cox-Ingersoll-Ross model

The Cox-Ingersoll-Ross (1985) model, commonly known as the CIR model was presented as an alternative to the Vasicek model. This model is defined by the following SDE:

$$dr_t = \alpha(\beta - r_t)dt + \sigma\sqrt{r_t}dW(t) \tag{4.8}$$

where α, β and σ are non-negative constants. By imposing the condition $2\alpha\beta > \sigma^2$, we ensure that a zero valued short rate is avoided. The CIR model is mean-reverting, but ...

The mean and variance of r_t conditional on $\mathcal{F}_v$ are given by

$$\mathbb{E}^{\mathbb{Q}}\{r_t|\mathcal{F}_v\} = r(v)\xi_v^t + \beta(1 - \xi_v^t) \tag{4.9}$$

$$\mathrm{Var}\{r_t|\mathcal{F}_v\} = \frac{r(v)\sigma^2}{\alpha}(\xi_v^t - e^{-2\alpha(t-v)}) + \frac{\beta\sigma^2}{2\beta}(1 - \xi_v^t)^2 \tag{4.10}$$

where $\xi_v^t = e^{-\alpha(t-v)}$. As in the Vasicek model, β can thought of as a smoothed average short rate which is targeted by the central bank. The CIR model has the advantage over the Vasicek model that for the condition $2\alpha\beta > \sigma^2$ the interest rates can never be negative.

The exponential Vasicek model

Let us consider an Ornstein-Uhlenbeck process z under the risk-neutral measure $\mathbb{Q}$ given by

$$dz(t) = [\beta - \alpha z(t)]\,dt + \sigma dW(t), \quad z(0) = z_0$$

where β, α, and σ are non-negative constants and $z_0 \in \mathbb{R}$. By setting $r_t = e^{z(t)}$, for each time t and applying Ito's lemma, we have that the short rate r_t satisfies the following

SDE

$$dr_t = r_t[\beta + \frac{\sigma^2}{2} - \alpha \ln r_t]dt + \sigma r_t dW(t). \tag{4.11}$$

The explicit solution to (4.11) for all $v \leq t$, is obtained by

$$r_t = \exp\left\{\ln r(v)\xi_v^t + \frac{\beta}{\alpha}(1 - \xi_v^t) + \int_s^t \xi_v^t + \int_v^t e^{-\alpha(t-v)dW(u)}\right\}$$

so that r_t conditional on $\mathcal{F}_v$ has a distribution that is log-normal with normal distribution given by

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}\{r_t\} &= \exp\left\{\ln r(v)\xi_v^t + \frac{\beta}{\alpha}\left(1 - \xi_v^t\right)\frac{\sigma^2}{4\alpha}\left[1 - e^{-2\alpha(t-v)}\right]\right\} \\
\mathbb{E}^{\mathbb{Q}}\{r_t\} &= \exp\left\{2\ln r(v)\xi_v^t + 2\frac{\beta}{\alpha}\left(1 - \xi_v^t\right)\frac{\sigma^2}{\alpha}\left[1 - e^{-2\alpha(t-v)}\right]\right\}
\end{aligned}$$

where $\xi_v^t = e^{-\alpha(t-v)}$. Therefore, contrary to the Dothan (1978) process, this is always mean reverting since

$$\lim_{t\to\infty} \mathbb{E}^{\mathbb{Q}}\{r_t|\mathcal{F}_v\} = \exp\left(\frac{\beta}{\alpha} + \frac{\sigma^2}{4\alpha}\right). \tag{4.12}$$

Another advantage of this model is that it is lognormally distributed, which implies that the short rate $r_t > 0$ at all times.

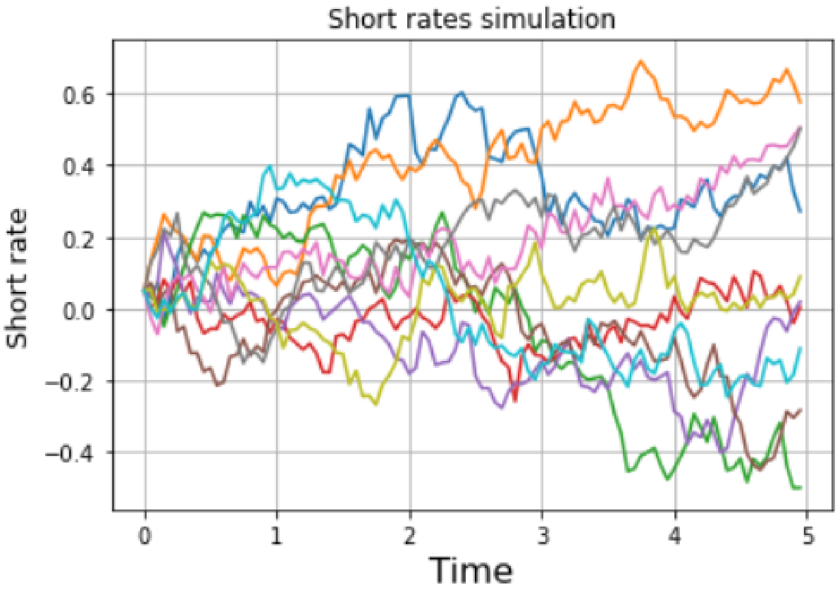


Figure 4.1: The Vasicek model simulation.

Figure 4.1 shows 10 Vasicek simulations where $r(0) = 0.01$, $T = 5$, $\sigma = 0.20$. We can see that five paths always converge to 0.01, which is the mean, as we approach $T = 5$ and 5 path is suggesting that we can expect negative interest rates, which are not usually observed in the market.

Generalized one-factor short rate models

It is possible to nest most of the models within the following stochastic process

$$dr_t = (\alpha + \beta r_t)dt + \rho r_t^{\gamma} dW(t) \tag{4.13}$$

where the parameters α, β, γ and ρ are constants. The Vasicek model corresponds to $\gamma = 0$ and the Cox-Ingersoll-Ross model corresponds to $\gamma = 1/2$. Note that when $\gamma > 0$, the volatility increases with the level of interest rate. This is called the *level effect.*

Equilibrium models generally assume that the drift and volatilities are constant parameters and this results in the term structure not matching the observable term structure in the market. Arbitrage-free models were developed in an attempt to address these shortcomings.

4.2 Arbitrage-free models

Arbitrage-free models were developed to address the shortcomings of equilibrium models. As already mentioned, the fact that equilibrium models assume constant parameters with the volatility parameter being the most crucial one when deriving the dynamics of the yield curve is not favored by a lot of market participants. In order to ensure that zero-coupon prices are exactly the same as what is observable in the market, we consider non-constant parameters for $\mu(r, t)$ and $\sigma(r, t)$. Arbitrage-free models were designed to ensure that the yield curve that is being outputted is consistent with what is observable in the markets. This implies that if we have a yield curve that has a positive slope, the short rate r_t will also have a positive slope, the same applies if the yield curve is decreasing or it is humped. The first arbitrage-free model we look at is the Ho-Lee model.

The Ho-Lee model

In 1986, Ho and Lee proposed the first term structure model. It has also a discrete time and continuous time versions. The continuous-time version is defined by the SDE

$$dr_t = \theta(t)dt + \sigma(r)dW(t)$$

where $\sigma(r)$ is constant. Here, $\theta(t)$ is a function of time and it is usually chosen by observing market data and calibrating it to the data. In choosing the drift term, we have to be very careful or we risk having negative interest rates which is not usually expected in financial applications.

The Hull-White model

In 1990, Hull and White published a paper exploring the extension of the Ho-Lee model to include mean reversion. This addition eliminates the problem of negative interest rates and has found wide application. that provide an exact fit to the initial term structure. The model is described by the following SDE

$$dr_t = [\theta(t) - \alpha r_t]dt + \sigma r_t^{\beta} dW(t)$$

where α and σ are constant. The Hull-White model contains many popular arbitrage-free models as special cases. For example, when $\alpha = 0$ and σ is constant we get the Ho-Lee model and when $\theta(t) = \theta$ we get the Vasicek model. It can be proved that, $f^M(0,T)$, the market instantaneous forward rate at time 0 for the maturity T i.e.

$$f^M(0,T) = -\frac{\partial \ln P_t^M(T)}{\partial T} \tag{4.14}$$

with $P_t^M(T)$ the market discount factor for the maturity T, we must have

$$\theta(t) = f_t(0,t) + \alpha f(0,t) + \frac{\sigma^2}{2\alpha}(1 - e^{-2\alpha t}).$$

We integrate (4.2) and get the following result

$$\begin{aligned} r_t &= r(v)\xi_v^t + \int_v^t \xi_v^t \theta(u)du + \sigma \int_v^t \xi_v^t dW(u), \\ &= r(v)\xi_v^t + \beta(t) - \beta(v)\xi_v^t + \sigma \int_v^t \xi_v^t dW(u) \end{aligned}$$

where

$$\beta(t) = F^M(0,t) + \frac{\sigma^2}{2\alpha^2}(1 - e^{-\alpha t})^2.$$

Therefore, r_t conditional on $\mathcal{F}_v$ has a normal distribution with mean and variance given respectively by

$$\mathbb{E}^{\mathbb{Q}}\{r_t|\mathcal{F}_v\} = r(v)\xi_v^t + \alpha(t) - \alpha(v)\xi_v^t$$
$$\mathrm{Var}\{r_t|\mathcal{F}_v\} = \frac{\sigma^2}{2\alpha}\left[1 - e^{-2\alpha(t-v)}\right].$$

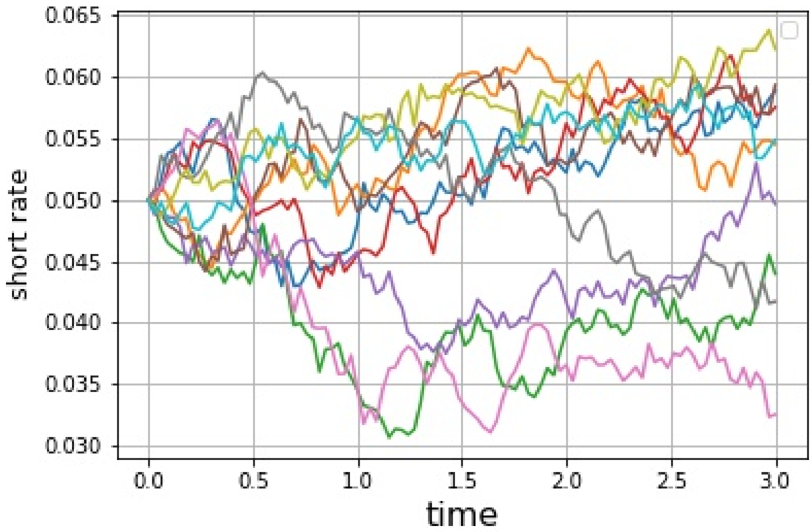


Figure 4.2: The Hull-White model simulations.

Figure 4.2, we computed 10 Hull-White model simulations with $R = 0.05,\ T = 3,\ \sigma = 0.25,\ \theta(t) = 0.05$. We notice that as we approach $T = 3$, the rate reverts back to $\theta(t) = 0.05$, except for one path.

Extension of the CIR model

Maghsoodi (1996), Jamshidian (1995), and Rogers (1995) proposed a representation of the extended CIR model as a sum of squares of Ornstein-Uhlenbeck processes when where the all the coefficients are functions of time. The short rate is defined by the SDE

$$dr_t = [\beta(t) - \alpha(t)r_t]dt + \sigma(t)\sqrt{r_t}dW(t), \tag{4.15}$$

where α, β, and σ are deterministic functions of time. The interest rate in this model follows the generalized chi-square distribution. It is difficult to derive the probability distribution of the squared Bessel processes with time-varying dimensions explicitly. No solution to this problem has been found yet. Just like in original CIR version, the bond price is determined by solving a Riccati equation. In general, there is no closed form formula for the bond price. Such an extension, however, is not analytically tractable.

The Black-Karasinski model

Another model that was proposed in an attempt to address the negative interest rates issue of the Hull White model is the Black and Karasinki (1991) model. In this model, lognormal dynamics of the short rate is defined by

$$d\ln(r_t) = [\beta(t) - \alpha(t)\ln(r_t)]dt + \sigma(t)dW(t), \quad r(0) = r_0 \tag{4.16}$$

where r_0 is a positive constant, $\beta(t)$, $\alpha(t)$ and $\sigma(t)$ are deterministic functions of time. Equation (4.16) shows that the instantaneous short rate follows the exponential Ornstein-Uhlenbeck process with time dependent coefficients. The function $\theta(t)$ is chosen so that the model fits the observed yield curve in the market. We choose α and $\sigma(t)$ so that the model can be fitted to the observed volatility curve in the market. We can also have a constant parameter version of (4.16) by setting $\alpha(t) = \alpha > 0$, $\sigma(t) = \sigma > 0$ which results in

$$d\ln(r_t) = [\beta(t) - \alpha\ln(r_t)]dt + \sigma dW(t), \quad r(0) = r_0. \tag{4.17}$$

Unlike the Hull-White model, the Black-Karasinski model does not yield analytical formulas for discount bonds and interest rate options. Therefore, under this model, one has to resort to numerical techniques to price interest rate derivatives.

From4.17, by Ito's lemma, we obtain

$$d(r_t) = r_t[\beta(t) + \frac{\sigma^2}{2} - \alpha(t)\ln(r_t)]dt + r_t\sigma(t)dW(t)$$

whose explicit solution satisfies, for each $v \leq t$,

$$r_t = \exp\left\{\ln r(v)\xi_v^t + \int_v^t \xi_v^t\theta(u)du + \sigma\int_v^t \xi_v^t dW(u)\right\}.$$

Therefore, r_t conditional on $\mathcal{F}_v$ has a log-normal distribution with first and second moments given respectively by

$$\begin{aligned}
\mathbb{E}_v^{\mathbb{Q}}\{r_t\} &= \exp\left\{\ln r(v)\xi_v^t + \int_v^t \xi_v^t\theta(u)du + \frac{\sigma^2}{4\alpha}\left[1 - e^{-2\alpha(t-v)}\right]\right\} \\
\mathbb{E}_v^{\mathbb{Q}}\{r_t^2\} &= \exp\left\{2\ln r(v)\xi_v^t + 2\int_v^t \xi_v^t\theta(u)du + \frac{\sigma^2}{\alpha}\left[1 - e^{-2\alpha(t-v)}\right]\right\}.
\end{aligned}$$

Moreover, setting

$$\theta(t) = \ln(r_0)e^{-\alpha t} + \int_0^t \xi_v^t\theta(u)du$$

we have that

$$\lim_{t\to\infty}\mathbb{E}^{\mathbb{Q}}(r_t) = \exp\left(\lim_{t\to\infty}\theta(t) + \frac{\sigma^2}{4\alpha}\right).$$

The equilibrium and arbitrage-free models discussed above focused on modelling r_t and other quantities in a multifactor framework. When looking at the Ho and Lee or the Hull and White model, one can ask how does the whole of the forward-rate curve evolve over time? This question gave inspiration to the development of the HJM framework. In the HJM, the main focus is the instantaneous forward-rate curve, $f(t, T)$ rather than r_t.

4.3 The Heath-Jarrow-Morton model

Multifactor short rate models have properties that make them more desirable over single-factor models with the one stand out property being that of being able to describe the intrinsic volatility structure and the Heath-Jarrow-Morton framework being the best multifactor model on the spectrum. The HJM model treats the forward rate curve at a point in time as an input, and produce the evolution of the curve according to a stochastic process. This process has the general form given by Equation (4.1) but it is more general than arbitrage-free models. Duffie and Kan proposed a multifactor model of the term structure, where the yield factors form a Markov process. The framework proposed by Heath, Jarrow, and Morton (1992) was a great leap forward in the perception and modelling of the term structure of interest rates. Their paper described the arbitrage-free conditions that a model of the yield curve must satisfy.

4.3.1 Construction of the Heath-Jarrow-Morton model

We now look at the construction of the HJM model, proposed in 1990 by Heath, Jarrow, and Morton.

The fact that virtually all the arbitrage-free interest models can be derived from the HJM framework makes the theory of the HJM more important in the literature. This model, however, had some drawbacks of its own. The first one being that they have an increase in dimensions compared to previously introduced short rate models. This increase in dimensions means that the lattice approach to pricing could not be used anymore and one has to resort to Monte Carlo techniques which are time consuming and are not the preferred option by market participants. This drawback was, however, addressed by the emergence of high dimensional low-discrepancy sequences. These sequences had the ability to significantly reduce the time needed for computation of this model, making the HJM approach a more practical proposition. The other two problems with this framework were that the instantaneous forward rate $f_{t,T}$ exploded with positive probability and that it is difficult to recover a set of caplets. This inspired the introduction of the BGM method, commonly referred to as the LIBOR market model.

In this chapter, we discussed classical short rate model such as the affine term structure models (Equilibrium models and Arbitrage-free models). We discussed their drawbacks and how each model inspired the development of the other. We closed off the chapter by discussing the unifying framework know as the HJM framework. As already mentioned, the HJM framework has two major drawbacks. In the next chapter, we are going to discuss a model aimed at resolving the drawbacks of the HJM model.

CHAPTER 5

LIBOR Market Model

Brace et al., 1997 and Miltersen et al., 1997 showed that it is possible to construct an arbitrage-free interest rate model in which the LIBOR rates follow a log-normal process leading to a Black-type pricing formulae for caps. Based on the Heath-Jarrow-Morton (HJM) forward rate approach, it builds a process for LIBOR interest rates, assuming a conditional lognormal process for LIBOR. In their approach, they modelled forward LIBOR rates that are observed in the market instead of instantaneous spot rates as in the HJM model. Since the LMM is a market model, it is defined by the volatilities imposed on various rates. Furthermore, the LMM prices interest rate derivatives such as caps and swaptions in such a way that is consistent with the Black formula. This makes calibrating the models to be easier. Thanks to LMM, practitioners can not only obtain consistent prices of ATM caps/floors and exotic interest rate derivatives, but also hedge the exotic derivatives by using ATM caps/floors as Vega hedging tools. Moreover, by joint calibration to cap/floor and swaption markets, traders are able to execute relative value trading between ATM caps/floors and ATM swaptions.

5.1 Introduction

In this section, we introduce the notation that will be used in the subsequent sections. We begin by setting the model:

- $L_i(t) := L(t; T_{i-1}, T_i)$ is the forward LIBOR rate observed at time t for the expiry-maturity period $[T_{i-1}, T_i]$ with the compounding period $\tau_i = T_i - T_{i-1}$;
- μ_i is the drift parameter;
- $\mathbb{Q}^{T_i}$ is the equivalent martingale measure where the numeraire is $P(\cdot, T_i)$, i.e. the T_i-forward measure;
- $W^i(t)$ is the i'th standard Brownian motion at time t.

Since we are still modelling interest rates, the dynamics of the LIBOR rate should follow the SDE given by Equation 4.1, i.e. it assumes the dynamics given by

$$dL_i(t) = \mu_i(L_i(t), t)dt + \sigma_i(t)L_i(t)dW_i^i(t), \quad t \leq T_{i-1}. \tag{5.1}$$

We now introduce the following lemma which gives us the dynamics of the forward rates under different measures.

Lemma 5.1. *For every $i \in [1, M]$, L_i is a martingale under the corresponding T_i-forward measure, on the interval $[0, T_{i-1}]$.*

Proof. From the definition of forward rates we have

$$L_i(t)P_t(T_i) = \frac{P_t(T_{i-1}) - P_t(T_i)}{\tau_i}.$$

We see that $L_i(t)P_t(T_i)$ is a tradable asset since it can be replicated by buying and selling two bonds. As such, it can be normalized by the numeraire $P_t(T_i)$ which by definition of a martingale measure, results in a martingale under $\mathbb{Q}_{T_j}$. $\square$

Under $\mathbb{Q}_{T_i}$, L_i has a driftless dynamics.

5.2 Construction of the LIBOR Market Model

We now look at the construction of the LIBOR Market Model.

Definition 5.1. *The forward rates of a discrete tenor LMM have dynamics that are given by the following SDE under their associated forward measure*

$$dL_i(t) = \sigma_i(t)L_i(t)dW_i^i(t), \quad t \leq T_{i-1} \tag{5.2}$$

for $i = 1, \ldots, M$. Here we assume that σ_i is deterministic scalar, and dW_i is the ith component of the $\mathbb{Q}_{T_I}$-Wiener process, hence is a standard Wiener process.

There exists an extension of this model $\sigma_i(t)$ are positive stochastic processes. If σ_i is bounded, the SDE has a unique strong solution, since it describes a geometric Wiener process. From Ito's formula,

$$\begin{aligned}
d\ln L_i(t) &= \sigma_i(t)dW_i^i(t) - \frac{\sigma_i(t)^2}{2}dt \\
\implies \ln L_i(T) &= \ln L_i(t) + \int_t^T \sigma_i(u)dW_i^i(u) - \int_t^T \frac{\sigma_i(u)^2}{2}du \\
\implies \quad L(T) &= L_i(t)e^{\int_t^T \sigma_i(u)dW_i^i(u) - \frac{1}{2}\int_t^T \sigma_i(u)^2du, \quad 0\leq u\leq T\leq T_{i-1}}.
\end{aligned}$$

The risk-neutral dynamics are needed to give the valuation of certain products in the market like Eurodollar futures. Let us start off by deriving the risk-neutral dynamics of the LMM.

Proposition 5.1. *The risk neutral dynamics of forward LIBOR rates in the LMM is:*

$$dL_n(t) = \tilde{\mu}_n(t)dt + \sigma_n(t)L_n(t)d\tilde{W}_n(t), \tag{5.3}$$

where

$$\begin{aligned}
\tilde{\mu}_n(t) &= \sum_{j=\eta(t)}^{n} \frac{\rho_{n,j}\tau_j\sigma_j(t)\sigma_n(t)L_j(t)}{1+\tau_j L_j(t)} + \underline{\sigma}_n(t)\rho \int_t^{T_{\eta(t)-1}} \sigma_f(t,u)'du \\
&= \sum_{j=\eta(t)}^{n} \frac{\rho_{n,j}\tau_j\sigma_j(t)\sigma_n(t)L_j(t)}{1+\tau_j L_j(t)} + \sum_{j=\eta(t)}^{n} \rho_{n,j}\sigma_n(t) \int_t^{T_{\eta(t)-1}} (\sigma_f)_j(t,u)du.
\end{aligned}$$

Proof. Let DC be the operator defined in Definition (B.1). We know that the dynamics of the cash account C_t follow the SDE:

$$dC_t = r_t C_t dt$$

where the volatility is zero. We can get the risk-neutral dynamics of L_n by using the Radon Nikodym theorem to change the numeraire from $P_t(T_n)$ to C_t. From this, we get the following dynamics:

$$dL_n(t) = -\underline{\sigma}_{P_t(T_n)}\rho\underline{\sigma}_n(t)'L_n(t)dt + \underline{\sigma}_n(t)'L_n(t)d\tilde{W}(t)$$

where $\underline{\sigma}_{P_t(T_n)}$ is the bond volatility. Since we can express bond prices as

$$P_t(T_n) = P_t(T_{\eta(t)-1}) \prod_{j=\eta(t)}^{n} \frac{1}{1+\tau_j L_j(t)}$$

The drift above can be computed as

$$
\begin{aligned}
\tilde{\mu}_n(t) :&= -\underline{\sigma}_{P_t(T_n)}\rho\underline{\sigma}_n(t)' \\
&= -\mathrm{DC}\left(\ln P_t(T_n)\right)\rho\underline{\sigma}_n(t)' \\
&= -\mathrm{DC}\left[\ln P_t(T_{\eta(t)-1}) + \sum_{j=\eta(t)}^{n} \ln\left(\frac{1}{1+\tau_j L_j(t)}\right)\right]\rho\underline{\sigma}_n(t)'.
\end{aligned}
$$

Now we see that since in general

$$
\ln P_{t,T} = -\int_t^T f(t,u)du
$$

the below follows intuitively

$$
\mathrm{DC}[P_{t,T}] = -\int_t^T \underline{\sigma}_f(t,u)du,
$$

where $\underline{\sigma}_f(t,u)$ is the absolute instantaneous vector volatility of the instantaneous forward rate $f(t,u)$. We therefore have that

$$
dL_n(t) = \tilde{\mu}_n(t)dt + \sigma_n(t)L_n(t)d\tilde{W}_n(t), \tag{5.4}
$$

where

$$
\begin{aligned}
\tilde{\mu}_n(t) &= \sum_{j=\eta(t)}^{n} \frac{\rho_{n,j}\tau_j\sigma_j(t)\sigma_n(t)L_j(t)}{1+\tau_j L_j(t)} + \underline{\sigma}_n(t)\rho\int_t^{T_{\eta(t)-1}} \sigma_f(t,u)'du \\
&= \sum_{j=\eta(t)}^{n} \frac{\rho_{n,j}\tau_j\sigma_j(t)\sigma_n(t)L_j(t)}{1+\tau_j L_j(t)} + \sum_{j=\eta(t)}^{n} \rho_{n,j}\sigma_n(t)\int_t^{T_{\eta(t)-1}} (\sigma_f)_j(t,u)du.
\end{aligned}
$$

□

The presentation of the drift in (5.4) does not look good because of the second summation. This summation is from the continuous tenor part coming from the evolution of $P_t(T_{\eta(t)-1})$, which is impossible to get from forward rates in our family. In order to close the equation, we are forced to simulate the dynamics of $\underline{\sigma}_f(t,u)$ but this is not consistent with the discrete tenor space that we are operating in. The spot measure was introduced in the literature as a discrete-time equivalent of the cash account.

Proposition 5.2. *The spot-LIBOR measure dynamics of forward LIBOR rates in the LMM is:*

$$dL_n(t) = \sigma_n(t)L_n(t)\sum_{j=\eta(t)}^{j}\frac{\rho_{n,j}\tau_j\sigma_j(t)L_j(t)}{1+\tau_jL_j(t)}dt + \sigma_n(t)L_n(t)dW_n^i(t). \tag{5.5}$$

Proof. Given a LIBOR short rate $L(t)$, its SDE has the general form given by

$$dL_n(t) = \mu_n(t)L_n(t)dt + \sigma_n(t)L_n(t)dW_n^i(t). \tag{5.6}$$

When the cash account is taken as the numeraire, we have that

$$C_t = P_t(T_{\eta(t)})\prod_{j=0}^{\eta(t)-1}[1+\tau_jL_j(t)]. \tag{5.7}$$

The deflated bond price, $D_t(T_n) = \frac{P_t(T_n)}{C_t}$ is a martingale w.r.t the spot measure, and

$$D_t(T_n) = \frac{P_t(T_n)}{C_t} = \prod_{j=\eta(t)}^{n-1}\frac{1}{1+\tau_jL_j(t)}\prod_{j=0}^{\eta(t)-1}\frac{1}{1+\tau_jL_j(t)}. \tag{5.8}$$

If $D_t(T_n)$ is a martingale that is positive, we have that $\frac{dD_t(T_{n+1})}{D_t(T_{n+1}} = V_{n+1}^i(t)dW(t)$ where $n = 1,\ldots,m$.

Applying Ito's lemma,

$$d\ln D_t(T_{n+1}) = -\frac{1}{2}||V_{n+1}(t)||dt + V_{n+1}^i(t)dW(t). \tag{5.9}$$

For a one factor model, we have that $\frac{dD_t(T_{n+1})}{D_t(T_{n+1}}=V_{n+1}(t)dW(t) \implies d\ln D_t(T_{n+1}) = -\frac{1}{2}||V_{n+1}(t)||dt + V^i_{n+1}(t)dW(t)$

$$d\ln D_t(T_{n+1}) = -\sum_{j=\eta(t)}^{n} d\ln(1+\tau_j L_j(t)). \tag{5.10}$$

We apply Ito's lemma and Equation (5.6), we get

$$V_{n+1}(t) = -\sum_{j=\eta(t)}^{n} \frac{\tau_j \sigma_j(t) L_j(t)}{1+\tau_j L_j(t)}. \tag{5.11}$$

This is the relation between the bond volatility and LIBOR. From this, we have that the drift term μ_n is given by

$$\mu_n = -\sigma_n V_{n+1}(t) = -\sum_{j=\eta(t)}^{n} \frac{\tau_j \sigma_j(t) L_j(t)}{1+\tau_j L_j(t)}\sigma_n \tag{5.12}$$

So from Equation (5.6), we have that the spot LIBOR dynamics are given by

$$dL_n(t) = \sigma_n(t) L_n(t) \sum_{j=\eta(t)}^{n} \frac{\rho_{n,j}\tau_j \sigma_j(t) L_j(t)}{1+\tau_j L_j(t)}dt + \sigma_n(t)L_n(t)dW^i_n(t). \tag{5.13}$$

□

We notice that under the numeraire C_d, no continuous-tenor term appears in the drift.

Proposition 5.3. *Terminal measure LMM dynamics.*

Under the assumptions of the LIBOR Market Model, the dynamics of each L_n, for

$n = 1, \ldots, M,$ *is*

$$n < i : dL_n(t) = \sigma_n(t)L_n(t) \sum_{j=n+1}^{j} \frac{\rho_{n,j}\tau_j\sigma_j(t)L_j(t)}{1+\tau_j L_j(t)} dt + \sigma_n(t)L_n(t)dW_n^i(t)$$

$$n = i : dL_i(t) = \sigma_n(t)L_n(t)dW_n^i(t)$$

$$n > i : dL_n(t) = -\sigma_n(t)L_n(t) \sum_{j=i+1}^{n} \frac{\rho_{n,j}\tau_j\sigma_j(t)L_j(t)}{1+\tau_j L_j(t)} dt + \sigma_n(t)L_n(t)dW_n^i(t) \quad \textit{for } t \le \min\{T_{n-1}, T_i\}.$$

Proof. We assume there exists a LMM that satisfies (5.2). The aim is to find the deterministic functions $\mu_n^i(t, L_n(t))$, where:

$$L(t) = (L_i(t), \ldots, L_M(t))'$$

such that the following is satisfied

$$dL_n(t) = \mu_n^i(t, L(t))L_n(t)dt + \sigma_n L_n dW_n^i. \tag{5.14}$$

For the $\mu_n^i(t, L(t))$ to be determined, we make use of the Radon-Nikodym theorem to move from $\mathbb{Q}^i$ to $\mathbb{Q}^n$, then impose that the $\mathbb{Q}^n$-resulting drift is null. Applying the Radon-Nikodym of $\mathbb{Q}^{i-1}$ with respect to $\mathbb{Q}^i$ at time t is

$$\frac{d\mathbb{Q}^{i-1}}{d\mathbb{Q}^i}|\mathcal{L}_t^W = \frac{P_t(T_{i-1})P_0(T)}{P_0(T_{i-1})P_t(T_i)} := \gamma_t^i \tag{5.15}$$

and by the fundamental law of asset pricing,

$$\gamma_t^i = \frac{P_0(T_i)}{P_0(T_{i-1})}(1 + L_i(t)\tau_i).$$

Therefore, if we assume (5.2), the SDE of γ_i under $\mathbb{Q}^i$ is

$$\begin{aligned} d\gamma_t^i &= \frac{P_0(T_i)}{P_0(T_{i-1})} dL_i(t)\tau = \frac{P_0(T_i)}{P_0(T_{i-1})} \tau_i \sigma_i(t) L_i(t) dW_i(t) \\ &= \frac{\gamma_t^i}{1 + L_i(t)\tau_i} \tau_i \sigma_i(t) L_i(t) dW_i(t). \end{aligned}$$

Thus, the density γ_i is in the form of an exponential martingale with associated process λ that is the d-dimensional $n \times n$ vector apart from the i-th component,

$$\lambda = \left(0, \dots, -\frac{\tau_i \sigma_i L_i}{1 + L_i \tau_i}, \dots 0\right)^T, \tag{5.16}$$

So that we can apply the formula of the change of drift with correlation:

$$dW^i(t) = dW^{i-1}(t) - \rho dt,$$

namely with components:

$$dW_j^i(t) = dW_j^{i-1}(t) + \rho^{j,i} \frac{\tau_i \sigma_i(t) L_i(t)}{1 + L_i(t)\tau_i} dt.$$

using induction, we get:

$$\begin{aligned} n < i : dW_j^i(t) &= dW_j^n(t) + \sum_{h=n+1}^{i} \rho^{jh} \frac{\tau_h \sigma_h(t) L_h(t)}{1 + L_h(t)\tau_h} dt, \\ n > i : dW_j^i(t) &= dW_j^n(t) - \sum_{h=n+1}^{i} \rho^{jh} \frac{\tau_h \sigma_h(t) L_h(t)}{1 + L_h(t)\tau_h} dt \end{aligned}$$

Then, substituting these into (5.14) and imposing the $\mathbb{Q}^n$-drift to be zero we obtain:

$$\begin{aligned}
n < i : L_n(t)(\mu_n^i(t, L(t))) + \sigma_i(t) \sum_{h=n+1}^{i} \rho^{jh} \frac{\tau_h \sigma_h(t) L_h(t)}{1 + L_h(t)\tau_h} dt = 0, \\
\implies \mu_n^i(t, L(t)) = -\sigma_i(t) \sum_{h=n+1}^{i} \rho^{jh} \frac{\tau_h \sigma_h(t) L_h(t)}{1 + L_h(t)\tau_h} dt \\
n > i : L_n(t)(\mu_n^i(t, L(t))) - \sigma_i(t) \sum_{h=n+1}^{i} \rho^{jh} \frac{\tau_h \sigma_h(t) L_h(t)}{1 + L_h(t)\tau_h} dt = 0, \\
\implies \mu_n^i(t, L(t)) = \sigma_i(t) \sum_{h=n+1}^{i} \rho^{jh} \frac{\tau_h \sigma_h(t) L_h(t)}{1 + L_h(t)\tau_h} dt.
\end{aligned}$$

□

At this point, we can turn around the argument to have the following existence result.

Proposition 5.4. *Consider $\{\sigma_i\}_{i=1}^M$, where $\sigma_i < \infty$ for all $i = 1, \dots, M$ and the terminal measure $\mathbb{Q}^M$ with associated Brownian motion W^M. If the process $L_1, \dots, L_M$ is defined by*

$$dL_i(t) = -\sigma_i(t) L(t) \sum_{j=i+1}^{M} \rho^{jh} \frac{\tau_h \sigma_h(t) L_h(t)}{1 + L_h(t)\tau_h} dt + \sigma_i(t) L_i(t) dW_i^M(t) \tag{5.17}$$

for $i = 1, \dots, M$, then the dynamics of L_i under $\mathbb{Q}^i$-dynamics of is given by (5.2).

Proof. First, we have to prove that the solution of (5.17) exists. For $i = M$ we simply have

$$dL_i(t) = \sigma_M(t) L_M(t) dW_M^M(t)$$

which is just an exponential martingale, where σ_M is bounded. This proves the existence of the solution to (5.17). Now, inductively: assume that (5.17) admits a solution for

$i+1,\dots,M$, then we present the i-th dynamics as

$$dL_i(t) = \mu(t, L_{i+1}(t), \dots, L_M(t))L_i dt + \sigma_i(t)L_i(t)dW_i^M(t)$$

where the fact that μ_i is only dependent on L_K for $n = i+1,\dots,M$ is key. So if we denote $L_{i+1}^M := (L_{i+1},\dots,L_M)^T$, we can get the explicit solution of the above SDE by using the Ito's formula:

$$\begin{aligned} d\ln L_i(t) &= \frac{dL_i(t)}{L_i(t)} - \frac{1}{2L_i(t)^2}\sigma_i(t)^2 L_i(t)^2 dt \\ &= \mu_i(t, L_{i+1}^M(t))dt + \sigma_i(t)dW_i^M(t) - \frac{1}{2}\sigma_i(t)^2 dt \\ \implies \ln L_(t) &= \ln L_i(0) + \int_0^t \left(\mu_i(u, L_{i+1}^M(u)) - \frac{\sigma_i(u)^2}{2}\right) du + \int_0^t \sigma_i(u)dW_i^i(u) \\ \implies L_i(t) &= L_i(t)\exp\left[\int_0^t \left(\mu_i(u, L_{i+1}^M(u)) - \frac{\sigma_i(t)^2}{2}\right) du\right] \exp\left[\int_0^t \sigma_i(u)dW_i^i(u)\right], \end{aligned}$$

for $0 \le t \le T_{i-1}$ and this proves existence.

Lastly, we have to show that the Novikov condition is satisfied by the process λ defined in (5.16), in which case the density process γ^i is a $\mathbb{Q}^i$-martingale and consequently we can apply the Girsanov Theorem, following the same procedure as in Proposition 5.3. If we have an initial LIBOR term structure that is non-negative, as it is $L(0) = (L_1(0),\dots,L_M(0))^T$, we see that all LIBOR rate processes will be always non-negative, thus the process λ in (5.16) is bounded and consequently satisfies the Novikov condition.

□

5.3 Valuation of caps in the LMM

Let us recap a bit on caps. In Chapter 3 we defined caps as a basket of caplets that all have the same strike price κ. Each caplet has a payoff at $\{T_1,\dots,T_N\}$. If we take the sum of all the caplets we get the price of the cap. The discounted payoff of a cap at time t is given by

$$\sum_{i=\alpha+1}^{\beta} D_f(t,T_i)N\tau_i\left(L(T_{i-1},T_i)-\kappa\right)_+ .$$

The LMM can also be used to price exotic caps such as ratchet caps and sticky caps. The discounted payoff of a ratchet cap is given by

$$\sum_{i=\alpha+1}^{\beta} D_f(t,T_i)N\tau_i(L(T_{i-1},T_i)-(L(T_{i-2},T_{i-1})+\epsilon))_+.$$

and the discounted payoff of a sticky cap is given by

$$\sum_{i=\alpha+1}^{\beta} D_f(t,T_i)N\tau_i(L(T_{i-1},T_i)-(\min(L(T_{i-2},T_{i-1}),\kappa)+\epsilon))_+.$$

Valuating caps with the LMM is interesting because we can use it for validations of the model and the calibrations by comparing the results from the analytical Black model commonly referred to as the Black76 model. Once we obtain the desirable results from the model, we can use the same parameters to compute the pricing of other models that are not analytically tractable. We now introduce a proposition that relates the LMM simulated caps with the Black76 cap prices. But we first give the following definition.

Definition 5.2. *Given market price data for caps with tenor structure as above mentioned, denoted by $Cap_m(t,T_i,\kappa)$ where $\{T_i\}_{i=0}^{j}=\{T_0,\dots,T_j\}$ the implied Black volatilities are defined as follows:*

- *The implied flat volatilities are the solutions $v_{T_1-cap},\dots,V_{T_M-cap}$ of the equations.*

Proposition 5.5. *The price of the i-th caplet implied by the LIBOR Market Model* *coincides with that given by the corresponding Black caplet formula:*

$$\begin{aligned}
capl_{LMM}(0, T_{i-1}, T_i, \kappa) &= capl_{Black}(0, T_{i-1}, T_i, \kappa, V_i) \\
&= P_j(t) Black(R_j(t), \kappa, \sum_j^F \sqrt{T_{j-1}}) \\
&= \tau_i P_0(T) Black(\kappa, L(0), T_{i-1}, T_i) v_i \sqrt{T_{i-1}}
\end{aligned}$$

where

$$(v_i)^2 = \frac{1}{T_{i-1}} \int_0^{T_{i-1}} \sigma_i(t)^2 dt. \tag{5.18}$$

Proof.

$$\text{capl}_{LMM}(t, T_{i-1}, T_i, \kappa) = \tau_i P_t(T_i) \mathbb{E}^{\mathbb{Q}_i} \{(L(T_{i-1}, T_i) - \kappa)_+ | \mathcal{F}_t\}$$

where $L(T_{i-1}, T_i) = L(T_{i-1}, T_{i-1}, T_i) = L_i(T_{i-1})$ and, for $T < T_{i-1}, L_i(T)$ is log-normal distributed under the forward measure $\mathbb{Q}^i$, indeed

$$L_i(T) = L_i(t) e^{\int_t^T \sigma_i(u) dW_i^i(u) - \frac{1}{2}\sigma_i^2(u) du} \qquad 0 \le t \le T \le T_{i-1}$$

with σ_i assumed to be deterministic. Let

$$X_i(t, T) := \int_i^T \sigma_i(u) dW_i^i(u) - \frac{1}{2}\sigma_i(u)^2 du,$$

we have

$$L_i(T) = L_i(t) e^{X_i(t,T)}, \quad X_i(t, T) \sim \mathcal{N}(m_i(t, T), \sum_i^2 (t, T)),$$

where

$$m_i(t, T) = -\frac{1}{2} \int_t^T \sigma_i(u)^2 du, \quad \sum_i^2 (t, T) = \int_t^T \sigma_i(u)^2 du$$

and $L_i(t) \in \mathbb{R}$ at time t. Thus the proof follows from $B.8$.

$\square$

The quantity v_i in (5.18), denote the average instantaneous percentage variance of the forward rate $L_i(t)$ for $t \in [0, T_{i-1}]$, that is its average volatility.

5.4 Calibrating the LMM to Caplets

It is standard practice to calibrate the LMM to the traded caplets in the market since they are among the most liquid ones. The traders are responsible for converting caplets to caps, then using Proposition 5.5 and market quoted volatilities, calibrating the parameters of the LMM to caps and floors becomes trivial.

Assume that we are at $t = 0$. If we have an empirical term structure of implied volatilities, then from Proposition 5.5, we have that the calibration of the LMM to traded caplets is done by choosing the deterministic LIBOR volatilities of forward rates $\sigma_1, \ldots, \sigma_N$ such that

$$(v_i)^2 = \frac{1}{T_{i-1}} \int_0^{T_{i-1}} \sigma_i(t)^2 dt, \qquad i = 1, \ldots, N. \tag{5.19}$$

5.4.1 Parameterizations of Volatility of Forward rates

Equation 5.19 is very undetermined, meaning when applying it, it is important to make some assumptions about the structure of the volatility functions. A lot of specifications have been proposed in the literature, with the two most popular ones being the piecewise-constant functions and the functional parameterized forms. In this dissertation, we will only use the former.

- General Piecewise-constant volatilities (GPC):

$$\sigma_i(t) = \sigma_{i,\eta(t)}(t), \qquad 0 < t \leq T_{i-1}, \tag{5.20}$$

where

$$\eta(t) = n \text{ if } T_{n-2} < t \leq T_{n-1}, \quad n \geq 1. \tag{5.21}$$

The maturity of the first LIBOR rate that is yet to expire at time t is indicated by $\eta(t)$. We can organize these σ's in Table , where the columns are the time intervals and the LIBOR rates are in the rows.

T	$T_0 = 0$	$T_1 = 0.5$	$T_2 = 1.0$	$\cdots$	$T_4 = 2.0$
$L_0(t)$	$\sigma_{0,0}$	dead	dead	$\cdots$	dead
$L_1(t)$	$\sigma_{1,0}$	$\sigma_{1,1}$	dead	$\cdots$	dead
$L_2(t)$	$\sigma_{2,0}$	$\sigma_{2,1}$	$\sigma_{2,2}$	$\cdots$	dead
$\vdots$	$\ldots$	$\ldots$	$\ldots$	$\ldots$	$\ldots$
$L_N(t)$	$\sigma_{N,0}$	$\sigma_{N,1}$	$\sigma_{N,2}$	$\cdots$	$\sigma_{N,N}$

Table 5.1: GPC volatilities.

The LIBOR rates only exist until the date where they are due to start applying. This implies that the instantaneous volatility that corresponds to the starting period ceases to exist when we are past the period.

To calibrate the model, we insert Equation 5.20 into Equation 5.19, and thus impose:

$$T_{i-1}(v_i)^2 = \int_0^{T_{i-1}} \sigma_{i,\eta}(t)^2 dt = \sum_{k=1}^{i} \tau_{k-2,k-1}\sigma_{i,k}, \qquad i = 1, \ldots, N. \tag{5.22}$$

We have a lot of configurations that can fit caplets in a perfect manner. It is a fact that:

$$\begin{aligned}
\sqrt{(T_0 - 0)}\sigma_{0,0} &\iff \sigma_{0,0} = v \\
\sqrt{(T_0 - 0)\sigma_{1,0}^2 + (T_1 - T_0)\sigma_{1,1}^2} &= \sqrt{T_1}v \\
\sqrt{(T_0 - 0)\sigma_{3,0}^2 + (T_1 - T_0)\sigma_{3,2}^2 + (T_2 - T_1)\sigma_{3,3}^2} &= \sqrt{T_1}v \\
&\vdots
\end{aligned}$$

Year,i:	1	2	3	4	5	6	7	8	9	10
v_i	15.50	18.25	17.91	17.74	17.27	16.79	16.30	16.01	15.76	15.54
$\sigma_{10,i-1}$	15.50	20.64	17.21	17.22	15.25	14.15	12.98	13.81	13.60	13.40

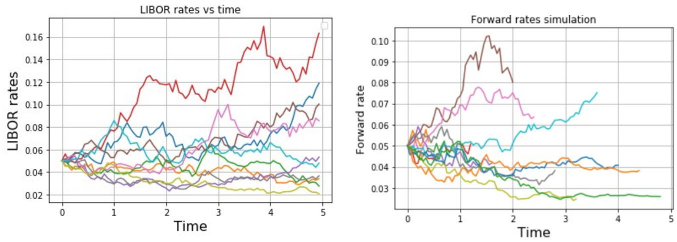
Table 5.2: Volatility data, accrual period= 1 year.

Lastly, its is important to specify the instantaneous correlation. A lot of interest derivatives are dependent on multiple LIBOR rates. There is some correlation between these rates. Since the payoff in the caplets are only dependent on just one LIBOR rates, we cannot use them to calibrate the instantaneous correlation between the LIBOR rates. Instead, we use European swaptions to calibrate the instantaneous correlations. The swaption payoff is depended on multiple forwards. We will assume the following structure of instantaneous correlation

$$\rho_{i,j} = \prod_{n,n+1}^{j-1}$$

$$\rho_{i,j} = \rho_{j,i}.$$

Definition 5.3. *Given a time structure $\{t_i\}_{i=0}^{N+1} = \{t_0, \dots t_n\}$. The terminal measure $\mathbb{Q}^{N+1}$, is the equivalent martingale measure that has the zero-coupon bond $P_t(t, t_{N+1})$ as the numeraire. The drift of all forward rates with resets times up to t_{N-1} under this measure just like in the case where $i < j$ and the forward rate $F(t, T_N, t_{N+1})$ has a zero drift.*



(a) Forward rates under the same application periods. **(b)** Forward rates under different application periods.

Figure 5.1: Forward rates simulations.

We simulated forward rates for $T = 5$, $\sigma = 0.25$ with the same application period in Figure 5.1a. In Figure 5.1b, we simulate the forward rates with different application periods. When simulating the LIBORs, we start at the initial time, $t = 0$, and move forward in time. As we move forward in time, more and more forwards reach their maturity dates and then cease to exist for times after their maturity. This means that only a few forward rates will still be alive as we move forward.

Now if we let $N(t) = P_t(T_{N+1})$ in Equation (3.15) and $\pi(t) = V(t)$ be the value of a derivative, then it follows that

$$\frac{V(t)}{P_t(t_{N+1})}$$

is a martingale and hence

$$V(t_0) = P_t(T_{N+1})\mathbb{E}^{\mathbb{Q}^{N+1}}\left\{\frac{V(t)}{P_t(t_{N+1})}\middle|\mathcal{F}_t\right\}$$

If we let

$$V'(t_n) = \frac{V(t)}{P_t(t_{N+1})},$$

then we have that

$$V(t_0) = P_t(T_{N+1})\mathbb{E}^{\mathbb{Q}^{N+1}}\left\{V'(t_n)\middle|\mathcal{F}_t\right\}$$

Let $V(t)$ to be the value of a caplet whose payoff is determined at t_n and paid at

t_{n+1}. If one is interested in getting the value of its payoff using the formula above, then it is important to decide its relative price. There are several ways of doing this, with the most appealing one, perhaps, being the relative price process at time t_{n+1}

$$V'(t_{n+1}) = \frac{V(t_{n+1})}{P_t(t_{N+1})}. \tag{5.23}$$

Another method is determining the relative price at t_n since this is where the payoff is determined. This can be expressed as

$$V'(t_n) = V(t_{n+1})\frac{P_{t_n}(t_{n+1})}{P_{t_n}(t_{N+1})} \tag{5.24}$$

It is worth noting that for a give path of simulated LIBOR rates, these expressions should not necessarily be the same, but they are equivalent in martingale pricing sense,

$$V'(t_n) = \mathbb{E}^{\mathbb{Q}^{N+1}}\left\{V'(t_{n+1})\middle|\mathcal{F}_{t_n}\right\}.$$

Another way is to reinvest the payoff from t_{n+1} to t_{N+1} into the deposit rate,

$$V(t_{N+1}) = V(t_{n+1})\prod_{i=n+1}^{N}(1+\tau_i F_{t_i}(t_i, t_{i+1})) = V'(t_{N+1}). \tag{5.25}$$

Again, this is equivalent to the other two in martingale pricing sense

$$V'(t_n) = \mathbb{E}^{\mathbb{Q}^{N+1}}\left\{V'(t_{N+1})\middle|\mathcal{F}_{t_n}\right\}.$$

We now consider an example of a cap simulation with the following parameters, $L(0) = 0.05$, $\kappa = 0.5$, $\tau = 0.5$, $\sigma = 0.20$. Summing all these caplets results in a cap. Using N number of realisations gives an estimate of the Monte Carlo computed value of a cap $\bar{V}^{cap}$ as

$$\bar{V}^{cap} = \frac{1}{N}\sum_{i=1}^{N} V_i^{cap} \tag{5.26}$$

We can also obtain the standard error of the simulation using Equation (A.3).

Standard cap valuation

Caplet	Black76	LMM Sim
$C(T_1, T_2)$	0.00082	0.00069
$C(T_2, T_3)$	0.00103	0.00102
$C(T_3, T_4)$	0.00127	0.00131
$C(T_4, T_5)$	0.00154	0.00133
$C(T_5, T_6)$	0.00164	0.00164
$C(T_6, T_7)$	0.00162	0.00182
$C(T_7, T_8)$	0.00184	0.00186
$C(T_8, T_9)$	0.00212	0.00209
$C(T_9, T_{10})$	0.00208	0.00221
Cap	0.01397	0.01397

Table 5.3: Caplet prices.

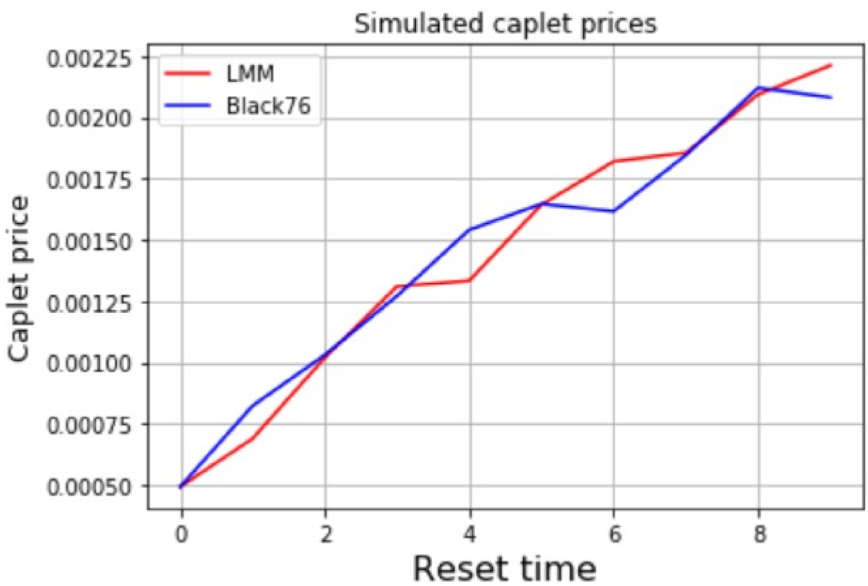


Figure 5.2: Simulated caplet prices.

In Table 5.3 and Figure 5.2, we valuated the payoff of 9 caplets using the analytical Black76 formula and simulated payoff. The caplets are of different maturities, starting from 1 to 9, where the notional is 1 and $\kappa = 0.05$. We used 2000 Monte Carlo simulations. As we move forward in time, the caplet payoffs increase. The two methods yielded similar caplet payoffs for different maturities.

Ratchet and sticky cap valuation

Caplet	Ratchet	Sticky
$C(T_1, T_2)$	0.00001	0.00001
$C(T_2, T_3)$	0.00004	0.00005
$C(T_3, T_4)$	0.00011	0.00012
$C(T_4, T_5)$	0.00019	0.00022
$C(T_5, T_6)$	0.00028	0.00034
$C(T_6, T_7)$	0.00036	0.00042
$C(T_7, T_8)$	0.00044	0.00053
$C(T_8, T_9)$	0.00053	0.00063
$C(T_9, T_{10})$	0.00058	0.00069
Cap	0.00254	0.00301

Table 5.4: Ratchet and sticky Caplet prices.

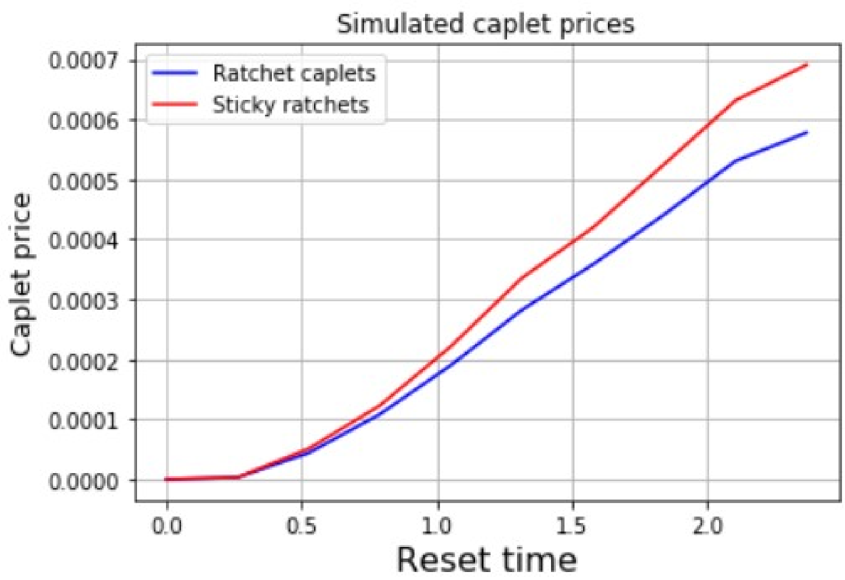


Figure 5.3: Simulated caplet prices.

In Table 5.4 and Figure 5.3 we valuated 9 Ratchet and sticky caplet prices. The caplets are of different maturities, starting three months from now until 5 years. We set $\kappa = 0.05$ and $\epsilon = 0.01$ and 2000 Monte Carlo simulations. As the maturities increase, the caplet prices also increase, but the Sticky caplets are generally higher than those of ratchet caplets. The standard caplets are always higher than both these nonstandard caplet at each maturity, and hence the cap has a higher payoff.

5.5 Valuation of swaptions in the LMM

We now look at the valuation of swaptions in the LMM framework. In Chapter 3 we introduced IRS as a financial contract where one party pays the floating leg $L_i(T_{i-1})$ and the other party pays a constant rate κ. As already mentioned, we have two types of IRS, a payer and a receiver IRS. In the case of the payer IRS, the payoff at time T_ϕ for the payer party of such an IRS can be expressed as

$$\sum_{i=\phi+1}^{\eta} D_f(T_\phi, T_i)\tau_i(L_i(T_{i-1}) - \kappa). \tag{5.27}$$

Accordingly, the discounted payoff at a time $t < T_\phi$ is

$$\sum_{i=\phi+1}^{\eta} D_f(t, T_i)\tau_i(L_i(T_{i-1}) - \kappa). \tag{5.28}$$

The value of such a contract can be computed as follows:

$$
\begin{aligned}
PFS(t,[T_\phi,\ldots,T_\eta],\kappa) &= \mathbb{E}^{\mathbb{Q}}\left\{\sum_{i=\phi+1}^{\eta} D_f(t,T_i)\tau_i(L_i(T_{i-1})-\kappa)\right\} && (5.29)\\
&= \sum_{i=\phi+1}^{\eta} P_t(T_i)\tau_i\mathbb{E}^{\mathbb{Q}^i}\left\{(L_i(T_{i-1})-\kappa)\right\} && (5.30)\\
&= \sum_{i=\phi+1}^{\eta} P_t(T_i)\tau_i(L_i(t)-\kappa) && (5.31)\\
&= \sum_{i=\phi+1}^{\eta} [P_t(T_{i-1})-(1+\tau_i\kappa)P_t(T_i)]. && (5.32)
\end{aligned}
$$

Notice that the same value is obtained by defining as T_ϕ-payoff

$$\sum_{i=\phi+1}^{\eta} P_{T_\phi}(T_i)\tau_i(L_i(T_\phi)-\kappa)$$

which results in the t-discounted payoff

$$D_f(t,T_\phi)\sum_{i=\phi+1}^{\eta} P_{T_\phi}(T_i)\tau_i(L_i(T_\phi)-\kappa)$$

and by taking the risk neutral expectation.

We now define the forward swap rate before moving on to the swaption definition.

Definition 5.4. *The forward swap rate $S_{\alpha,\beta}(t)$ at time t for the sets of times T and year fractions τ is the rate in the rate in the fixed leg of the above IRS that makes the IRS a fair contract at the present time t. The value that makes the contract fair is given by*

$$S_{\phi,\eta}(t) = \frac{P_t(T_\phi)-P_t(T_\eta)}{\sum_{j=c+1}^{d}\tau_j' P_t(T_j')} = \frac{1-\prod_{j=\phi+1}^{\eta}\frac{1}{1+\tau_j L_j(t)}}{\sum_{i=\phi+1}^{\eta}\tau_i\prod_{j=\phi+1}^{i}\frac{1}{1+\tau_j L_j(t)}}.$$

Finally we notice that the IRS discounted payoff (5.28) for a κ different from the swap rate can be expressed, at time $t = 0$, also in terms of swap rates as

$$D_f(0, T_\phi)(S_{\phi,\eta}(T_\phi) - \kappa) \sum_{i=\phi+1}^{\eta} \tau_i. \tag{5.33}$$

We now move on to define a swaption. As already mentioned in Chapter 3, a swaption is a contract that gives its holder the right to enter at a future time $T_\phi > 0$ an IRS, whose first reset time usually coincides with T_ϕ, with payments occurring at dates $T_{\phi+1},\ T_{\phi+2}, \ldots,\ T_\eta$. The fixed rate κ of the underlying swap is usually called *swaption strike*. As explained in Chapter 3, for the payer swaption this right will be exercised only when the swap rate at the exercise time T_ϕ is larger than the IRS fixed rate κ. Consequently, if we assume unit notional amount, the payer swaption payoff can be written as

$$D_f(0, T_\phi)(S_{\phi,\eta}(T_\phi) - \kappa)_+ \sum_{i=\phi+1}^{\eta} \tau_i. \tag{5.34}$$

Up to this point, we have assumed that only one yield curve is needed to price an interest rate derivative. Meaning that the payoff and the discount factor can all be obtained from one yield curve. This was the case prior to the 2007-09 financial crisis. It was thought that the it is impossible for AA rated banks to default on their loans. During the financial crisis of 2007-09, there were reports of banks manipulating LIBOR and these reports were later investigated by both the UK and US governments. After the financial crisis, market participants started searching for a more risk free rate that can be used for discounting payoffs. The Overnight Index Swap rate was elected as the best candidate to be the risk free rate. This implied that for pricing derivatives whose payoff is depended on LIBOR, two zero curves need to be modelled. The LIBOR zero curve for the payoff, and the OIS zero curve for discounting.

The modelling of these two zero rates simultaneously introduces arbitrage in the market. If we make the assumption that market participants can borrow or lend without risk, at either rate given by the zero curves, arbitrage opportunities emerge. For instance, a participant (bank) can lend at the OIS rate and borrow at the LIBOR rate to lock in a profit. In the next chapter, we introduce a model proposed by Andrei Lyashenko and Fabio Mercurio (2018) that aims to address the drawbacks of the LMM.

CHAPTER 6

The generalized Forward Market Model

The financial crisis of 2007-09 prompted the search for an alternative risk free rate, to transition away from LIBOR. The overnight rates have been chosen as this risk-free rates in all major economies. In the LMM framework, two zero rates must be modelled in order to price derivatives that are dependent on LIBOR. This, however, resulted in arbitrage opportunities in the market.

Andrei Lyashenko and Fabio Mercurio (2018) proposed a model that aims at addressing this issue. They show that by modelling the dynamics of term rates directly, it is possible to simulate both forward-looking and backward-looking rates using a single stochastic process for both. This joint modelling, for all given application periods, leads to an extension of the classic single-curve LMM, which we call the generalized Forward Market Model (FMM). The FMM is based on the concept of extended zero-coupon bonds, which is simple and natural, and is convenient when dealing with backward-looking setting -in-arrears. This concept has been used before by Glasserman and Zhao (2000) or Andersen and Piterbarg (2010) to define hybrid numeraires and measures but it has never been used to extend the definition of term rates from forward-looking to backward-looking, or to provide a generalization of a single-curve LMM. The FMM has some properties that the LMM does not have such as modelling the dynamics of the forward-rate under $\mathbb{Q}$ so that we do not have to approximate the dynamics under $\mathbb{Q}^d$. We first introduce the concept of backwards rates.

6.1 Introduction

The generalized Forward market model extends on the LIBOR market model, which we introduced in Chapter 5. The key idea in this model is that of the extended zero coupon bonds, first used by Glasserman and Zhao (2000) forms the basis of this LMM extension. The main difference between the LMM and the FMM is that in the LMM, the rates are only defined for times until the beginning of the accrual period and the rate is defined for all times, including the accrual period and the time after the accrual period. As we have seen in Chapter 5, the LIBOR rates evolve until T_{j-1}, which is the beginning of the accrual period. We say that this is a forward-looking rate. In the transition from LIBOR, we use backward-looking rates. Unlike the forward-looking rates, one has to wait until the end of the accrual period, T_j, to know their fixing, and even for times after the end of the accrual period. Figure 6.1 gives an illustration of the difference between the two rates.

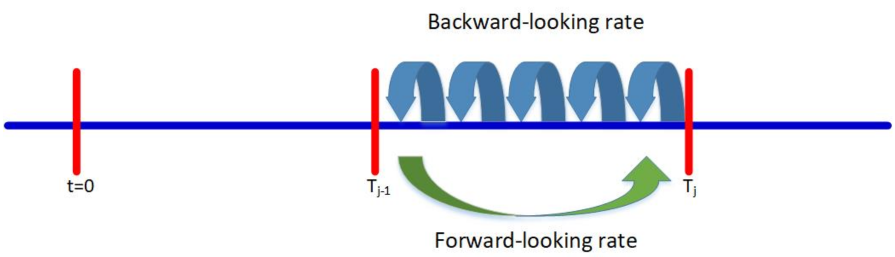


Figure 6.1: Forward-looking rate vs Backward-looking rate.

It is worth noting that financial institutions still have contracts that are dependent on the forward rates with maturities up to 30 years or even longer, so the new model still has to cater for that. As we will see in the coming chapters, the FMM models both the forward-looking and the backward-looking rates, and it can still be used to cater for the parties who are still interested in forward-looking rates.

6.2 Fundamentals of the generalized Forward Market Model

In the FMM, we will assume the same tenor structure as we did in Chapter 2, $0 = T_0, T_1 .., T_M$, and denote by τ_j the year fraction for the interval $[T_{j-1}, T_j)$. We now define it as $\tau_j = \tau_j - \tau_{j-1}$. For each time t, we define $\eta(t) = \min j : T_j \geq t$, which is the index of the element of the time structure that is the closest to time t being equal to or greater than t. Modelling RFRs such as SOFR or reformed SONIA is of great importance. We also consider a continuous-time financial market with an instantaneous risk-free rate r_t. If the same RFR is used for discounting and for deriving term rates, this implies a return to the classic single-curve modelling environment. This rate r_t has an associated cash account C_t such that $C(0) = 1$ as shown before, and the price at time t of the risk-free zero-coupon bond with maturity T, $P_{t,T}$, is given by:

$$P_{t,T} = \mathbb{E}^{\mathbb{Q}} \left\{ e^{-\int_t^T r(u)du} | \mathcal{F}_t \right\} \tag{6.1}$$

which is defined for $t \leq T$. We now introduce an important proposition.

Proposition 6.1. *The definition of $P_{t,T}$ can be extended to times $t > T$.*

Proof. Using 6.1 and the definition of C_t, when can write:

$$P_{t,T} = \mathbb{E}^{\mathbb{Q}} \left\{ e^{\int_T^t r(u)du} | \mathcal{F}_t \right\} = e^{\int_T^t r(u)du} = \frac{C_t}{C_t} \tag{6.2}$$

since $\int_T^t r(u)du$ is $\mathcal{F}_t$-measurable. □

Remark 6.1. *If we set the maturity to be $T = 0$, we have that $P_t(0) = C_t$, implying that the cash account can be viewed as a zero-coupon bond expiring immediately.*

Consider the self-financing strategy Y_T that consists of buying the zero-coupon bond with maturity T, and reinvesting the proceeds of the bond's unit notional received at T at the risk-free rate r_t from time T on-wards. Denoting by $Y(t)$ the time-t value of this strategy, we have:

$$Y_T(t) = \begin{cases} P_{t,T} & t \leq T \\ e^{\int_T^t r(u)du} = \frac{C_t}{C_t} & t > T \end{cases} \tag{6.3}$$

From here we see that, for $t > T$, $Y_T(t)$ is exactly the extended bond price defined by 6.1. We can now conclude that, for each given T, $Y_T(t) = P_{t,T}$ for all times t. Therefore, hereafter, we will refer to strategy Y_T as to the extended zero-coupon bond with maturity T, and bond prices $P_{t,T}$ will be meant in the extended sense. The extended zero-coupon bond defined by (6.1) can still be considered as a viable numeraire since it is the value of a self financing strategy and is strictly positive. Therefore, we can define the extended T-forward measure $\mathbb{Q}^T$ as follows:

Definition 6.1. *The extended T-forward measure $\mathbb{Q}^T$ can be defined the usual way as the equivalent martingale measure associated with the extended bond price $P_{t,T}$ for any time t. Since, for $t > T$,*

$$P_{t,T} = \frac{C_t}{C_t}.$$

The extended T-forward measure $\mathbb{Q}^T$ is a hybrid measure that combines the classic T-forward measure up to the maturity time T with the risk neutral cash account measure $\mathbb{Q}$ after T. Such hybrid T-forward measures have been discussed in the literature. We note that $P_t(0) = C_t$, so the risk-neutral cash-account measure is a particular case of the extended T-forward measure, where T is equal to zero: $\mathbb{Q} = \mathbb{Q}^0$.

6.3 Backward-looking rates and Backward-looking forward rates

In this section, we introduce backward-looking rates and the backward-looking forward rate. It is important that the reader understands the fundamental differences between these two rates as they play a key role in the generalized FMM. Before we introduce the mathematical concepts pertaining to these two rates, we will first explain the difference using the timeline in Figure 6.2 so as to ease the reader into the mathematical definitions.

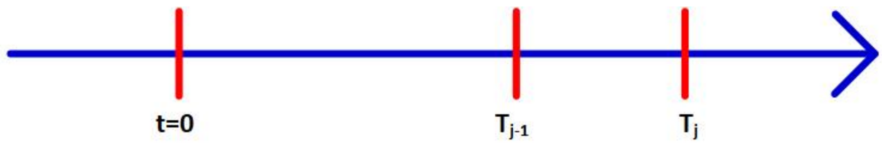


Figure 6.2: Backwards-looking rates vs backwards-looking forward rates.

The backward-looking rate $R(T_{j-1}, T_j)$ can be defined as the rate that will be observed in the market at the end of an accrual period. One can also refer to this rate as the market risk-free term rate (RFRs). Assume that $T_j - T_{j-1} = 3$ months in Figure 6.2 and we are standing at T_j. The compounded overnight rates from T_{j-1} to T_j can be defined as the backward-looking rate. The overnight rates are only known at T_j for every period $[T_{j-1}, T_j]$ and are supplied by the market. So in order for us to get this backward-looking rate, it is best to use the compounded setting-in-arrears method.

The backward-looking forward rate, on the other hand, is the backward-looking rate observed at $t = 0$ for a future accrual period $[T_{j-1}, T_j]$. So, if we are yet to arrive at the end of the accrual period for the backward-looking rate, i.e. $t < T_j$, we must give an expectation of the backward-looking rate over the accrual period. This backward-looking forward rate will be simulated by the generalized FMM for $t \in [T_0, T_j)$. It is possible to simulate the expected backward-looking forward rate until T_j. It is only logical that we expect the backward-looking rate to be the same as the backward-looking forward rate at T_j.

In this section, we introduced the backward-looking rate and the backward-looking forward rate and explained how they are different. A thorough understanding of these two rates is important in order to understand the concepts that will be introduced in the coming sections.

6.3.1 Backward-looking rate

In this section we define backward-looking rate which we previously denoted by $R(T_{j-1}, T_j)$. This backward-looking rate is the daily compounded setting-in-arrears rate based on the overnight risk-free rates. We first define r_i as the overnight rate on day i with δ being the corresponding day-count fraction over the period $[T_{j-1}, T_j]$. We assume that it takes n days in the accrual period. For each $j = 1, ..., M$, we approximate the daily-compounded setting-in-arrears rate for the interval $[T_{j-1}, T_j)$, which we denote by $R(T_{j-1}, T_j)$ as follows:

$$R(T_{j-1}, T_j) = \frac{1}{\tau_j}\left[\prod_{i=1}^{n}(1 + r_i\delta_i) - 1\right] \tag{6.4}$$

$$R(T_{j-1}, T_j) \approx \frac{1}{\tau_j}\left(e^{\int_{T_{j-1}}^{T_j} r(u)du} - 1\right) = \frac{1}{\tau_j}\left[P_{j-1}(T_j) - 1\right] \tag{6.5}$$

$$\tag{6.6}$$

To get the approximation, we take the limit of the day-count fraction to zero. We see that extended zero coupon definition is used in Equation 6.5. In-arrears rates are backward-looking in nature because one has to wait until the end of their accrual period to know their fixing value. Alternatively one can define forward looking rates, which are set at the beginning of their accrual period. Let $F(T_{j-1}, T_j)$ denote such a rate and by arbitrage-free, we have:

$$F(T_{j-1}, T_j) = \mathbb{E}^{\mathbb{Q}_{T_j}} \left\{ R(T_{j-1}, T_j) | \mathcal{F}_{T_{j-1}} \right\} \tag{6.7}$$

for each $j = 1, ..., M$. If Equation 6.7 does not hold, it would imply that the forward rate which is fixed at T_{j-1} is not equal to the expectation of the backward-looking rate over the accrual period $[T_{j-1}, T_j]$.

6.3.2 Backward-looking forward rates

In this section, we define the backward-looking forward rate $R(t, T_{j-1}, T_j)$. For ease of read, we will use the shorthand notation $R_j(t)$. For instance, the expected backward-looking forward rate in the future over the accrual period $[T_{j-1}, T_j)$ observed at $t = 0$ is denoted by $R_j(0)$. Since we are still at $t = 0$, we still have to simulate this rate until the end of the accrual period.

Definition 6.2. *The backward-looking forward rate $R_j(t)$ at time t is the value of the fixed rate κ_R in the swaplet paying $\tau_j[R(T_{j-1}, T_j) - \kappa_R]$ at time T_j, such that the swaplet has zero value at time t. By arbitrage-free, we have:*

$$R_j(t) = \mathbb{E}^{\mathbb{Q}_{T_j}} \left\{ R(T_{j-1}, T_j) | \mathcal{F}_{T_{j-1}}] \right\}. \tag{6.8}$$

From (6.4) and (6.7) we see that, for each $j = 1, ..., M$, the forward-looking spot rate $F(T_{j-1}, T_j) = R_j(T_{j-1})$. We can also derive the dynamics of the backward-looking forward rate dynamics under $\mathbb{Q}$, which is the risk neutral measure. We start from Equation 6.8

$$R_j(t) = \mathbb{E}^{\mathbb{Q}_{T_j}}\left\{R(T_{j-1},T_j)|\mathcal{F}_{T_{j-1}}]\right\} = \mathbb{E}^{\mathbb{Q}_{T_j}}\left\{\frac{1}{\tau_j}e^{\int_{T_{j-1}}^{T_j} r(u)du} - 1\middle|\mathcal{F}_{T_{j-1}}\right\}, \tag{6.9}$$

$$1 + \frac{1}{\tau_j}R_j(t) = E^{\mathbb{Q}_{T_j}}\left\{e^{\int_{T_{j-1}}^{T_j} r(u)du}\middle|\mathcal{F}_{T_{j-1}}\right\}. \tag{6.10}$$

We use Definition 3.8 to move from one measure to another:

$$\frac{d\mathbb{Q}}{d\mathbb{Q}_{T_j}} = \frac{C(T_j)}{C_t}\frac{P_t(T_j)}{P_{T_j}(T_j)}. \tag{6.11}$$

Using the definition of an expectation, from Equation 6.10, we have that the RHS of the equation can be rewritten as

$$\mathbb{E}^{\mathbb{Q}_{T_j}}\left\{e^{\int_{T_{j-1}}^{T_j} r(u)du}\middle|\mathcal{F}_{T_{j-1}}\right\} = \int e^{\int_{T_{j-1}}^{T_j} r(u)du} d\mathbb{Q}_{T_j} = \int e^{\int_{T_{j-1}}^{T_j} r(u)du}\frac{C(T_j)}{C_t}\frac{P_t(T_j)}{P_{T_j}(T_j)}d\mathbb{Q} \tag{6.12}$$

$$= \frac{1}{P_j(t)}\mathbb{E}^{\mathbb{Q}}\left\{e^{-\int_t^{T_j} r(u)du}e^{\int_{T_{j-1}}^{T_j} r(u)du}\middle|\mathcal{F}_{T_{j-1}}\right\} = \frac{1}{P_j(t)}\mathbb{E}^{\mathbb{Q}}\left\{e^{-\int_t^{T_{j-1}} r(u)du}\middle|\mathcal{F}_{T_{j-1}}\right\} \tag{6.13}$$

$$= \frac{P_{j-1}(t)}{P_j(t)}. \tag{6.14}$$

So we can write

$$R_j(t) = \frac{1}{\tau_j}\left[\frac{P_{j-1}(t)}{P_j(t)} - 1\right]. \tag{6.15}$$

This is the classic, simply-compounded, forward-rate formula, which thanks to our definition of extended bond price, holds for each time t, even those after T_j. From Equation 6.15, we observe four key properties of the backward-looking forward rate:

(i) $R_j(t)$ is a martingale under the T_j-forward measure. To prove this, we can rewrite Equation 6.15 as

$$P_j(t)R_j(t) = \frac{1}{\tau_j}\left[P_{j-1}(t) - P_j(t)\right].$$

When we examine the movement of $R_j(t)$ under the $\mathbb{Q}_{T_j}$, we see that it is a martingale

$$\frac{R_j(t)P_j(t)}{P_j(t)} = \mathbb{E}^{\mathbb{Q}_{T_j}}\left\{\frac{R_j(U)P_j(U)}{P_j(U)}\middle|\mathcal{F}_{T_{j-1}}\right\},$$
$$R_j(t) = \mathbb{E}^{\mathbb{Q}_{T_j}}\left\{R(U)|\mathcal{F}_{T_{j-1}}\right\}.$$

This derivation is similar to Lemma 5.1 for the LMM. Again, only one $R_j(t)$ is a martingale under the $\mathbb{Q}_{T_j}$, all the other rate are martingales under their respective the $\mathbb{Q}_{T_j}$'s.

(ii) $R_j(t)$ is equal to the forward-looking spot rate at time $T_{j-1} : R_j(T_{j-1}) = F(T_{j-1}, T_j)$. To prove this, we look at Equation 6.15.

$$R_j(t) = \frac{1}{\tau_j}\left[\frac{P_{j-1}(t)}{P_j(t)} - 1\right] = \frac{1}{\tau_j}\left[\frac{P_{j-1}(t)}{P_j(t)} - \frac{P_j(T_{j-1})}{P_j(T_{j-1})}\right] = \frac{1}{\tau_j}\left[\frac{P_{j-1}(T_{j-1}) - P_j(T_{j-1})}{P_j(T_{j-1})}\right]$$
$$= F(T_{j-1}, T_j).$$

(iii) $R_j(t)$ is equal to the realized backward-looking rate at time $T_j : R_j(T_j) = R(T_{j-1}, T_j)$. This follows from the definition of backward-looking forward rates. By setting $t = T_j$ in Equation 6.15 and compare it with Equation 6.5,

$$R_j(T_j) = \frac{1}{\tau_j}\left[\frac{P_{j-1}(T_j)}{P_j(T_j)} - 1\right] = \frac{1}{\tau_j}\left[P_{j-1}(T_j) - 1\right],$$
$$R(T_{j-1}, T_j) = \frac{1}{\tau_j}\left[P_{j-1}(T_j) - 1\right].$$

(iv) $R_j(t)$ stops evolving after $T_j : R_j(t) = R(T_{j-1}, T_j), t > T_j$. This is a key property because want the dynamics of $R_j(t)$ to be fixed after the accrual period. From Equation 6.15, for $t > T_j$, where we use the definition of the extended zero-coupon bonds, we get

$$\begin{aligned}R_j(t) &= \frac{1}{\tau_j}\left[\frac{P_{j-1}(t)}{P_j(t)}-1\right] = R_j(t) = \frac{1}{\tau_j}\left[\frac{P(t,T_{j-1})}{P(t,T_j)}-1\right] = \frac{1}{\tau_j}\left[\frac{C_t}{C(T_{j-1})}\frac{C(T_j)}{C_t}-1\right]\\ &= \frac{1}{\tau_j}\left[\frac{C(T_j)}{C(T_{j-1})}-1\right].\end{aligned}$$

6.3.3 Forward-looking forward rates

We already claimed that the generalized FMM models both the forward-looking forward rates and the backward-looking forward rates. In this section, we look to provide a mathematical argument to support this claim. We first recap on forward-looking forward rates.

Definition 6.3. *The forward-looking forward rate $F_j(t)$ at time t is the value of the fixed rate κ_F in the swaplet that pays $\tau[F(T_{j-1},T_j)-\kappa_F]$ at time T_j, such that the swaplet has zero value at time t. By arbitrage-free, we have for $t \leq T_{j-1}$:*

$$\begin{aligned}F_j(t) &= \mathbb{E}^{\mathbb{Q}_{T_j}}\{F(T_{j-1},T_j)|\mathcal{F}_t\} =\\ &= \mathbb{E}^{\mathbb{Q}_{T_j}}\{\mathbb{E}^{\mathbb{Q}_{T_j}}\{R(T_{j-1},T_j)|\mathcal{F}_{T_{j-1}}\}|\mathcal{F}_t\}\\ &= \mathbb{E}^{\mathbb{Q}_{T_j}}\{F(T_{j-1},T_j)|\mathcal{F}_t\} = R_j(t).\end{aligned}$$

From this last equality, we can see that $F_j(t) = R_j(t)$ for $t \leq T_{j-1}$. At time T_{j-1}, $F_j(t)$ fixes, while $R_j(t)$ continues its journey until it fixes at time T_j. Finally, $R_j(t) = R_j(T_j),\ t \geq T_j$. So in the new generalized FMM, we can get the dynamics of $R_j(t)$ over the accrual period $[0, T_{j-1}]$, which is the same as that of $F_j(t)$ over the same accrual period.

6.4 The generalized FMM

In this section we will model the joint evolution of rates $R_j(t)$ for $j = 1, ..., M$, and introduce a natural extension of the classic single-curve LMM to the case where rates are set in arrears. The advantage of this approach is that we can obtain both forward-looking and backward-looking fixings using a single process. Just as in the LMM case, the generalized FMM has the $\mathbb{Q}_{T_j}$ dynamics given by

$$dR_j(t) = \mu_j(t)dt + \hat{\sigma}dW_j(t), \tag{6.16}$$

where we have the freedom to choose the dynamics of $\hat{\sigma}$. In the previous chapter, we showed that $R_j(t)$ is a martingale under its associated extended T_j-forward measure (Property i). This implies under that specific measure, the dynamics have a zero drift. We assume that the rates have a log-normal distribution, just as we did with the LMM. This means that we will be considering the log-normal FMM. We choose the log-normal dynamics because of the possibility of pricing the interest rate derivatives such as caps/floors in the same manner they are priced in the market. We therefore, choose to specify the volatility parameter in the following manner

$$dR_j(t) = \sigma_j(t)R_j(t)1_{\{t\leq T_j\}}dW_j(t) \tag{6.17}$$

where, for each $j = 1, ..., M$, $\sigma_j(t)$ is an adapted process and $W_j(t)$ is a standard Wiener such that $dW_i(t)dW_j(t) = \rho_{i,j}dt$. To ensure that the process is defined, and is constant, for times larger than T_j (Property iv), we introduce the indication function $1_{\{t\leq T_j\}}$.

Again, $\sigma_j(t)$ is the instantaneous volatility. It is worth examining the behaviour of the instantaneous volatility of the new $R_j(t)$, with its behaviour over the accrual period $[T_{j-1}, T_j]$ being more interesting. For this examination of the instantaneous volatility, we will consider Figure 6.3. We assume that the accrual period is 3 months or 90 days.

Looking at the upper picture and assuming that we are at T_{j-1}, it means that we are yet to know the overnight rates which we need to compound in order to get the backward-looking rate. Each line represents 5.625 days. In the bottom picture, we already know the overnight rates of 78.75 days and they have been fixed. Since already know most of most of the overnight rates, the remaining days will not have much of an impact on what the realized backward-looking rate at T_j will be. We see that as we cover more days, the volatility becomes less and less. So we can conclude that the volatility of $R_j(t)$ should decrease more and more over $[T_{j-1}, T_j]$.

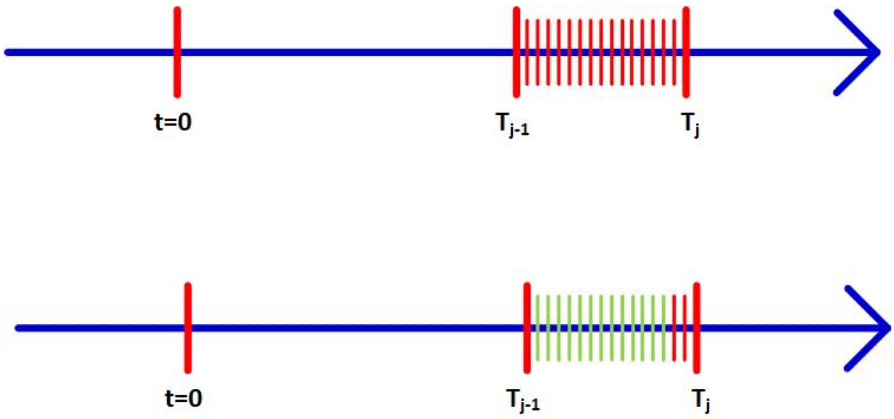


Figure 6.3: Volatility in the accrual period.

We then choose a piece-wise differentiable (deterministic) function g_j such that the dynamics of $R_j(t)$ are given by

$$dR_j(t) = \sigma_j(t) R_j(t) g_j(t) dW_j(t). \tag{6.18}$$

Since we are free to choose this g_j function, we let

$$g_j(t) = \min\left\{ \frac{(T_j - t)_+}{T_j - T_{j-1}}, 1 \right\}. \tag{6.19}$$

The behaviour of this $g_j(t)$ over $[0, T_j]$ can be described as follows:

(i) For $0 \leq t < T_{j-1}$, $g_j(t)$ since t is much farther away from T_j compared to T_{j-1}. This implies that $R_j(t) = F_j(t)$ in this period.

(ii) For $T_{j-1} \leq t < T_j$, $g_j(t)$ is monotonically decreasing. This implies that the we incorporate the decrease in volatility of the dynamics already explained above.

(iii) For $t \geq T_j$, $g_j(t) = 0$. This means that (Property iv) holds. So we can do away with the indicator function.

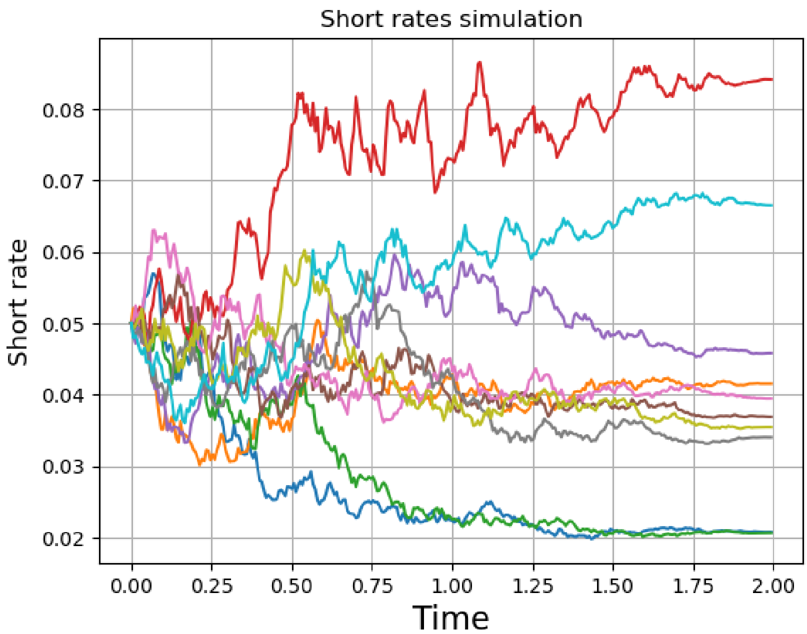


Figure 6.4: 10 paths of the FMM dynamics under the same application period.

In Figure 6.4 we compute 10 paths of the backward-looking forward rates. We notice that unlike in the LMM case, these dynamics are defined for any time t. The rate does not stop at $T = 1.75$, but continues to evolve in a stochastic manner until $T = 2$, and proceeds to remain constant thereafter. In this application period, $T = [1.75, 2]$, the effect of the decaying volatility in the rate accrual period is clearly visible.

6.4.1 Construction of the generalized FMM

The dynamics of $R_j(t)$ is given by Equation 6.18 under the corresponding T_j-forward measure. A market model where all forward rates, for $j = 1, ..., M$, are modelled jointly can be defined by deriving the dynamics of each forward under a common probability measure. We apply the change of numeraire formula relating the drifts of a given process under two measures with known numeraires. In our case, we know the dynamics of $R_j(t)$ under the T_j-forward measure, and we want to derive its dynamics under a measure $\mathbb{Q}^N$, which is associated with numeraire $N(t)$. Assuming continuous dynamics, the drift of R_j under $\mathbb{Q}^N$, as a function of time t, is given by:

$$\mu(R_j; \mathbb{Q}^N) = \frac{dR_j(t) d\ln(N(t)/P_t(T_j))}{dt}. \tag{6.20}$$

Proposition 6.2. *The forward rate dynamics under the risk-neutral measure $\mathbb{Q}$ is given by the following SDE:*

$$dR_j(t) = \sigma_j(t) g_j(t) R_j(t) \sum_{i=1}^{j} \rho_{i,j} \frac{\tau_i \sigma_i(t) g_i(t) R_i(t)}{1 + \tau_i R_i(t)} dt + \sigma_j(t) g_j(t) R_i(t) dW_j^{\mathbb{Q}} \tag{6.21}$$

where $W_j^{\mathbb{Q}}$ is a $\mathbb{Q}$-Wiener process.

Proof. We first apply (6.20) to the case where $N(t) = C_t$ and $\mathbb{Q}^N = \mathbb{Q}$ so that the drift of R_j under $\mathbb{Q}$ is given by

$$\mu(R_j; \mathbb{Q})(t) = \frac{dR_j(t) d\ln(C_t/P_t(T_j))}{dt}. \tag{6.22}$$

Thanks to the definition of extended bond prices, we can write:

$$
\begin{aligned}
\ln \frac{C_t}{P_t(T_j)} &= \ln \frac{P_t(0)}{P_t(T_j)} \\
&= \ln \prod_{i=1}^{j} \frac{P_t(T_{i-1})}{P_t(T_i)} \\
&= \ln \prod_{i=1}^{j} [1 + \tau_i R_i(t)] \\
&= \sum_{i=1}^{j} \ln[1 + \tau_i R_i(t)].
\end{aligned}
$$

We then have:

$$
\begin{aligned}
\mu(R_j; \mathbb{Q}) &= \frac{dR_j(t) d \sum_{i=1}^{j} \ln[1 + \tau_i R_i(t)]}{dt} \\
&= \sum_{i=1}^{j} \frac{dR_j(t) d \ln[1 + \tau_i R_i(t)]}{dt} \\
&= \sum_{i=1}^{j} \frac{\tau_i}{1 + \tau_i R_i(t)} \frac{dR_j(t) dR_i(t)}{dt}.
\end{aligned}
$$

From (6.18), We have that:

$$
\mu(R_j; \mathbb{Q}) = \sigma_j(t) g_j(t) \sum_{i=1}^{j} \rho_{i,j} \frac{\tau_i \sigma_i(t) g_i(t)}{1 + \tau_i R_i(t)}
$$

The $\mathbb{Q}$-dynamics of R_j then becomes:

$$
dR_j(t) = \sigma_j(t) g_j(t) \sum_{i=1}^{j} \rho_{i,j} \frac{\tau_i \sigma_i(t) g_i(t)}{1 + \tau_i R_i(t)} dt + \sigma_j(t) g_j(t) dW_j^{\mathbb{Q}} \tag{6.23}
$$

where $W_j^{\mathbb{Q}}$ is a $\mathbb{Q}$-Wiener process.

$\square$

Remark 6.2. *Because of the definition of $g_i(t)$, which is zero for $t \geq T_i$, that is for $\eta(t) > i$, the drift term in* (6.21) *can be expressed in terms of the index function $\eta(t)$. We have*

$$dR_j(t) = \sigma_j(t)g_j(t)\sum_{i=\eta(t)}^{j}\rho_{i,j}\frac{\tau_i\sigma_i(t)g_i(t)}{1+\tau_iR_i(t)}dt + \sigma_j(t)g_j(t)dW_j^{\mathbb{Q}}. \tag{6.24}$$

Proposition 6.3. *The forward rate dynamics under the classic spot-LIBOR measure $\mathbb{Q}^d$ is given by the following SDE:*

$$dR_j(t) = \sigma_j(t)g_j(t)\sum_{i=\eta(t)+1}^{j}\rho_{i,j}\frac{\tau_i\sigma_i(t)g_i(t)}{1+\tau_iR_i(t)}dt + \sigma_j(t)g_j(t)dW_j^{d} \tag{6.25}$$

where W_j^d is a standard Wiener process under $\mathbb{Q}^d$, and where the drift is zero when $j \leq \eta(t)$.

Proof. We now apply(6.20) to the case where $N(t) = C_d(t)$ and $\mathbb{Q}^N = \mathbb{Q}^d$. The drift of R_j under $\mathbb{Q}^d$ is given by:

$$\begin{aligned}\mu(R_j;\mathbb{Q}^d)(t) &= \frac{dR_j(t)d\ln(B_d(t)/P_t(T_j))}{dt}\\ &= \frac{dR_j(t)d\ln(P(t,T_{\eta(t)})/P_t(T_j))}{dt}\end{aligned}$$

Following steps similar to those in the previous section, it can be shown that the $\mathbb{Q}^d$-dynamics are given by:

$$dR_j(t) = \sigma_j(t)g_j(t)\sum_{i=\eta(t)+1}^{j}\rho_{i,j}\frac{\tau_i\sigma_i(t)g_i(t)}{1+\tau_iR_i(t)}dt + \sigma_j(t)g_j(t)dW_j^{d} \tag{6.26}$$

where W_j^d is a standard Wiener process under $\mathbb{Q}^d$, and where the drift is zero when $j \leq \eta(t)$. $\square$

Remark 6.3. *Comparing the* (6.24) *and* (6.26) *we see that the difference between rate* *dynamics under $\mathbb{Q}$ and $\mathbb{Q}^d$ is given by:*

$$\mu(R_j;\mathbb{Q})(t) - \mu(R_j;\mathbb{Q}^d)(t) = \frac{\tau_\eta(t)\sigma_{\eta(t)}(t)g_{\eta(t)}(t)}{1+\tau_{\eta(t)}R_{\eta(t)}(t)}\sigma_j(t)g_j(t). \tag{6.27}$$

In the classic LMM, when approximating $\mathbb{Q}$ with $\mathbb{Q}^d$, one cannot quantify the magnitude of the approximation. The generalized FMM addresses this issue. (6.27) *gives us the impact of moving from $\mathbb{Q}$ to $\mathbb{Q}^d$. Note also, since $g_i(T_i) = 0$, the $\mathbb{Q}$-drift term in* (6.24) *does not experience a jump when time t moves right past T_i and index $\eta(t)$ jumps from i to $i+1$. On the other hand, the $\mathbb{Q}^d$-drift in* (6.26) *does jump since it loses the* $(i+1)$*-term.*

Proposition 6.4. *The forward rate dynamics under the T_k-forward measure is given by the following SDE:*

$$dR_j(t) = \sigma_j(t)g_j(t)\sum_{i=K+1}^{j}\rho_{i,j}\frac{\tau_i\sigma_i(t)g_i(t)}{1+\tau_iR_i(t)}dt + \sigma_j(t)g_j(t)dW_j^K \tag{6.28}$$

for $j > k$ and

$$dR_j(t) = -\sigma_j(t)g_j(t)\sum_{i=j+1}^{k}\rho_{i,j}\frac{\tau_i\sigma_i(t)g_i(t)}{1+\tau_iR_i(t)}dt + \sigma_j(t)g_j(t)dW_j^K \tag{6.29}$$

when $j < k$ and where W_j^k is a standard Wiener process under $\mathbb{Q}^{T_k}$.

Finally, when $N(t) = P_t(T_k)$ and $\mathbb{Q}^N = \mathbb{Q}^{T_k}$ the drift of R_j under $\mathbb{Q}^{T_k}$ is given by:

$$\mu(R_j;\mathbb{Q}^{T_k})(t) = \frac{dR_j(t)d\ln(P_t(T_k)(t)/P_t(T_j))}{dt}. \tag{6.30}$$

Repeating the same procedure as before, we then get

$$dR_j(t) = \sigma_j(t)g_j(t)\sum_{i=K+1}^{j}\rho_{i,j}\frac{\tau_i\sigma_i(t)g_i(t)}{1+\tau_iR_i(t)}dt + \sigma_j(t)g_j(t)dW_j^K \tag{6.31}$$

for $j > k$ and

$$dR_j(t) = -\sigma_j(t)g_j(t)\sum_{i=j+1}^{k}\rho_{i,j}\frac{\tau_i\sigma_i(t)g_i(t)}{1+\tau_i R_i(t)}dt + \sigma_j(t)g_j(t)dW_j^K \tag{6.32}$$

when $j < k$ and where W_j^k is a standard Wiener process under $\mathbb{Q}^{T_k}$.

Remark 6.4. *Since $\mathbb{Q}^{T_k}$ is a hybrid measure that consists of the classic T_k-forward measure up to time T_k and of the risk-neutral measure $\mathbb{Q}$ after T_k, we should have:*

$$\mu(R_j; \mathbb{Q}^{T_k}(t)) = \mu(R_j; \mathbb{Q})(t) \ for \ t \geq T_k.$$

In fact, using the drifts formulas from the previous sections, we get for $t \geq T_k$ and $j > k$

$$\begin{aligned}\mu(R_j, \mathbb{Q})(t) &= \sigma_j(t)g_j(t)\sum_{i=1}^{j}\frac{\tau_i\sigma_i(t)g_i(t)}{1+\tau_i R_i(t)}\rho_{i,j} \\ &= \sigma_j(t)g_j(t)\sum_{i=k+1}^{j}\frac{\tau_i\sigma_i(t)g_i(t)}{1+\tau_i R_i(t)}\rho_{i,j} = \mu(R_j, \mathbb{Q}^{T_k})(t)\end{aligned}$$

where we used the property that $g_i(t) = 0$ for $t > T_i$. In particular, when $k = 0$, we can confirm that

$$\mu(R_j, \mathbb{Q}^{T_d})(t) = \mu(R_j, \mathbb{Q})(t).$$

6.5 Differences between FMM and LMM

The FMM is essentially an extension of the classic single-curve LMM since it models the joint dynamics of both forward-looking rates $F_j(t)$ and backward-looking forward rates $R_j(t)$, since $F_j(t) = R_j(t)$ for all times t before the expiry time T_{j-1} of $F_j(t)$. The FMM also has additional properties summarized below:

(i) **Model Completeness**

The generalized forward rates $R_j(t)$ are distinguished from IBORs by their property of being complete in terms of in terms of spanning the periods defined by the time grid $[T_0, ..., T_m]$. For any index $j = 1, \ldots, M$ and for any time t. The price of zero-coupon bond with maturity T_j can be expressed in terms of the cash account C_t and forward rates as follows:

$$P_t(T_j) = C_t \prod_{i=1}^{j} \frac{1}{1 + \tau_i R_i(t)} \tag{6.33}$$

with the equality holding for all t, including $t > T_j$ which implies

$$\frac{dP_t(T_j)}{P_t(T_j)} = r_t dt - \sum_{i=1}^{j} \frac{\tau_i}{1 + \tau_i R_i(t)} \sigma_i(t) g_i(t) dW_i^{\mathbb{Q}}(t). \tag{6.34}$$

So the volatility of all bonds $P_t(T_j)$ is known and is a function of rates as well as their instantaneous co-variance structure. Analogous representations are not available in the LMM when using forward looking rates $F_j(t)$. It is for this reason why we have a closed-form drift term representation under the $\mathbb{Q}$-measure for FMM, but not for LMM. In this respect, FMM is a complete model while LMM is not.

(ii) **Better pricing of futures contracts**

The availability of $\mathbb{Q}$-dynamics of term forward rates presents a number of advantages when it comes to the valuation of contracts. The time-t future price of a contract that pays out H_T at time $T > t$ is given by:

$$f_t = \mathbb{E}^{\mathbb{Q}}\{H_T | \mathcal{F}_t\}.$$

Since no $\mathbb{Q}$-dynamics are directly available in a classic LMM, one typically approximates $\mathbb{Q}$ with $\mathbb{Q}^d$ to explicitly calculate the future price f_t:

$$f(t) \approx \mathbb{E}^{\mathbb{Q}^d}\{H_T | \mathcal{F}_t\}$$

with $\mathbb{E}^{\mathbb{Q}^d}$ denoting the expectation under $\mathbb{Q}^d$. We no longer need this approximation for the FMM, since we know the forward-rate dynamics under $\mathbb{Q}$, so the former formula can be used.

6.5.1 Easier extension to a cross-currency interest-rate model

Let us assume that we have a two-currency economy where domestic and foreign rates are driven by corresponding FMMs, and denote by $X(t)$ the spot exchange rate at time t, meaning that one unit of foreign currency can be purchased with $X(t)$ units of domestic currency. If we model the dynamics of $X(t)$, we can easily derive the dynamics of the foreign FMM under the domestic measure $\mathbb{Q}$, as well as the dynamics of the domestic FMM under the foreign money-market risk-neutral measure $\mathbb{Q}^f$. In fact, assuming continuous dynamics, the drift of the foreign rate R_j^f under the domestic measure $\mathbb{Q}$ is given by:

$$\begin{aligned}\mu(R_j^f;\mathbb{Q})(t) &= \mu(R_j^f;\mathbb{Q}^f)(t) + \frac{dR_j^f(t)d\ln(C_t/[C^f(t)X(t)])}{2} \\ &= \mu(R_j^f;\mathbb{Q}^f)(t) - \frac{dR_j^f(t)d\ln(X(t))}{dt}\end{aligned} \tag{6.35}$$

where $C^f(t)$ is the price at time t of the foreign cash account. An analogous formula applies for the drift of $R_j(t)$ under $\mathbb{Q}^f$. Again, similar formulas are not available in the classic LMM.

6.5.2 More natural hybrid modelling

For a hybrid equity-IR model, the equity risk-neutral drift rate is equal to r_t, which is not known in the classic LMM, and nor are its integrals. When using an FMM instead, we can express the integral of the drift rate in terms of rates $R_j(t)$, which are by definition known in the model. To show this, assume that the $\mathbb{Q}$-dynamics of a given stock S is:

$$dS(t) = r_t S(t)dt + \sigma S(t)dW(t).$$

Then, for each pair of indexes $j < k$, we can write:

$$\begin{aligned} S(T_k) &= S(T_j)e^{\int_{T_j}^{T_k}} e^{-\frac{1}{2}\sigma^2(T_k - T_j) + \sigma[W(T_k) - W(T_j)]} \\ &= S(T_j) \prod_{i=j+1}^{k} e^{\int_{T_j}^{T_k}} e^{-\frac{1}{2}\sigma^2(T_k - T_j) + \sigma[W(T_k) - W(T_j)]} \\ &= S(T_j) \prod_{i=j+1}^{k} e^{\int_{T_j}^{T_k}} e^{-\frac{1}{2}\sigma^2(T_k - T_j) + \sigma[W(T_k) - W(T_j)]}. \end{aligned} \tag{6.36}$$

Therefore, $S(t)$ can be simulated by simulating the joint evolution of rates $R_j(t)$ and Wiener process $W(t)$, assuming a correlation structure among them.

6.6 Pricing using the Forward Market Model

6.6.1 The valuation of an RFR fixed-floating swap

We consider a swap where the floating leg pays at each time $T_j,\ j = \phi + 1, \ldots,\ \eta$, a rate that is obtained by compounding the daily fixings of the RFR from T_{j-1} to T_j, and where the fixed leg pays the fixed rate κ on dates $T'_{c+1}, \ldots, T'_d$, with $T'_c = T_\phi$ and $T'_d = T_\eta$. We approximate the floating-leg payment at each T_j by $R(T_{j-1}, T_j$ times the corresponding year fraction, that is

$$\tau_j R(T_{j-1}, T_j) = e^{\int_{T_{j-1}}^{T_j} r(u)du} - 1.$$

The swap value to the fixed-rate payer at time $t < T_{\phi+1}$ is thus given by:

$$\sum_{j=\phi+1}^{b} \tau_j P_t(T_j) R_j(t) - \kappa \sum_{j=\phi+1}^{d} \tau'_j P_t(T'_j)$$

where τ'_j denotes the year fraction for the fixed-leg interval $[T'_{j-1}, T'_j)$. Note that for $t \leq T_\phi$, the price of the swap remains the same when we switch from forward-looking to backward-looking rates. The corresponding forward swap rate is defined as the fixed rate κ that makes the swap value equal to zero at time t, that is

$$S(t) = \frac{\sum_{j=\phi+1}^{\eta} \tau_j P_t(T_j) R_j(t)}{\sum_{j=c+1}^{d} \tau'_j P_t(T'_j)} = \frac{P_t(T_\phi) - P_t(T_\eta)}{\sum_{j=c+1}^{d} \tau'_j P_t(T'_j)}$$

where the second equality follows from (6.15). This formula coincides with the classic one in the single-curve framework.

Remark 6.5. *Since the extended zero coupon prices $P_t(T_j)$, and forward rates $R_j(t)$ are defined for all values of time t, the swap price formula above is also defined for all values of t. Similar to the case of zero-coupon bond, for $t > T_\phi$ the swap value given by the formula above represents the time-t value of a self-financing investment strategy where all cash flows are reinvested at the risk-free rate to roll them forward to the present time. This can be considered as a total present value of the strategy, which is inclusive of past cash flows, and can be used to compare current performance of different investments.*

6.6.2 The valuation of an Risk Free Rate cap

In the generalized FMM framework, we have two types of caplets which we can construct for every accrual period $[T_{j-1}, T_j]$. The first one is just like the caplet used in the LMM framework, which is called the forward-looking caplet. Given a strike price κ, this caplet has a time T_j payoff given by

$$[R(T_{j-1}; T_{j-1}, T_j) - \kappa]_+. \tag{6.37}$$

We showed that $R_j(t) = F_j(t)$ for $t \leq T_{j-1}$, so we can model this caplet in the same way as modelled the LMM caplet. From property ii of $R_j(t)$, we get the assurance that this caplet will be the same as the caplet observed in today's market at T_{j-1}.

The second caplet is known as a backward-looking caplet. Given a strike price κ, this caplet has a time T_j payoff given by

$$[R(T_j; T_{j-1}, T_j) - \kappa]_+. \tag{6.38}$$

In this caplet, the payoff is only known at time T_j, which is the payoff date. There is more uncertainty with the payoff of this type of caplet because the backward rate evolves over more time.

The difference in the two payoffs lies in that the former is known at the beginning of the application period, T_{j-1}, whereas the latter is known at the end, T_j. The valuation of the two caplets relies on the modelling of the forward rate $R_j(t)$ in the T_j-forward measure. Since the rate for the backward-looking caplet is known only at the end of the accrual period, it means that there is more uncertainty with this caplet and therefore, the price of this caplet should be higher than that of the forward-looking caplet. We can also prove this using the tower property of conditional expectations and the Jensen inequality. For $t \leq T_{j-1}$,

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}_{T_j}}\{(R_j(T_j) - \kappa)_+ | \mathcal{F}_t\} &= \mathbb{E}^{\mathbb{Q}_{T_j}}\{\mathbb{E}^{\mathbb{Q}_{T_j}}[(R_j(T_j) - \kappa)_+ | \mathcal{F}_{\mathcal{T}_{|-\infty}}]\mathcal{F}_t\} \\
&\geq \mathbb{E}^{\mathbb{Q}_{T_j}}\{\mathbb{E}^{\mathbb{Q}_{T_j}}\{(R_j(T_j)\mathcal{F}_{T_{j-1}}) - \kappa)\}_+\mathcal{F}_t\} \\
&= \mathbb{E}^{\mathbb{Q}_{T_j}}\{(R_j(T_{j-1}) - \kappa))_+\mathcal{F}_t\}
\end{aligned}$$

where we applied the martingale property of R_j, that is: $\mathbb{E}^{\mathbb{Q}_{T_j}}[R_j(T_j) | \mathcal{F}_{T_{j-1}}] = R_j(T_{j-1})$. This proves that the backward-looking caplet is always more expensive than the forward-looking one. For example, let us choose the dynamics of (6.18) to be lognormal with constant volatility, which with some abuse of notation we denote by σ_j:

$$dR_j(t) = \sigma_j R_j(t) g_j(t) dW_j(t) \tag{6.39}$$

where we assume the function g_j to be the piece-wise linear function (6.19). This dynamics results in Black-like prices for both caplets. In fact, let us denote the price at time t of the forward-looking and backward-looking caplets by $C_j^F(t)$ and $C_j^B(t)$, respectively. Then, we denote the Black forward price of the caplet by

$$\text{Black}(R, \kappa, v) = R\Phi\left(\frac{\ln(R/\kappa) + \frac{1}{2}v^2}{v}\right) - \kappa\Phi\left(\frac{\ln(R/\kappa) - \frac{1}{2}v^2}{v}\right)$$

and by Φ the standard normal distribution function, we have:

$$\begin{aligned}
C_j^F(t) &= P_j(t)\text{Black}(R_j(t), \kappa, \sum_j^F \sqrt{T_{j-1}}) \\
C_j^B(t) &= P_j(t)\text{Black}(R_j(t), \kappa, \sum_j^R \sqrt{T_j})
\end{aligned}$$

where

$$\begin{aligned}
\sum_j^F &= \sigma_j \\
\sum_j^R &= \sigma_j \sqrt{\frac{1}{3} + \frac{2}{3}\frac{T_{j-1}}{T_j}}.
\end{aligned}$$

The value of $\sum_j^R$ is obtained by calculating the integrated variance of $R_j(t)$ up to time T_j:

$$\left(\sum_{j}^{R}\right)^2 T_j = \sigma_j^2 T_{j-1} + \int_{T_{j-1}}^{T_j} \sigma_j^2 \left(\frac{T_j - t}{T_j - T_{j-1}}\right)^2 dt.$$

Since $(\sum_j^R)^2 T_j \geq \sigma_j^2 T_{j-1}$, we can confirm that $C_j^B(t) \geq C_j^F(t)$.

Caplet	Backward-looking	Forward-looking
$C(T_1, T_2)$	0.00086	0.00066
$C(T_2, T_3)$	0.00114	0.00010
$C(T_3, T_4)$	0.00137	0.00120
$C(T_4, T_5)$	0.00176	0.00157
$C(T_5, T_6)$	0.00179	0.00167
$C(T_6, T_7)$	0.00185	0.00170
$C(T_7, T_8)$	0.00200	0.00197
$C(T_8, T_9)$	0.00195	0.00187
$C(T_9, T_{10})$	0.00204	0.00195
Cap	0.0257	0.0251

Table 6.1: Forward-looking caplets vs Backward-looking caplets.

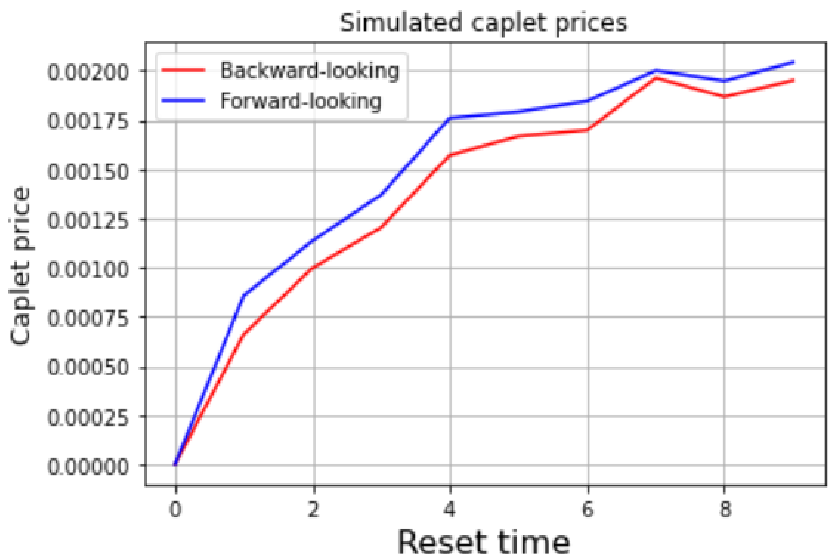


Figure 6.5: Forward-looking caplets vs Backward-looking caplets.

In Table 6.1 and Figure 6.5, we valuated the payoff of 9 caplets using the analytical Black76 formula and simulated payoff as we did in the LMM case. The caplets are of different maturities, starting from 1 to 9, where the notional is 1 and $\kappa = 0.05$. We used 2000 Monte Carlo simulations. As we move forward in time, the caplet payoffs increase. The backward-looking rates yielded payoffs that are higher than the forward-looking caplets. This is consistent with the theory discussed above.

6.6.3 The valuation of an Risk Free rate ratchet and sticky cap

In Chapter 5, we introduced ratchet and sticky caps. Since the FMM is just an extension of the LMM, it can therefore be used to price these exotic options. Since in this framework we have two types of caps, it implies that we also have two types of ratchet and sticky caps. The first ratchet cap is a forward-looking ratchet-cap, which is analogous to the ratchet cap defined in the LMM framework. The second type is a backward-looking ratchet cap.

Definition 6.4. *An interest rate ratchet cap is a financial contract consisting of a series of caplets paying at $\{T_{\alpha+1}, \ldots, T_\beta\}$ where the strike is equal to the sum of the LIBOR rate at the previous reset date a specified spread $\epsilon \in \mathbb{R}$. The discounted payoff is given by*

$$\sum_{i=\alpha+1}^{\beta} D_f(t, T_i) N \tau_i (R(T_{i-1}, T_i) - (R(T_{i-2}, T_{i-1}) + \epsilon))_+.$$

Definition 6.5. *An interest rate sticky cap is a financial contract consisting of a series of caplets paying at $\{T_{\alpha+1}, \ldots, T_\beta\}$ where the strike is equal to the sum of the cap rate at the previous reset date a specified spread $\epsilon \in \mathbb{R}$. The discounted payoff is given by*

$$\sum_{i=\alpha+1}^{\beta} D_f(t, T_i) N \tau_i (R(T_{i-1}, T_i) - (\min(R(T_{i-2}, T_{i-1}), \kappa) + \epsilon))_+.$$

Caplet	Ratchet	Sticky
$C(T_1, T_2)$	0.00025	0.00031
$C(T_2, T_3)$	0.00062	0.00069
$C(T_3, T_4)$	0.00088	0.00101
$C(T_4, T_5)$	0.00101	0.00118
$C(T_5, T_6)$	0.00135	0.00159
$C(T_6, T_7)$	0.00143	0.00165
$C(T_7, T_8)$	0.00160	0.00185
$C(T_8, T_9)$	0.00177	0.00205
$C(T_9, T_{10})$	0.00200	0.00229
Cap	0.0251	0.0257

Table 6.2: Pricing of the caps using the FMM.

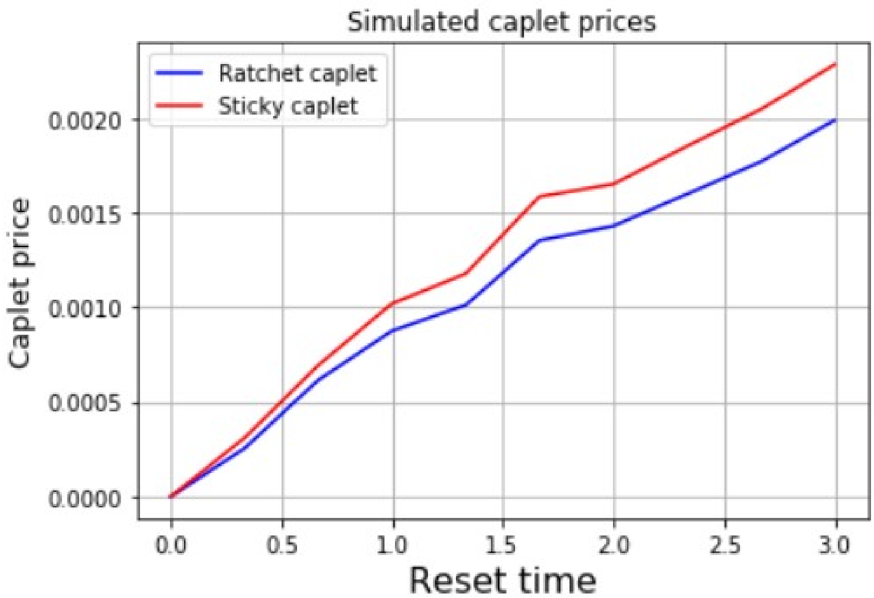


Figure 6.6: Forward-looking caplets vs Backward-looking caplets.

In Table 6.2 and Figure 6.6 we simulated both ratchet and sticky caplet prices for different maturities. The caplet are of different maturities, starting three months from now until 5 years. We set $\kappa = 0.05$ and $\epsilon = 0.01$ and used 2000 Monte Carlo simulations. Just like the standard caplets, as the maturities increase, both the caplet payoff increase. The sticky caplets are generally higher than the ratchet caplets. We notice that both these caplets have payoffs that are less than that given by the standard caplet payoffs.

6.6.4 The valuation of a RFR swaption

In this section we address the RFR swaptions. We first give a definition of the swaption in the FMM framework.

Definition 6.6. *An RFR swaption, either payer or receiver, can be defined as the option to enter a spot RFR swap on the swaption's maturity date. We denote by T_ϕ the swaption's maturity, by κ the strike price, $T_j,\ j = \phi + 1, \dots, \eta$ the floating-leg dates, and by $T'_{c+1}, \dots, T'_d$ the fixed-leg ones. We assume $T'_c = T_\phi$ and $T'_d = T_\eta$. The (payer) swaption payoff at time T_ϕ is thus given by:*

$$[S(T_\phi) - \kappa]_+ \sum_{j=c+1}^{d} \tau'_j P_t(T'_j)$$

where

$$S(t) = \frac{\sum_{j=\phi+1}^{\eta} \tau_j P_t(T_j) R_j(t)}{\sum_{j=c+1}^{d} \tau'_j P_t(T'_j)} = \frac{P_t(T_\phi) - P_t(T_\eta)}{\sum_{j=c+1}^{d} \tau'_j P_t(T'_j)}$$

Similar to the LIBOR-based swap rate, a RFR is a martingale under the forward swap measure associated with annuity numeraire, $\sum_{j=c+1}^{d} \tau'_j P_t(T'_j)$. When valuing swaptions in the FMM, where each forward rate evolves according to (6.18), is equivalent to the valuation of LIBOR-based swaptions in the single-curve LMM.

In this chapter, we gave an introduction of the generalized FMM. In the LIBOR-transition, we noticed that the rates do not stop evolving at the beginning of their accrual period, but continue to evolve until the end of the accrual period. The FMM model these new rates, called the backward-looking rates. The concept of the extended zero coupon bond is the main foundation to this new model. In this chapter, we first gave a definition of the backward-looking rate and then gave a mathematical expression of it. We discussed the properties of this new backward-looking rate. A key takeaway from the theory we discussed in this chapter is that the forward-looking forward rates and the backward-looking forward rates were the equal for all times before the accrual period and the forward-looking rate freezes at the beginning of the accrual period while the backward-looking rates continue to evolve until the end of the accrual period.

We also derived the dynamics of the FMM under three different measures, one of the measures being the risk neutral measure. In the LIBOR framework, we could not derive the dynamics of the rates under the risk-neutral measure. We also saw that in the accrual period, the volatility decreases as more and more overnight rates are realized, thus reducing the uncertainty. The FMM incorporates this with the g function. We concluded the chapter by pricing caplets suing 2000 Monte Carlo simulations. In the post-LIBOR world, two types of caps exist, the forward-looking cap and the backward-looking cap. The latter can be shown that it is more expensive that the forward-looking cap. We also introduced swaptions in the new FMM framework. In the next chapter, we will compare all caps using more Monte Carlo simulations and also under different times. We will also compare swaptions using both the forward-looking rates and the backward-looking rates.

CHAPTER 7

Numerical Results

In this chapter we look at pricing the different derivatives that we introduced in the previous chapters. We first consider the standard caplets and simulate the payoff using the forward-looking rates and the backward-looking rates. We then look at the nonstandard caplets and simulate their payoffs using the forward-looking and the backward-looking rates. We conclude by comparing the payoffs of swaptions given by the forward-looking rates and the backward-looking rates.

7.1 Valuation of the standard caps

In this section, we compare the caplet payoff given by the forward-looking and the backward-looking rates. We compare the results implied by the FMM and the analytical Black76 formula. We compare the results for two tenors, $T = 5$ and $T = 10$, with the rates resetting every three months.

Caplet	Maturity	Forward caplets	Backward Caplets	CB	CF
$C(T_1, T_2)$	0.5	0.00072	0.00094	0.00106	0.00088
$C(T_2, T_3)$	0.75	0.00104	0.00118	0.00128	0.00116
$C(T_3, T_4)$	1	0.00131	0.00145	0.00154	0.00141
$C(T_4, T_5)$	1.25	0.00151	0.00163	0.00170	0.00160
$C(T_5, T_6)$	1.5	0.00165	0.00178	0.00185	0.00173
$C(T_6, T_7)$	1.75	0.00183	0.00194	0.00200	0.00189
$C(T_7, T_8)$	2	0.00196	0.00203	0.00209	0.00202
$C(T_8, T_9)$	2.25	0.00219	0.00228	0.00233	0.00225
$C(T_9, T_{10})$	2.5	0.00217	0.00224	0.00229	0.00223
$C(T_{10}, T_{11})$	2.75	0.00239	0.00248	0.00252	0.00244
$C(T_{11}T_{12})$	3	0.00241	0.00245	0.00250	0.00246
$C(T_{12}, T_{13})$	3.25	0.00254	0.00260	0.00265	0.00259
$C(T_{13}, T_{14})$	3.5	0.00256	0.00261	0.00266	0.00260
$C(T_{14}, T_{15})$	3.75	0.00259	0.00266	0.00270	0.00264
$C(T_{15}, T_{16})$	4	0.00278	0.00285	0.00289	0.00282
$C(T_{16}, T_{17})$	4.25	0.00277	0.00283	0.00287	0.00281
$C(T_{17}, T_{18})$	4.5	0.00291	0.00297	0.00301	0.00295
$C(T_{18}, T_{19})$	4.75	0.00299	0.00306	0.00310	0.00302
$C(T_{19}, T_{20})$	5	0.00291	0.00293	0.00296	0.00295
Cap		0.041248	0.042886	0.043997	0.042462

Table 7.1: Forward-looking vs Backward-looking caplets where $T = 5$.

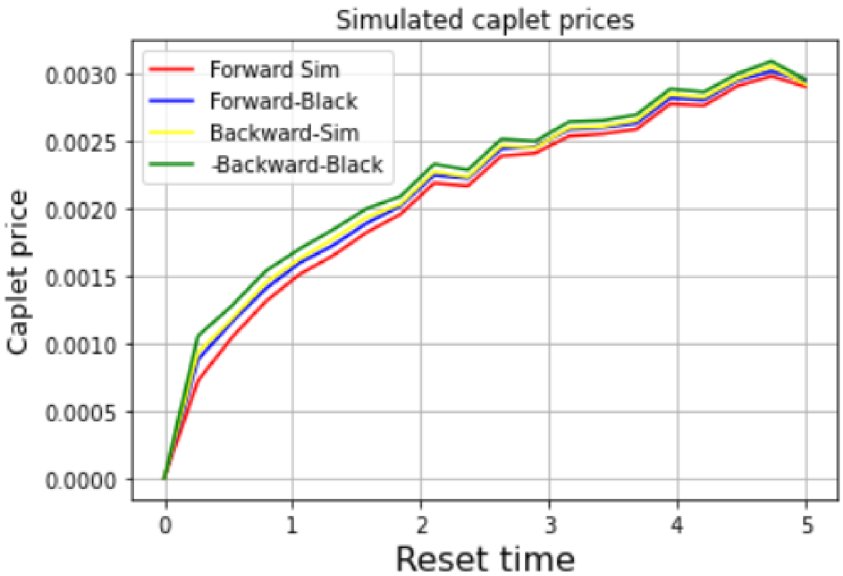


Figure 7.1: Forward-looking vs Backward-looking caplets where $T = 5$.

In Table 7.1 and Figure 7.1, we compared the forward-looking caplets with the backward-looking caplets using the Black76 formula and the option payoff definition. After 10 000 trials, we see that the backward-looking caplets are higher than the forward-looking caplets. As the maturities increase, so do the caplet prices. This is due to the fact that the rates evolve over more time.

Caplet	Maturity	Forward Caplets	Backward Caplets	CF	CB
$C(T_1, T_2)$	0.5	0.00100	0.00112	0.00127	0.00136
$C(T_2, T_3)$	0.75	0.00150	0.00159	0.00173	0.00180
$C(T_3, T_4)$	1	0.00201	0.00208	0.00216	0.00222
$C(T_4, T_5)$	1.25	0.00252	0.00257	0.00265	0.00271
$C(T_5, T_6)$	1.5	0.00270	0.00276	0.00283	0.00288
$C(T_6, T_7)$	1.75	0.00306	0.00310	0.00318	0.00322
$C(T_7, T_8)$	2	0.00374	0.00379	0.00376	0.00381
$C(T_8, T_9)$	2.25	0.00387	0.00391	0.00404	0.00408
$C(T_9, T_{10})$	2.5	0.00409	0.00413	0.00425	0.00429
$C(T_{10}, T_{11})$	2.75	0.00399	0.00403	0.00421	0.00425
$C(T_{11} T_{12})$	3	0.00433	0.00436	0.00453	0.00456
$C(T_{12}, T_{13})$	3.25	0.00512	0.00516	0.00519	0.00522
$C(T_{13}, T_{14})$	3.5	0.00525	0.00528	0.00537	0.00540
$C(T_{14}, T_{15})$	3.75	0.00542	0.00545	0.00549	0.00552
$C(T_{15}, T_{16})$	4	0.00459	0.00462	0.00462	0.00465
$C(T_{16}, T_{17})$	4.25	0.00487	0.00490	0.00498	0.00501
$C(T_{17}, T_{18})$	4.5	0.00540	0.00544	0.00554	0.00558
$C(T_{18}, T_{19})$	4.75	0.00567	0.00569	0.00591	0.00594
$C(T_{19}, T_{20})$	5	0.00679	0.00683	0.00696	0.00698
$C(T_{20}, T_{21})$	5.25	0.00594	0.00596	0.00600	0.00603
$C(T_{21}, T_{22})$	5.5	0.00596	0.00598	0.00603	0.00605
$C(T_{22}, T_{23})$	5.75	0.00617	0.00619	0.00633	0.00636
$C(T_{23}, T_{24})$	6	0.00669	0.00672	0.00646	0.00648
$C(T_{24}, T_{25})$	6.25	0.00595	0.00598	0.00610	0.00612
$C(T_{25}, T_{26})$	6.5	0.00589	0.00591	0.00602	0.00604
$C(T_{26}, T_{27})$	6.75	0.00648	0.00651	0.00643	0.00646
$C(T_{27}, T_{28})$	7	0.00658	0.00660	0.00670	0.00672
$C(T_{28}, T_{29})$	7.25	0.00622	0.00625	0.00659	0.00661
$C(T_{29}, T_{30})$	7.5	0.00602	0.00605	0.00600	0.00602
$C(T_{30}, T_{31})$	7.75	0.00656	0.00659	0.00660	0.00662
$C(T_{31}, T_{32})$	8	0.00579	0.00582	0.00593	0.00595
$C(T_{32}, T_{33})$	8.25	0.00560	0.00562	0.00565	0.00567
$C(T_{33}, T_{34})$	8.5	0.00559	0.00562	0.00561	0.00563
$C(T_{34}, T_{35})$	8.75	0.00572	0.00574	0.00563	0.00565
$C(T_{35}, T_{36})$	9	0.00558	0.00560	0.00565	0.00567
$C(T_{36}, T_{37})$	9.25	0.00683	0.00685	0.00680	0.00682
$C(T_{37}, T_{38})$	9.5	0.00559	0.00562	0.00547	0.00549
$C(T_{38}, T_{39})$	9.75	0.00515	0.00517	0.00525	0.00527
$C(T_{39}, T_{40})$	10	0.00524	0.00523	0.00533	0.00535
Cap		0.19547	0.19682	0.199253	0.20049

Table 7.2: Forward-looking vs Backward-looking standard caplets where $T = 10$.

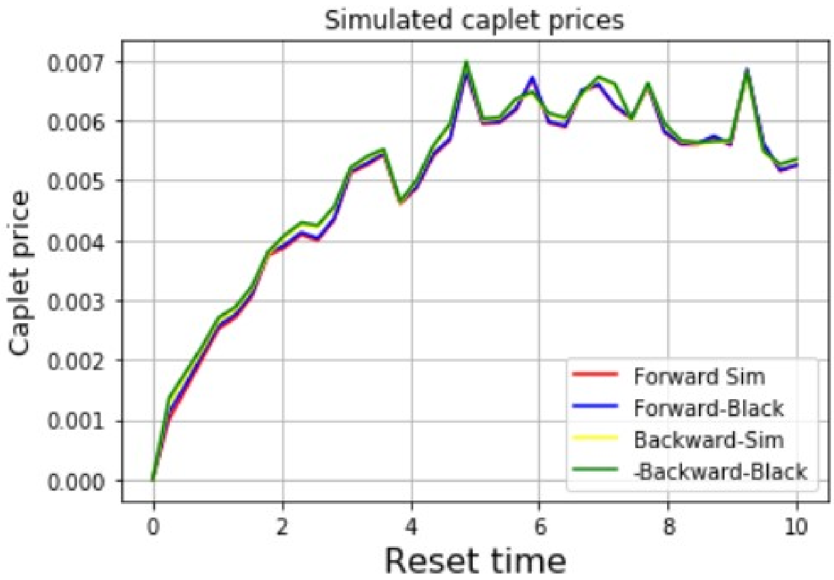


Figure 7.2: Forward-looking vs Backward-looking caplets where $T = 10$.

In Table 7.2 and Figure 7.2, we compared the forward-looking caplets with the backward-looking caplets using the Black76 formula and the option payoff definition. After 10 000 trials, we see that the backward-looking caplets are higher than the forward-looking caplets. As the maturities increase, so do the caplet prices. This is due to the fact that the rates evolve over more time.

7.2 Valuation of the nonstandard caps

In this section, we compare the nonstandard caplet payoff given by the forward-looking and the backward-looking rates. We compare the results implied by the FMM and the analytical Black76 formula. We compare the results for two tenors, $T = 5$ and $T = 10$, with the rates resetting every three months.

Caplet	Maturity	Forward Ratchet	Backward Ratchet	Forward Sticky	Backward Sticky
$C(T_1, T_2)$	0.5	0.00028	0.00028	0.00068	0.00079
$C(T_2, T_3)$	0.75	0.00088	0.00103	0.00119	0.00139
$C(T_3, T_4)$	1	0.00134	0.00156	0.00159	0.00185
$C(T_4, T_5)$	1.25	0.00164	0.00192	0.00186	0.00218
$C(T_5, T_6)$	1.5	0.00203	0.00239	0.00221	0.00259
$C(T_6, T_7)$	1.75	0.00227	0.00265	0.00241	0.00281
$C(T_7, T_8)$	2	0.00259	0.00300	0.00276	0.00319
$C(T_8, T_9)$	2.25	0.00284	0.00328	0.00300	0.00345
$C(T_9, T_{10})$	2.5	0.00306	0.00354	0.00322	0.00368
$C(T_{10}, T_{11})$	2.75	0.00316	0.00365	0.00332	0.00381
$C(T_{11} T_{12})$	3	0.00342	0.00389	0.00354	0.00403
$C(T_{12}, T_{13})$	3.25	0.00351	0.00397	0.00364	0.00412
$C(T_{13}, T_{14})$	3.5	0.00381	0.00430	0.00397	0.00448
$C(T_{14}, T_{15})$	3.75	0.00402	0.00456	0.00418	0.00472
$C(T_{15}, T_{16})$	4	0.00397	0.00449	0.00407	0.00458
$C(T_{16}, T_{17})$	4.25	0.00426	0.00478	0.00437	0.00488
$C(T_{17}, T_{18})$	4.5	0.00427	0.00479	0.00440	0.00494
$C(T_{18}, T_{19})$	4.75	0.00431	0.00483	0.00444	0.00497
$C(T_{19}, T_{20})$	5	0.00440	0.00491	0.00449	0.00500
Cap		0.041248	0.042886	0.043997	0.042462

Table 7.3: Forward-looking vs Backward-looking caplets where $T = 5$.

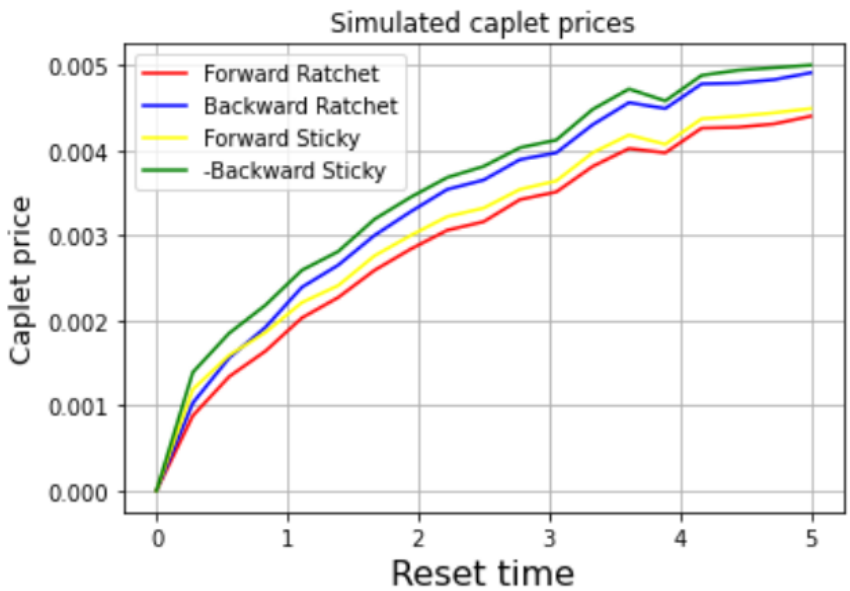


Figure 7.3: Forward-looking vs Backward-looking caplets where $T = 5$.

In Table 7.3 and Figure 7.3 we valuated Ratchet and sticky caplet prices where the tenor is $T = 5$. We set $\kappa = 0.05$ and $\epsilon = 0.01$ and 10 000 Monte Carlo simulations. We notice that the backward-looking sticky caplets yield prices that are more than those of the forward-looking sticky caplets and the ratchet caplets. Just as in the case of the standard caplets, as we move through the tenor points, the payoffs increase. We notice that these nonstandard caplets yield payoffs that are less than those of the standard caplets at the corresponding tenor points.

Caplet	Maturity	Forward Ratchet	Backward Ratchet	Forward Sticky	Backward Sticky
$C(T_1, T_2)$	0.5	0.000296	0.00110	0.00071	0.00147
$C(T_2, T_3)$	0.75	0.00081	0.00162	0.00111	0.00192
$C(T_3, T_4)$	1	0.00133	0.00217	0.00160	0.00246
$C(T_4, T_5)$	1.25	0.00167	0.00260	0.00194	0.00283
$C(T_5, T_6)$	1.5	0.00230	0.00327	0.00246	0.00343
$C(T_6, T_7)$	1.75	0.00263	0.00378	0.00288	0.00402
$C(T_7, T_8)$	2	0.00264	0.00379	0.00295	0.00410
$C(T_8, T_9)$	2.25	0.00314	0.00425	0.00338	0.00442
$C(T_9, T_{10})$	2.5	0.00326	0.00453	0.00337	0.00462
$C(T_{10}, T_{11})$	2.75	0.00364	0.00499	0.00372	0.00503
$C(T_{11}T_{12})$	3	0.00430	0.00567	0.00443	0.00582
$C(T_{12}, T_{13})$	3.25	0.00414	0.00551	0.00428	0.00571
$C(T_{13}, T_{14})$	3.5	0.00406	0.00526	0.00405	0.00526
$C(T_{15}, T_{16})$	4	0.00427	0.00562	0.00451	0.00580
$C(T_{16}, T_{17})$	4.25	0.00469	0.00596	0.00480	0.00607
$C(T_{17}, T_{18})$	4.5	0.00527	0.00676	0.00538	0.00685
$C(T_{18}, T_{19})$	4.75	0.00515	0.00643	0.00531	0.00659
$C(T_{19}, T_{20})$	5	0.00549	0.00674	0.00574	0.00696
$C(T_{20}, T_{21})$	5.25	0.00556	0.00670	0.00566	0.00678
$C(T_{21}, T_{22})$	5.5	0.00566	0.00707	0.00583	0.00723
$C(T_{22}, T_{23})$	5.75	0.00566	0.00708	0.00575	0.00718
$C(T_{23}, T_{24})$	6	0.00590	0.00729	0.00593	0.00729
$C(T_{24}, T_{25})$	6.25	0.00578	0.00706	0.00586	0.00715
$C(T_{25}, T_{26})$	6.5	0.00545	0.00657	0.00560	0.00668
$C(T_{26}, T_{27})$	6.75	0.00620	0.00728	0.00622	0.00732
$C(T_{27}, T_{28})$	7	0.00603	0.00742	0.00616	0.00757
$C(T_{28}, T_{29})$	7.25	0.00596	0.00711	0.00603	0.00715
$C(T_{29}, T_{30})$	7.5	0.00574	0.00686	0.00596	0.00707
$C(T_{30}, T_{31})$	7.75	0.00663	0.00777	0.00676	0.00793
$C(T_{31}, T_{32})$	8	0.00645	0.00751	0.00646	0.00750
$C(T_{32}, T_{33})$	8.25	0.00665	0.00766	0.00675	0.00772
$C(T_{33}, T_{34})$	8.5	0.00632	0.00725	0.00646	0.00734
$C(T_{34}, T_{35})$	8.75	0.00661	0.00743	0.00672	0.00756
$C(T_{35}, T_{36})$	9	0.00611	0.00697	0.00601	0.00684
$C(T_{36}, T_{37})$	9.25	0.00641	0.00723	0.00654	0.00734
$C(T_{37}, T_{38})$	9.5	0.00618	0.00694	0.00624	0.00703
$C(T_{38}, T_{39})$	9.75	0.00637	0.00716	0.00652	0.00703
$C(T_{39}, T_{40})$	10	0.00579	0.00645	0.00589	0.00703
Cap		0.18051	0.22288	0.18587	0.22792

Table 7.4: Forward-looking vs Backward-looking nonstandard caplets where $T = 10$.

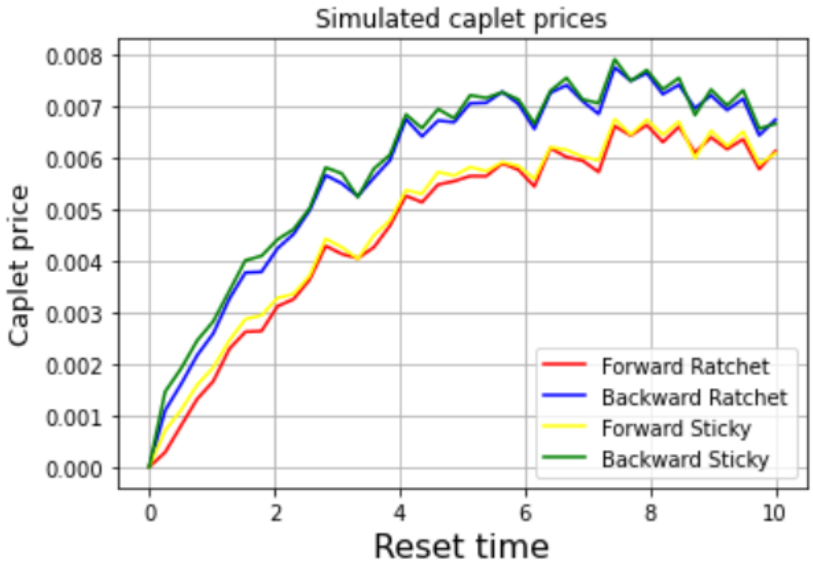


Figure 7.4: Forward-looking vs Backward-looking nonstandard caplets where $T = 10$.

In Table 7.4 and Figure 7.4 we valuated Ratchet and sticky caplet prices where the tenor is $T = 10$. We set $\kappa = 0.05$ and $\epsilon = 0.01$ and 10 000 Monte Carlo simulations. We notice that the backward-looking sticky caplets yield prices that are more than those of the forward-looking sticky caplets and the ratchet caplets. Just as in the case of the standard caplets, as we move through the tenor points, the payoffs increase. We notice that these nonstandard caplets yield payoffs that are less than those of the standard caplets at the corresponding tenor points.

7.3 Valuation of swaptions

In this section we compare the swaption payoffs given by the forward-looking rates and the backward-looking rates with tenors from 1 to 10 years and swaption maturities from 1 to 5 years.

Swaption maturity (Years)					
Length (Years)	1	2	3	4	5
1	0.33877	0.28893	0.23752	0.19252	0.16537
2	0.62753	0.54543	0.49127	0.36536	0.31860
3	0.88480	0.78203	0.71550	0.52729	0.46591
4	1.12042	1.00427	1.00027	0.68316	0.61411
5	1.34470	1.22486	1.15817	0.84090	0.76696
6	1.56379	1.44671	1.39136	1.00041	1.0001
7	1.78649	1.68069	1.64532	1.17106	1.11099
8	2.02093	1.93380	1.92843	1.36121	1.31995
9	2.27243	2.21902	1.83419	1.57726	1.56915
10	2.55804	2.31424	2.10183	1.83520	1.67802

Table 7.5: A table of swaption payoffs given by the forward-looking rates.

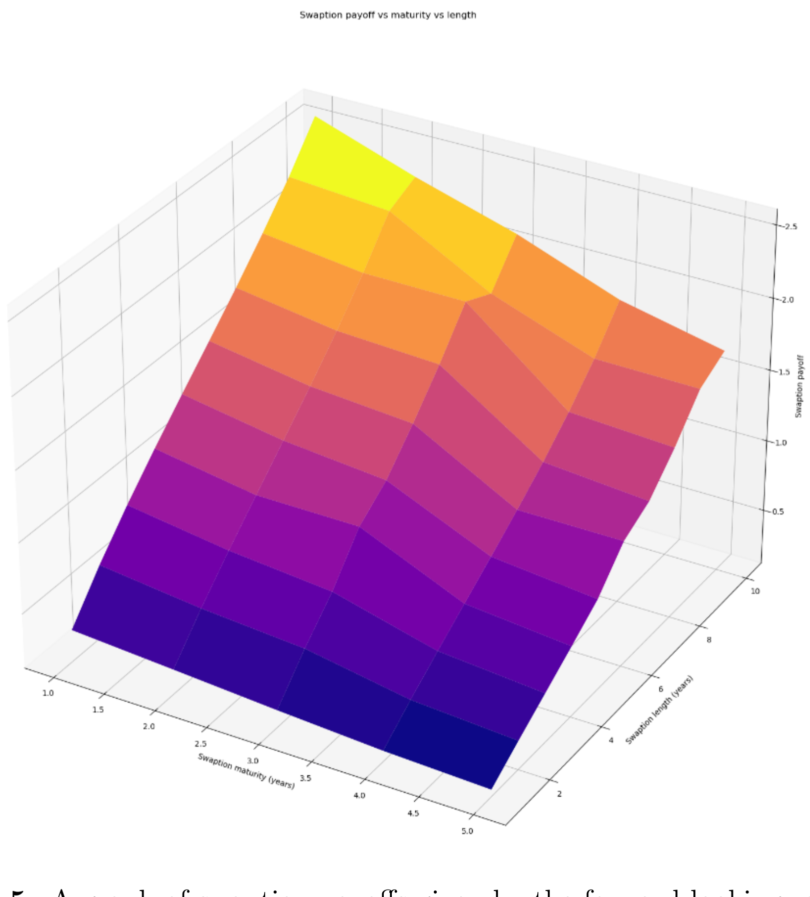


Figure 7.5: A graph of swaption payoffs given by the forward-looking rates.

In Table 7.5 and Figure 7.5, we set the $\kappa = 0.05$ and $\sigma = 0.20$ and conducted 10 000

Monte Carlo trials. We notice that the swaption payoff decreases as maturity increases. For a given maturity, the swaption payoff increases with the tenor.

	Swaption maturity (Years)				
Length (Years)	1	2	3	4	5
1	0.28739	0.24629	0.21099	0.18577	0.16099
2	0.53827	0.46397	0.40065	0.35756	0.31147
3	0.76464	0.66062	0.57967	0.52027	0.45507
4	0.96968	0.84752	0.75465	0.67883	0.59693
5	1.17136	1.02728	0.92272	0.83359	0.74086
6	1.36859	1.20817	1.10142	0.99701	0.88871
7	1.56431	1.39322	1.29028	1.16611	1.05054
8	1.76654	1.58401	1.42663	1.34903	1.22460
9	2.00247	1.79920	1.63372	1.55559	1.42469
10	2.23389	2 .04676	1.87311	1.78794	1.65019

Table 7.6: A table of swaption payoffs given by the backward-looking rates.

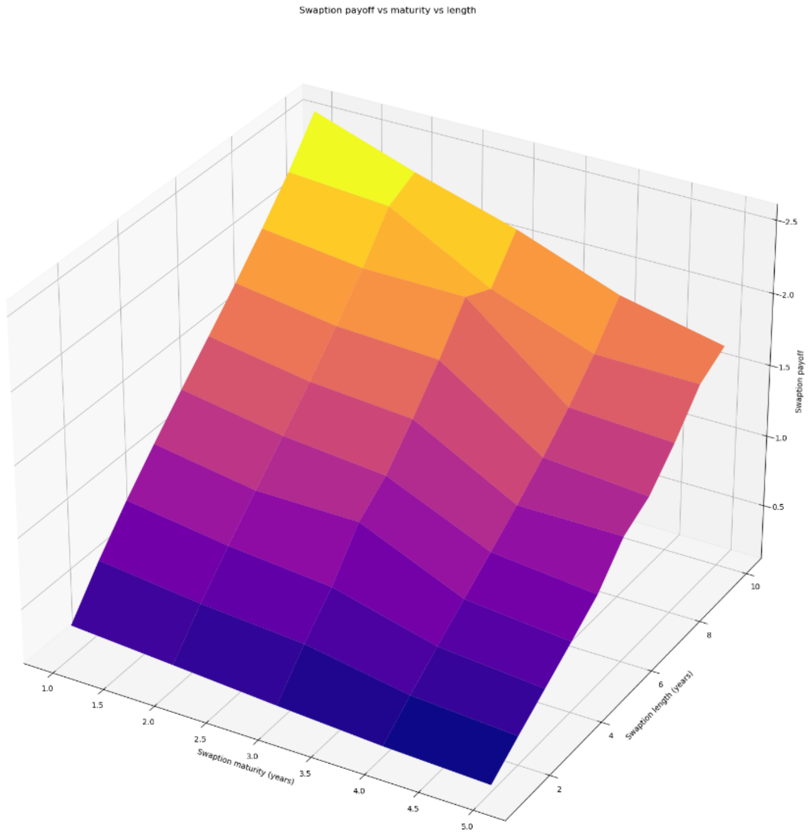


Figure 7.6: A graph of swaption payoffs given by the backward-looking rates.

In Table 7.6 and Figure 7.6, we set the $\kappa = 0.05$ and $\sigma = 0.20$ and conducted 10 000 Monte Carlo trials. We notice that the swaption payoff decreases as maturity increases. For a given maturity, the swaption payoff increases with the tenor. We notice that the swaptions given by the backward-looking rates are generally lower than those given by forward-looking rates. Both these swaptions, however, depict a similar trend as shown by the two graphs above.

CHAPTER 8

Conclusion and further research

The financial crisis of 2007-09 exposed the notion of too big to fall idea that was prevalent in the market and ever since then, the interbank borrowing activity has declined rapidly. This had an impact on the LIBOR rates setting since this rate relied on a few AA rated banks and above borrowings submissions. The market participants have since settled on alternate risk free rates that will be replacing the LIBOR rates at the beginning of 2022. Most countries have already settled on their alternative risk free rates. These new risk free rates should first be converted into term rates, and the best approaching of doing this is by using the setting-in-arrears method. The LIBOR market model, which has been the most favored model for pricing the products in the market such as the caps, swaptions, and even exotic products does not have the capability of modelling the new risk free rates, and thus a new model is needed. This is due to the fact that the LMM only model the forward rates until the beginning of their accrual periods and then stops evolving afterwards. These new rates, however, keep evolving until the end of their accrual period. A new model which is essentially an extension of the LIBOR market model was proposed in the literature in 2018 which has the capability of modelling the new risk free rates.

In this dissertation, we first gave a background of the interest rate models and how they progressed from equilibrium models to the market models. In the LIBOR market model frame work, we derived the dynamics of the forward rates under the spot measure and the terminal measure. We noticed that the risk-neutral dynamics were hard to derive under this framework. When implementing the LIBOR market model, we made use of Ito's lemma and made the assumption of constant drift per step (pricer predictor method). When using this method, we can move from one tenor point to another. We then proceeded to simulate the payoffs of the standard caplet using the analytical Black76 formula and using the option payoff formula. We concluded that the payoffs were very close to each other after using 2000 Monte Carlo simulations. We also simulated of two nonstandard caplets, namely, the ratchet cap and the sticky cap. We noticed that the sticky caps had a higher payoff compared to the ratchet caps. We concluded by discussing swaptions in the LMM framework.

We then introduced the generalized Forward market model. One of the key differences between the LIBOR market model and the generalized Forward market model is that the former is based on the normal zero-coupon bonds and the latter is based on the extended zero-coupon bonds. The two zero-coupon bonds differ in that one is only defined for certain times t while the latter is defined for all times t, including for times $t > T$. The generalized Forward market model can therefore have dynamics that are defined for all times t. This model also preserves the dynamics of the forward-looking rates previously given by the LIBOR market model and also provide some extra properties that cannot be found in the LMM framework. The fact that the generalized forward market model can model both the forward-looking and backward-looking rates should make it favorable over the LIBOR market model. We introduced the definition of the new backward-looking forward rates along with its mathematical expression. We also introduced the properties of this new backward-looking rates. We noticed that just as in the LIBOR market model framework, this new backward-looking rate is a martingale under the corresponding T_j-measure. For times $t \leq T$, the forward-looking rate and the backward-looking rate are the same under the new generalized Forward market model framework. We also introduced the dynamics of the backward-looking dynamics under the martingale measure and realized that the volatility had special characteristics. We noticed that the volatility of the new backward-looking rates decreased as more and more overnight rates were set. We then proceeded to derive the dynamics of the new rates under different measures just like we did in the LIBOR market model framework using the change-of-measure technique. We derived the dynamics under the risk neutral measure, spot measure, and the terminal measure. It is worth noting that the risk-neutral dynamics could not be obtained in the LIBOR market model framework. When implementing the generalized Forward market model, we used a combination of the pricer predictor method and Euler's method. The former was used for times $t \leq T_j$ whereas the latter was used for the accrual period. Just as we did in the LIBOR market model framework, we proceeded to simulate the payoffs of the standard caplet using the analytical Black76 formula and using the option payoff formula. We concluded that the payoffs were very close to each other after using 2000 Monte Carlo simulations. We also simulated of two nonstandard caplets, namely, the ratchet cap and the sticky cap. We noticed that the sticky caps had a higher payoff compared to the ratchet caps. We concluded the chapter by discussing swaptions in the FMM framework and discovered that it is the same as in the LMM framework.

In the last chapter, we proceeded to test the generalized Forward market model caplets payoffs under different tenors and a larger number of Monte Carlo simulations, 10 000 simulations to validate the model. We compared the forward-looking caplets with the backward-looking caplets and saw that the backward-looking payoff was always higher than the forward-looking payoff. We also compared the swaption payoffs given by forward-looking rates with the backward-looking rates. We noticed that the forward-looking rates yielded a higher payoff than the backward-looking swaption payoff.

Further research

In this dissertation, we priced European caplets, two exotic caplets, namely the ratchet and the sticky caplets and swaptions. Another area that can be explored to expand upon the research conducted in this dissertation is to consider pricing non-simple exotic options such as Barrier options and American options. One might also be interested in calibrating different parameters of this model to different model parameters. Since swaptions are dependent on multiple forward rates, it would be interesting to observe the behavior of the model to the calibration to swaption volatilities. When pricing caplets, the favored analytical formula is the Black76 formula. Baaquie has proposed an alternative to this theory, called the quantum field theory. In the Black76 formula, each caplet has its own volatility function, whereas in the quantum field, a single volatility function can be used to price a bond option. This field looks very attractive and it would be very interesting to see how the new backward-looking caplets would behave under this theory.

APPENDIX A

Pricing caps using Monte Carlo

In 1976, Phelim.P.Boyle provided an alternative way of obtaining the value of derivatives known as the Monte Carlo method. This method are generally used to evaluate integrals of the form

$$M(f) = \int_R f(Y)dY \tag{A.1}$$

where f is an L^2 arbitrary function, R is the range of integration, and $X \in R$.

We can use this method to find an estimate of the integral as follows

$$\hat{M}(f) = \frac{1}{n}\sum_{i=1}^{n} f(Y_i). \tag{A.2}$$

We can obtain the volatility v of the estimate as follows

$$v = \frac{1}{n-1}\sum_{i=1}^{n}(I(Y_i - \hat{I})^2.$$

Monte Carlo is used to find an approximation of Equation (A.1) the more preferred method as we increase the dimension of the function increases. This approximation is equivalent to the mean of the function f over n points that are picked randomly. The law of large numbers says that this converges to the true value of the integral as $n \to \infty$. Furthermore, the Central limit theorem guarantees that the sample mean distribution of $f(Y)$ will assume a near-normal or normal distribution if the n is large enough.

The standard error, denoted by se of the estimate as follows

$$\frac{\sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(I(Y_i - \hat{I})^2.}}{\sqrt{n}} \tag{A.3}$$

APPENDIX B

Stochastic Calculus

Definition B.1. *A diffusion coefficient operator $D^C(\cdot)$ can be defined as an operator that extracts the vector diffusion coefficient from the dynamics of a given diffusion process. Consider indeed $D^{C_t}(Z)$ as the row vector v_t in*

$$dZ(t) = \mu(t)Z(t)dt + \sigma_t Z(t)dW_t$$

for diffusion processes. Consider the following SDE $dZ_1(t) = \sigma_1 Z_1(t)(CdW)_1(t)$, then

$$D^{C_t}(Z_1) = [\sigma_1 Z_1(t),\ 0,\ 0, \ldots,\ 0] = \sigma_1 Z_1(t) e_1 \tag{B.1}$$

where $e_1 = [1,\ 0 \ldots\ , 0]$ is the first vector in the canonical basis.

Definition B.2. *A stochastic process $V = (V(t);\ t \geq 0)$ is called a generalized Ornstein-*

Uhlenbeck type process if it is given by

$$V(t) = V_0\rho(t) + \int_0^t \rho(t-u)dL(u), \quad \forall t \geq 0, \tag{B.2}$$

where

$$\frac{d\rho(t)}{dt} = -\int_0^t \rho(u)d\mu_t(u), \quad with \rho(0) = 1. \tag{B.3}$$

In (B.3), μ_t is a signed measure for each $t \geq 0$, $V(0) = V_0$ is the initial condition and it can either be a random variable or deterministic real constant. The function $\rho(\cdot)$ is deterministic and it is referred to as the *memory kernel function.*

Lemma B.1. *Let S be a random variable with a log-normal distribution that has a mean μ and a variance σ^2, i.e $S = Ce^X$ with $X \sim \mathcal{N}(\mu, \sigma^2)$ and $C \in \mathbb{R}$, and $K \in \mathbb{R}^+$. Then:*

$$\mathbb{E}\{(S-K)^+\} = Ce^{\frac{\sigma^2}{2}+\mu}\Phi\left(\frac{\ln\frac{C}{K} + \mu + \sigma^2}{\sigma}\right) - K\Phi\left(\frac{\ln\frac{C}{K} + \mu}{\sigma}\right) \tag{B.4}$$

where Φ is the cdf for the $\mathcal{N}(0,1)$ distribution.

Proof.

$$
\begin{aligned}
\mathbb{E}\{(S-K)^+\} &= \int_{\mathbb{R}} \max(Ce^x - K, 0) - \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \\
&= \frac{1}{\sqrt{2\pi}\sigma} \int_{(Ce^x > K)} (Ce^x - K) e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \\
&= \frac{1}{\sqrt{2\pi}\sigma} \int_{(x > \ln\frac{K}{C})} Ce^x e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx - K \int_{(x > \ln\frac{K}{C})} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \\
&= \frac{1}{\sqrt{2\pi}\sigma} \int_{(\ln\frac{K}{C})}^{+\infty} Ce^{-\frac{(x^2 + (\mu)^2 - 2x\mu - 2x\sigma^2)}{2\sigma^2}} dx - K \int_{(\ln\frac{K}{C})}^{+\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \\
&= \frac{1}{\sqrt{2\pi}\sigma} \int_{\frac{(\ln\frac{K}{C}) - \mu - \sigma^2}{\sigma}}^{+\infty} Ce^{-\frac{z^2}{2}} e^{\frac{\sigma^2}{2} + \mu} \sigma dz - K \int_{\frac{(\ln\frac{K}{C}) - \mu}{\sigma}}^{+\infty} e^{-\frac{u^2}{2}} \sigma dx \qquad \text{(B.5)} \\
&= \frac{1}{\sqrt{2\pi}\sigma} \left(Ce^{\frac{\sigma^2}{2} + \mu} \int_{\frac{(\ln\frac{K}{C}) - \mu - \sigma^2}{\sigma}}^{+\infty} e^{-\frac{z^2}{2}} \sigma dz - K \int_{\frac{(\ln\frac{K}{C}) - \mu}{\sigma}}^{+\infty} e^{-\frac{u^2}{2}} \sigma dx \right) \qquad \text{(B.6)} \\
&= Ce^{\frac{\sigma^2}{2} + \mu} \Phi\left(\frac{\ln\frac{C}{K} + \mu + \sigma^2}{\sigma} \right) - K\Phi\left(\frac{\ln\frac{C}{K} + \mu}{\sigma} \right) \qquad \text{(B.7)}
\end{aligned}
$$

where the two summands in B.5-B.6 are obtained respectively with the following change of variable:

$$
z = \frac{x - \mu - \sigma^2}{\sigma}, u = \frac{x - \mu}{\sigma}.
$$

□

Corollary 1. *If S is a random variable that has a log-normal distribution with mean $-\frac{1}{2}\sigma^2$ and variance σ^2, i.e. $S = Ce^X$ with $X \sim \mathcal{N}(-\frac{1}{2}\sigma^2, \sigma^2)$ and $C \in \mathbb{R}$ and $K \in \mathbb{R}^+$, then:*

$$
\mathbb{E}\{(S-K)^+\} = C\Phi(d1(K, C, \sigma^2)) - K\Phi(d2(K, C, \sigma^2)), \qquad \text{(B.8)}
$$

where

$$d1(K, S, \sigma^2) := \frac{\ln(\frac{S}{K}) + \frac{\sigma^2}{2}}{\sigma}, \tag{B.9}$$

$$d2(K, S, \sigma^2) := d1(K, S, \sigma^2) - \sigma. \tag{B.10}$$

Proof. By substituting μ with $-\frac{1}{2}\sigma^2$ from B.4, the result follows. $\square$

APPENDIX C

Code

```
def vasicek_model(r0, theta, sig, k, n, m, T):
    r = np.zeros((n, m))
    r[:, 0] = r0
    dt = T/m
    for it in range(0, n):
        for j in range(0, m-1):
            r[it, j+1] = r[it, j] + k*(theta-r[it, j])*dt + sig*dw(dt)
    return r
```

```
def ho_lee_model(r0, sig, n, m, T):
    r = np.zeros((n, m))
    r[:, 0] = r0
    dt = T/m
    thetat =np.linspace(0.02*standard_normal(), 0.5*standard_normal(), m)
    for it in range(0, n):
        for j in range(0, m-1):
            r[it, j+1] = r[it, j] + thetat[j]*r[it, j]*dt + sig*r[it, j]*
```

```
dw(dt)
    return r

```

```
def gFMM2(r0, dlt, sig, m, T,K):
    Time = np.linspace(0, T, m) #Time
    n=len(Time)                 #Length of time
    r = np.zeros((n, n))        #matrix of the LIBORs
    dt = T/n                    #Steps
    r[:, 0] = r0                #Initial Values
    RFR=np.zeros((n,32))        #Matrix of backwardlooking rates
    RFR[0,0]=r0                 #First backward-looking rate
    D=np.zeros(n)

    #Simulating the forward-looking rates
    for j in range(0, n):
        for it in range(j+1, n):
            summation = 0
            for k in range(it+1, n):
                        summation+=(sig*dlt*r[k, j])/(1+dlt*r[k, j])
            r[it, j+1] = r[it, j]*np.exp((-summation*sig-0.5*sig**2)*dt+
   sig*dw(dt)) #Simulating the LIBOR forward rates
            if j+1==it:
                RFR[it,0]=r[it,j+1]    #Storing the LIBORs

    #Simulating the accrual period
    X=np.linspace(0,0.25,32)  #This spans the accrual period
    g=[]
    for j in range(len(X)):
        g.append(min( max(0.25-X[j],0)/0.25,1))

    for it in range(n):
        for j in range(1,len(X)):
            summation=0
```

```
            for k in range(0,j):
                summation+=((sig/4)*sig*RFR[it,k]*g[k])/(1+sig*RFR[it,k])
            RFR[it,j]=RFR[it,j-1]+(-summation*RFR[it,j-1]*g[j-1]*(sig/4)
*0.025+RFR[it,j-1]*(sig/4)*g[j-1]*dw(dt))


    #Calculating the bonds
        P  = np.zeros((n, n))
    for j in range(n):
            for it in range(j, n):
                Prd = 1
                for k in range(j, it+1):
                    Prd = Prd * (1 + dlt * r[k][j])**-1
                P[it][j] = Prd
                if it==j:
                    D[j]=P[it][j]
    ##Cap payoff
    C=np.zeros(n)
    C2=np.zeros(n)
    for it in range(n):
        C[it]=D[it]*0.25*max(RFR[it,0]-K,0) #Forward looking cap
        C2[it]=D[it]*0.25*max(RFR[it,-1]-K,0) #Backward looking cap\


      #Black76
    Caps=np.zeros(n) #This matrix stores the black caplets
    d1=np.zeros(n)
    d2=np.zeros(n)
    T=np.arange(0,n,0.25)
    for it in range(n):
        d1[it]=(log(RFR[it,0]/K)+0.5*0.25*((sig*sqrt(T[0]))**2))/(sig*
sqrt(T[0]))
        d2[it]=d1[it]-sig*sqrt(0.25)
```

```
    for it in range(n):
        Caps[it]=max(D[it]*0.25*((RFR[it,0]*scipy.stats.norm.cdf(d1[it])-K*scipy.stats.norm.cdf(d2[it]))),0)
    #Backward-looking Black 76
    Caps2=np.zeros(n) #This matrix stores the black caplets
    d1=np.zeros(n)
    d2=np.zeros(n)
    T=np.arange(0,n,0.25)
    for it in range(n):
        d1[it]=(log(RFR[it,-1]/K)+0.5*0.25*((sig*sqrt(1/3+(2/3)*(T[it])/(T[it]+0.25))*sqrt(0.5))**2))/(sig*sqrt(1/3+(2/3)*(T[it])/(T[it]+0.25))*sqrt(0.5))
        d2[it]=d1[it]-sig*sqrt(0.25)
    for it in range(n):
        Caps2[it]=max(D[it]*0.25*((RFR[it,-1]*scipy.stats.norm.cdf(d1[it])-K*scipy.stats.norm.cdf(d2[it]))),0)
    BCap2=sum(Caps2)
     ##Ratchet Cap payoff
    Ratchet=np.zeros(n)
    Ratchet2=np.zeros(n)
    e=0.01
    for it in range(1, n):
        Ratchet[it]=D[it]*0.25*max(RFR[it,0]-(RFR[it-1,0]+e),0) #Forward looking cap
        Ratchet2[it]=D[it]*0.25*max(RFR[it,-1]-(RFR[it-1,-1]+e),0) #Backward looking cap

    ##Sticky Cap payoff
    Sticky=np.zeros(n)
    Sticky2=np.zeros(n)
    e=0.01
    for it in range(1, n):
        Sticky[it]=D[it]*0.25*max(RFR[it,0]-(min(RFR[it-1,0],K)+e),0) #
```

```
Forward looking cap
            Sticky2[it]=D[it]*0.25*max(RFR[it,-1]-(min(RFR[it-1,-1],K)+e),0)
#Backward looking cap



     #Calculating the swaption payoff
    S=np.zeros(n)
    for it in range(16,56):
        S[it]=D[it]*0.25*(F[it,-1]-K)
    Swap=D[-1]*max(sum(S),0)
    return RFR, Caps, Caps2, Ratchet,Ratchet2, Sticky, Sticky2

```

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